(a) $\phi(A) = SAS^{-1}$, $A \in SL(n,\mathbb{C})$ or

(b) $\phi(A) = S\overline{A}S^{-1}$, $A \in SL(n,\mathbb{C})$.

Here, $\overline{A}$ denotes the matrix obtained from A by applying the complex conjugation entrywise.

3. [PS98] Let $n \geq 3$ and let $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ be a continuous mapping. Then ϕ preserves spectrum, commutativity, and rank one matrices (no linearity, additivity, or multiplicativity is assumed) if and only if there exists an invertible matrix $R \in \mathbb{C}^{n\times n}$ such that ϕ is an (R, R^{-1})-standard map.
4. [BR00] Let $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ be a spectrum preserving C^1-diffeomorphism (again, we do not assume that ϕ is additive or multiplicative). Then ϕ is a local similarity.
5. [HHW04] Let $n \geq 2$. Then $\phi : \mathcal{H}_n \to \mathcal{H}_n$ is a bijective map such that $\phi(A)$ and $\phi(B)$ are adjacent for every adjacent pair $A, B \in \mathcal{H}_n$ if and only if there exist a nonzero real number c, an invertible $R \in \mathbb{C}^{n\times n}$, and $S \in \mathcal{H}_n$ such that either

   (a) $\phi(A) = cRAR^* + S$, $A \in \mathcal{H}_n$ or

   (b) $\phi(A) = cR\overline{A}R^* + S$, $A \in \mathcal{H}_n$.
6. [Mol01] Let $n \geq 2$ be an integer and $\phi : \mathcal{H}_n \to \mathcal{H}_n$ a bijective map such that $\phi(A) \leq \phi(B)$ if and only if $A \leq B$, $A, B \in \mathcal{H}_n$ (here, $A \leq B$ if and only if $B - A$ is a positive semidefinite matrix). Then there exist an invertible $R \in \mathbb{C}^{n\times n}$ and $S \in \mathcal{H}_n$ such that either

   (a) $\phi(A) = RAR^* + S$, $A \in \mathcal{H}_n$ or

   (b) $\phi(A) = R\overline{A}R^* + S$, $A \in \mathcal{H}_n$.

   This result has an infinite-dimensional analog important in quantum mechanics. In the language of quantum mechanics, the relation $A \leq B$ means that the expected value of the bounded observable A in any state is less than or equal to the expected value of B in the same state.

Examples:

1. We define a mapping $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ in the following way. For a diagonal matrix A with distinct diagonal entries, we define $\phi(A)$ to be the diagonal matrix obtained from A by interchanging the first two diagonal elements. Otherwise, let $\phi(A)$ be equal to A. Clearly, ϕ is a bijective mapping preserving spectrum, rank, and commutativity in both directions. This shows that the continuity assumption is indispensable in Fact 3 above.
2. Let $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ be a map defined by $\phi(0) = E_{12}$, $\phi(E_{12}) = 0$, and $\phi(A) = A$ for all $A \in \mathbb{C}^{n\times n} \setminus \{0, E_{12}\}$. Then ϕ is a bijective spectrum preserving map that is not a local similarity. More generally, we can decompose $\mathbb{C}^{n\times n}$ into the disjoint union of the classes of matrices having the same spectrum and then any bijection leaving each of this classes invariant preserves spectrum. Thus, the assumption on differentiability is essential in Fact 4 above.

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23

Matrices over Integral Domains

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In this chapter, we present some results on matrices over integral domains, which extend the well-known results for matrices over the fields discussed in Chapter 1 of this book. The general theory of linear algebra over commutative rings is extensively studied in the book [McD84]. It is mostly intended for readers with a thorough training in ring theory. The aim of this chapter is to give a brief survey of notions and facts about matrices over classical domains that come up in applications. Namely over the ring of integers, the ring of polynomials over the field, the ring of analytic functions in one variable on an open connected set, and germs of analytic functions in one variable at the origin. The last section of this chapter is devoted to the notion of strict equivalence of pencils.

Most of the results in this chapter are well known to the experts. A few new results are taken from the book in progress [Frixx], which are mostly contained in the preprint [Fri81].

23.1 Certain Integral Domains

Definitions:

A commutative ring without zero divisors and containing identity 1 is an **integral domain** and denoted by $\mathbb{D}$.

The **quotient field** F of a given integral domain $\mathbb{D}$ is formed by the set of equivalence classes of all quotients $\frac{a}{b}, b \neq 0$, where $\frac{a}{b} \equiv \frac{c}{d}$ if and only if $ad = bc$, such that

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b}\frac{c}{d} = \frac{ac}{bd}, \quad b, d \neq 0.$$

For $\mathbf{x} = [x_1, \ldots, x_n]^T \in \mathbb{D}^n, \alpha = [\alpha_1, \ldots, \alpha_n]^T \in \mathbb{Z}_+^n$ we define $\mathbf{x}^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ and $|\alpha| = \sum_{i=1}^n |\alpha_i|$.

$\mathbb{D}[\mathbf{x}] = \mathbb{D}[x_1, \ldots, x_n]$ is the ring of all **polynomials** $p(\mathbf{x}) = p(x_1, \ldots, x_n)$ in n variables with coefficients in $\mathbb{D}$:

$$p(\boldsymbol{x}) = \sum_{|\alpha| \leq m} a_\alpha \boldsymbol{x}^\alpha.$$

The **total degree**, or simply the **degree** of $p(\mathbf{x}) \neq 0$, denoted by deg p, is the maximum $m \in \mathbb{Z}_+$ such that there exists $a_\alpha \neq 0$ such that $|\alpha| = m$. (deg $0 = -\infty$.)

A polynomial p is **homogeneous** if $a_\alpha = 0$ for all $|\alpha| <$ deg p.

A polynomial $p(x) = \sum_{i=0}^{n} a_i x^i \in \mathbb{D}[x]$ is **monic** if $a_n = 1$.

$F(\mathbf{x})$ denotes the quotient field of $F[\mathbf{x}]$, and is the **field of rational functions** over F in n variables.

Let $\Omega \subset \mathbb{C}^n$ be a nonempty path-connected set. Then $\mathrm{H}(\Omega)$ denotes the ring of analytic functions $f(\mathbf{z})$, such that for each $\zeta \in \Omega$ there exists an open neighborhood $O(\zeta, f)$ of ζ such that f is analytic on $O(f, \zeta)$. The addition and the product of functions are given by the standard identities: $(f+g)(\zeta) = f(\zeta)+g(\zeta), (fg)(\zeta) = f(\zeta)g(\zeta)$. If Ω is an open set, we assume that f is defined only on Ω. If Ω consists of one point ζ, then H_ζ stands for $\mathrm{H}(\{\zeta\})$.

Denote by $\mathcal{M}(\Omega), \mathcal{M}_\zeta$ the quotient fields of $\mathrm{H}(\Omega), \mathrm{H}_\zeta$ respectively.

For $a, d \in \mathbb{D}$, d **divides** a (or d is a **divisor** of a), denoted by $d|a$, if $a = db$ for some $b \in \mathbb{D}$.

$a \in \mathbb{D}$ is **unit** if $a|1$.

$a, b \in \mathbb{D}$ are **associates**, denoted by $a \equiv b$, if $a|b$ and $b|a$. Denote $\{\{a\}\} = \{b \in \mathbb{D} : \ b \equiv a\}$.

The associates of $a \in \mathbb{D}$ and the units are called **improper** divisors of a.

$a \in \mathbb{D}$ is **irreducible** if it is not a unit and every divisor of a is improper.

A nonzero, nonunit element $p \in \mathbb{D}$ is **prime** if for any $a, b \in \mathbb{D}$, $p|ab$ implies $p|a$ or $p|b$.

Let $a_1, \ldots, a_n \in \mathbb{D}$. Assume first that not all of $a_1, \ldots, a_n$ are equal to zero. An element $d \in \mathbb{D}$ is a **greatest common divisor** (g.c.d) of $a_1, \ldots, a_n$ if $d|a_i$ for $i = 1, \ldots, n$, and for any d' such that $d'|a_i, i = 1, \ldots, n$, $d'|d$. Denote by $(a_1, \ldots, a_n)$ any g.c.d. of $a_1, \ldots, a_n$. Then $\{\{(a_1, \ldots, a_n)\}\}$ is the equivalence class of all g.c.d. of $a_1, \ldots, a_n$. For $a_1 = \ldots = a_n = 0$, we define 0 to be the g.c.d. of $a_1, \ldots, a_n$, i.e. $(a_1, \ldots, a_n) = 0$.

$a_1, \ldots, a_n \in \mathbb{D}$ are **coprime** if $\{\{(a_1, \ldots, a_n)\}\} = \{\{1\}\}$.

$I \subseteq \mathbb{D}$ is an **ideal** if for any $a, b \in I$ and $p, q \in \mathbb{D}$ the element $pa + qb$ belongs to I.

An ideal $I \neq \mathbb{D}$ is **prime** if $ab \in I$ implies that either a or b is in I.

An ideal $I \neq \mathbb{D}$ is **maximal** if the only ideals that contain I are I and $\mathbb{D}$.

An ideal I is **finitely generated** if there exists k elements (**generators**) $p_1, \ldots, p_k \in I$ such that any $i \in I$ is of the form $i = a_1 p_1 + \cdots + a_k p_k$ for some $a_1, \ldots, a_k \in \mathbb{D}$.

An ideal is **principal** ideal if it is generated by one element p.

$\mathbb{D}$ is a **greatest common divisor** domain (GCDD), denoted by $\mathbb{D}_g$, if any two elements in $\mathbb{D}$ have a g.c.d.

$\mathbb{D}$ is a **unique factorization** domain (UFD), denoted by $\mathbb{D}_u$, if any nonzero, nonunit element a can be factored as a product of irreducible elements $a = p_1 \cdots p_r$, and this factorization is unique within order and unit factors.

$\mathbb{D}$ is a **principal ideal** domain (PID), denoted $\mathbb{D}_p$, if any ideal of $\mathbb{D}$ is principal.

$\mathbb{D}$ is a **Euclidean** domain (ED), denoted $\mathbb{D}_e$, if there exists a function $d : \mathbb{D}\backslash\{0\} \to \mathbb{Z}_+$ such that:

$$\begin{aligned}
&\text{for all } a, b \in \mathbb{D},\ ab \neq 0, \quad d(a) \leq d(ab);\\
&\text{for any } a, b \in \mathbb{D},\ ab \neq 0,\ \text{there exists } t, r \in \mathbb{D} \text{ such that}\\
&a = tb + r,\ \text{where either } r = 0 \text{ or } d(r) < d(b).
\end{aligned}$$

It is convenient to define $d(0) = \infty$. Let $a_1, a_2 \in \mathbb{D}_e$ and assume that $\infty > d(a_1) \geq d(a_2)$. **Euclid's algorithm** consists of a sequence $a_1, \ldots, a_{k+1}$, where $(a_1 \ldots a_k) \neq 0$, which is defined recursively as follows:

$$a_i = t_i a_{i+1} + a_{i+2}, \quad a_{i+2} = 0 \text{ or } d(a_{i+2}) < d(a_{i+1}) \quad \text{for} \quad i = 1, \ldots k-1.$$

[Hel43], [Kap49] A GCDD $\mathbb{D}$ is an **elementary divisor** domain (EDD), denoted by $\mathbb{D}_{ed}$, if for any three elements $a, b, c \in \mathbb{D}$ there exists $p, q, x, y \in \mathbb{D}$ such that $(a, b, c) = (px)a + (py)b + (qy)c$.

A GCDD $\mathbb{D}$ is a **Bezout** domain (BD), denoted by $\mathbb{D}_b$, if for any two elements $a, b \in \mathbb{D}$ $(a, b) = pa + qb$, for some $p, q \in \mathbb{D}$.

$p(x) = \sum_{i=0}^{m} a_i x^{m-i} \in \mathbb{Z}[x], a_0 \neq 0, m \geq 1$ is **primitive** if 1 is a g.c.d. of $a_0, \ldots, a_m$.

For $m \in \mathbb{N}$, the set of integers modulo m is denoted by $\mathbb{Z}_m$.

Facts:

Most of the facts about domains can be found in [ZS58] and [DF04]. More special results and references on the elementary divisor domains and rings are in [McD84]. The standard results on the domains of analytic functions can be found in [GR65]. More special results on analytic functions in one complex variable are in [Rud74].

1. Any integral domain satisfies *cancellation laws*: if $ab = ac$ or $ba = ca$ and $a \neq 0$, then $b = c$.
2. An integral domain such that any nonzero element is unit is a field F, and any field is an integral domain in which any nonzero element is unit.
3. A finite integral domain is a field.
4. $\mathbb{D}[\mathbf{x}]$ is an integral domain.
5. $\mathrm{H}(\Omega)$ is an integral domain.
6. Any prime element in $\mathbb{D}$ is irreducible.
7. In a UFD, any irreducible element is prime. This is not true in all integral domains.
8. Let $\mathbb{D}$ be a UFD. Then $\mathbb{D}[x]$ is a UFD. Hence, $\mathbb{D}[\mathbf{x}]$ is a UFD.
9. Let $a_1, a_2 \in \mathbb{D}_e$ and assume that $\infty > d(a_1) \geq d(a_2)$. Then Euclid's algorithm terminates in a finite number of steps, i.e., there exists $k \geq 3$ such that $a_1 \neq 0, \ldots, a_k \neq 0$ and $a_{k+1} = 0$. Hence, $a_k = (a_1, a_2)$.
10. An ED is a PID.
11. A PID is an EDD.
12. A PID is a UFD.
13. An EDD is a BD.
14. A BD is a GCDD.
15. A UFD is a GCDD.
16. The converses of Facts 10, 11, 12, 14, 15 are false (see Facts 28, 27, 21, 22).
17. [DF04, Chap. 8] An integral domain that is both a BD and a UFD is a PID.
18. An integral domain is a Bezout domain if and only if any finitely generated ideal is principal.
19. $\mathbb{Z}$ is an ED with $d(a) = |a|$.
20. Let $p, q \in \mathbb{Z}[x]$ be primitive polynomials. Then pq is primitive.
21. $F[x]$ is an ED with $d(p)$ = the degree of a nonzero polynomial. Hence, $F[x_1, \ldots, x_n]$ is a UFD. But $F[x_1, \ldots, x_n]$ is not a PID for $n \geq 2$.
22. $\mathbb{Z}[x_1, \ldots, x_m]$, $F[x_1, \ldots, x_n]$, and $\mathrm{H}(\Omega)$ (for a connected set $\Omega \subset \mathbb{C}^n$) are GCDDs, but for $m \geq 1$ and $n \geq 2$ these domains are not BDs.
23. [Frixx] (See Example 17 below.) Let $\Omega \subset \mathbb{C}$ be an open connected set. Then for $a, b \in \mathrm{H}(\Omega)$ there exists $p \in \mathrm{H}(\Omega)$ such that $(a, b) = pa + b$.
24. H_ζ, $\zeta \in \mathbb{C}$, is a UFD.
25. If $\Omega \subset \mathbb{C}^n$ is a connected open set, then $\mathrm{H}(\Omega)$ is not a UFD. (For $n = 1$ there is no prime factorization of an analytic function $f \in \mathrm{H}(\Omega)$ with an infinite countable number of zeros.)
26. Let $\Omega \subset \mathbb{C}$ be a compact connected set. Then $\mathrm{H}(\Omega)$ is an ED. Here, $d(a)$ is the number of zeros of a nonzero function $a \in \mathrm{H}(\Omega)$ counted with their multiplicities.
27. [Frixx] If $\Omega \subset \mathbb{C}$ is an open connected set, then $\mathrm{H}(\Omega)$ is an EDD. (See Example 17.) As $\mathrm{H}(\Omega)$ is not a UFD, it follows that $\mathrm{H}(\Omega)$ is not a PID. (Contrary to [McD84, Exc. II.E.10 (b), p. 144].)
28. [DF04, Chap. 8] $\mathbb{Z}[(1 + \sqrt{-19})/2]$ is a PID that is not an ED.

Examples:

1. $\{1, -1\}$ is the set of units in $\mathbb{Z}$. A g.c.d. of $a_1, \ldots, a_k \in \mathbb{Z}$ is **uniquely normalized** by the condition $(a_1, \ldots, a_k) \geq 0$.
2. A positive integer $p \in \mathbb{Z}$ is irreducible if and only if p is prime.
3. $\mathbb{Z}_m$ is an integral domain and, hence, a field with m elements if and only if p is a prime.
4. $\mathbb{Z} \supset I$ is a prime ideal if and only if all elements of I are divisible by some prime p.

5. $\{1, -1\}$ is the set of units in $\mathbb{Z}[x]$. A g.c.d. of $p_1, \dots, p_k \in \mathbb{Z}[x]$, is **uniquely normalized** by the condition $(p_1, \dots, p_k) = \sum_{i=0}^{m} a_i x^{m-i}$ and $a_0 > 0$.
6. Any prime element in $p(x) \in \mathbb{Z}[x]$, $\deg p \geq 1$, is a primitive polynomial.
7. Let $p(x) = 2x+3, q(x) = 5x-3 \in \mathbb{Z}[x]$. Clearly p, q are primitive polynomials and $(p(x), q(x)) = 1$. However, 1 cannot be expressed as $1 = a(x)p(x) + b(x)q(x)$, where $a(x), b(x) \in \mathbb{Z}[x]$. Indeed, if this was possible, then $1 = a(0)p(0) + b(0)q(0) = 3(a(0) - b(0))$, which is impossible for $a(0), b(0) \in \mathbb{Z}$. Hence, $\mathbb{Z}[x]$ is not BD.
8. The field of quotients of $\mathbb{Z}$ is the field of rational numbers $\mathbb{Q}$.
9. Let $p(x), q(x) \in \mathbb{Z}[x]$ be two nonzero polynomials. Let $(p(x), q(x))$ be the g.c.d of p, q in $\mathbb{Z}[x]$. Use the fact that $p(x), q(x) \in \mathbb{Q}[x]$ to deduce that there exists a positive integer m and $a(x), b(x) \in \mathbb{Z}[x]$ such that $a(x)p(x) + b(x)q(x) = m(p(x), q(x))$. Furthermore, if $c(x)p(x) + d(x)q(x) = l(p(x), q(x))$ for some $c(x), d(x) \in \mathbb{Z}[x]$ and $0 \neq l \in \mathbb{Z}$, then $m|l$.
10. The set of real numbers $\mathbb{R}$ and the set of complex numbers $\mathbb{C}$ are fields.
11. A g.c.d. of $a_1, \dots, a_k \in F[x]$ is **uniquely normalized** by the condition $(p_1, \dots, p_k)$ is a monic polynomial.
12. A linear polynomial in $\mathbb{D}[\mathbf{x}]$ is irreducible.
13. Let $\Omega \subset \mathbb{C}$ be a connected set. Then each irreducible element of $\mathrm{H}(\Omega)$ is an associate of $z - \zeta$ for some $\zeta \in \Omega$.
14. For $\zeta \in \mathbb{C}$ H_ζ, every irreducible element is of the form $a(z - \zeta)$ for some $0 \neq a \in \mathbb{C}$. A g.c.d. of $a_1, \dots, a_k \in \mathrm{H}_\zeta$ is **uniquely normalized** by the condition $(a_1, \dots, a_k) = (z - \zeta)^m$ for some nonnegative integer m.
15. In $\mathrm{H}(\Omega)$, the set of functions which vanishes on a prescribed set $U \subseteq \Omega$, i.e.,
$$I(U) := \{f \in \mathrm{H}(\Omega) : \quad f(\zeta) = 0,\ \zeta \in U\},$$
is an ideal.
16. Let Ω be an open connected set in $\mathbb{C}$. [Rud74, Theorems 15.11, 15.13] implies the following:
    - $I(U) \neq \{0\}$ if and only if U is a countable set, with no accumulations points in Ω.
    - Let U be a countable subset of Ω with no accumulation points in Ω. Assume that for each $\zeta \in U$ one is given a nonnegative integer $m(\zeta)$ and $m(\zeta) + 1$ complex numbers $w_{0,\zeta}, \dots, w_{m(\zeta),\zeta}$. Then there exists $f \in \mathrm{H}(\Omega)$ such that $f^{(n)}(\zeta) = n! w_{n,\zeta}, n = 0, \dots, m(\zeta)$, for all $\zeta \in U$. Furthermore, if all $w_{n,\zeta} = 0$, then there exists $g \in \mathrm{H}(\Omega)$ such that all zeros of g are in U and g has a zero of order $m(\zeta) + 1$ at each $\zeta \in U$.
    - Let $a, b \in \mathrm{H}(\Omega), ab \neq 0$. Then there exists $f \in \mathrm{H}(\Omega)$ such that $a = cf, b = df$, where $c, d \in \mathrm{H}(\Omega)$ and c, d do not have a common zero in Ω.
    - Let $c, d \in \mathrm{H}(\Omega)$ and assume that c, d do not have a common zero in Ω. Let U be the zero set of c in Ω, and denote by $m(\zeta) \geq 1$ the multiplicity of the zero $\zeta \in U$ of c. Then there exists $g \in \mathrm{H}(\Omega)$ such that $(e^g)^{(n)}(\zeta) = d^{(n)}(\zeta)$ for $n = 0, \dots, m(\zeta)$, for all $\zeta \in U$. Hence, $p = \frac{e^g - d}{c} \in \mathrm{H}(\Omega)$ and $e^g = pc + d$ is a unit in $\mathrm{H}(\Omega)$.
    - For $a, b \in \mathrm{H}(\Omega)$ there exists $p \in \mathrm{H}(\Omega)$ such $(a, b) = pa + b$.
    - For $a, b, c \in \mathrm{H}(\Omega)$ one has $(a, b, c) = p(a, b) + c = p(xa + b) + c$. Hence, $\mathrm{H}(\Omega)$ is EDD.
17. Let $I \subset \mathbb{C}[x, y]$ be the ideal given by given by the condition $p(0, 0) = 0$. Then I is generated by x and y, and $(x, y) = 1$. I is not principal and $\mathbb{C}[x, y]$ is not BD.
18. $\mathbb{D}[x, y]$ is not BD.
19. $\mathbb{Z}[\sqrt{-5}]$ is an integral domain that is not a GCDD, since $2 + 2\sqrt{-5}$ and 6 have no greatest common divisor. This can be seen by using the norm $N(a + b\sqrt{-5}) = a^2 + 5b^2$.

23.2 Equivalence of Matrices

In this section, we introduce matrices over an integral domain. Since any domain $\mathbb{D}$ can be viewed as a subset of its quotient field F, the notion of determinant, minor, rank, and adjugate in Chapters 1, 2, and 4 can be applied to these matrices. It is an interesting problem to determine whether one given matrix can be transformed to another by left multiplication, right multiplication, or multiplication on both sides, using only matrices invertible with the domain. These are equivalence relations and the problem is to characterize left (row) equivalence classes, right (columns) equivalence classes, and equivalence classes in $\mathbb{D}^{m\times n}$. For BD, the left equivalence classes are characterized by their Hermite normal form, which are attributed to Hermite. For EDD, the equivalence classes are characterized by their Smith normal form.

Definitions:

For a set S, denote by $S^{m\times n}$ the set of all $m\times n$ matrices $A=[a_{ij}]_{i=j=1}^{i=m,j=n}$, where each $a_{ij}\in S$.

For positive integers $p\le q$, denote by $Q_{p,q}$ the set of all subsets $\{i_1,\dots,i_p\}\subset\{1,2,\dots q\}$ of cardinality p, where we assume that $1\le i_1<\dots<i_p\le q$.

$U\in\mathbb{D}^{n\times n}$ is $\mathbb{D}$-**invertible** (**unimodular**), if there exists $V\in\mathbb{D}^{n\times n}$ such that $UV=VU=I_n$.

$\mathbf{GL}(n,\mathbb{D})$ denotes the group of $\mathbb{D}$-invertible matrices in $\mathbb{D}^{n\times n}$.

Let $A,B\in\mathbb{D}^{m\times n}$. Then A and B are **column equivalent**, **row equivalent**, and **equivalent** if the following conditions hold respectively:

$$\begin{aligned}
&B=AP \quad \text{for some } P\in\mathbf{GL}(n,\mathbb{D}) \quad (A\sim_c B),\\
&B=QA \quad \text{for some } Q\in\mathbf{GL}(m,\mathbb{D}) \quad (A\sim_r B),\\
&B=QAP \quad \text{for some } P\in\mathbf{GL}(n,\mathbb{D}),\ Q\in\mathbf{GL}(m,\mathbb{D}) \quad (A\sim B).
\end{aligned}$$

For $A\in\mathbb{D}_g^{m\times n}$, let

$$\begin{aligned}
\mu(\alpha,A)&=\text{g.c.d. }\{\det A[\alpha,\theta],\ \theta\in Q_{k,n}\},\quad \alpha\in Q_{k,m},\\
\nu(\beta,A)&=\text{g.c.d. }\{\det A[\phi,\beta],\ \phi\in Q_{k,m}\},\quad \beta\in Q_{k,n},\\
\delta_k(A)&=\text{g.c.d. }\{\det A[\phi,\theta],\ \phi\in Q_{k,m},\ \theta\in Q_{k,n}\}.
\end{aligned}$$

$\delta_k(A)$ is the k-th **determinant invariant** of A.

For $A\in\mathbb{D}_g^{m\times n}$,

$$\begin{aligned}
&i_j(A)=\frac{\delta_j(A)}{\delta_{j-1}(A)},\quad j=1,\dots,\text{rank } A,\quad (\delta_0(A)=1),\\
&i_j(A)=0 \quad \text{for rank } A<j\le\min(m,n),
\end{aligned}$$

are called the **invariant factors** of A. $i_j(A)$ is a **trivial** factor if $i_j(A)$ is unit in $\mathbb{D}_g$. We adopt the **normalization** $i_j(A)=1$ for any trivial factor of A. For $\mathbb{D}=\mathbb{Z},\mathbb{Z}[x],F[x]$, we adopt the normalizations given in the previous section in the Examples 1, 6, and 12, respectively.

Assume that $\mathbb{D}[x]$ is a GCDD. Then the invariant factors of $A\in\mathbb{D}[x]^{m\times n}$ are also called **invariant polynomials**.

$D=[d_{ij}]\in\mathbb{D}^{m\times n}$ is a **diagonal** matrix if $d_{ij}=0$ for all $i\ne j$. The entries $d_{11},\dots,d_{\ell\ell}$, $\ell=\min(m,n)$, are called the **diagonal entries** of D. D is denoted as $D=\operatorname{diag}(d_{11},\dots,d_{\ell\ell})\in\mathbb{D}^{m\times n}$.

Denote by $\Pi_n\subset\mathbf{GL}(n,\mathbb{D})$ the group of $n\times n$ permutation matrices.

An $\mathbb{D}$-invertible matrix $U\in\mathbf{GL}(n,\mathbb{D})$ is **simple** if there exists $P,Q\in\Pi_n$ such that $U=P\begin{bmatrix}V&0\\0&I_{n-2}\end{bmatrix}Q$, where $V=\begin{bmatrix}\alpha&\beta\\\gamma&\delta\end{bmatrix}\in\mathbf{GL}(2,\mathbb{D})$, i.e., $\alpha\delta-\beta\gamma$ is $\mathbb{D}$-invertible.

U is **elementary** if U is of the above form and $V=\begin{bmatrix}\alpha&0\\\gamma&\delta\end{bmatrix}\in\mathbf{GL}(2,\mathbb{D})$, i.e., α,δ are invertible.

For $A\in\mathbb{D}^{m\times n}$, the following row (column) operations are **elementary row operations**:

(a) Interchange any two rows (columns) of A.
(b) Multiply row (column) i by an invertible element a.
(c) Add to row (column) j b times row (column) i $(i\ne j)$.

For $A \in \mathbb{D}^{m\times n}$, the following row (column) operations are **simple row operations**:

(d) Replace row (column) i by a times row (column) i plus b times row (column) j, and row (column) j by c times row (column) i plus d times row (column) j, where $i \neq j$ and $ad - bc$ is invertible in $\mathbb{D}$.

$B = [b_{ij}] \in \mathbb{D}^{m\times n}$ is in **Hermite normal** form if the following conditions hold. Let $r = \text{rank } B$. First, the i-th row of B is a nonzero row if and only if $i \leq r$. Second, let b_{in_i} be the first nonzero entry in the i-th row for $i = 1, \ldots, r$. Then $1 \leq n_1 < n_2 < \cdots < n_r \leq n$.

$B \in \mathbb{D}_g^{m\times n}$ is in **Smith normal** form if B is a diagonal matrix $B = \text{diag}(b_1, \ldots, b_r, 0, \ldots, 0)$, $b_i \neq 0$, for $i = 1, \ldots, r$ and $b_{i-1}|b_i$ for $i = 2, \ldots, r$.

Facts:

Most of the results of this section can be found in [McD84]. Some special results of this section are given in [Fri81] and [Frixx]. For information about equivalence over fields, see Chapter 1 and Chapter 2.

1. The cardinality of $Q_{p,q}$ is $\binom{q}{p}$.
2. If U is $\mathbb{D}$-invertible then det U is a unit in $\mathbb{D}$. Conversely, if det U is a unit then U is $\mathbb{D}$-invertible, and its inverse U^{-1} is given by $U^{-1} = (\det U)^{-1}\text{adj } U$.
3. For $A \in \mathbb{D}^{m\times n}$, the rank of A is the maximal size of the nonvanishing minor. (The rank of zero matrix is 0.)
4. Column equivalence, row equivalence, and equivalence of matrices are equivalence relations in $\mathbb{D}^{m\times n}$.
5. For any $A, B \in \mathbb{D}^{m\times n}$, one has $A \sim_r B \iff A^T \sim_c B^T$. Hence, it is enough to consider the row equivalence relation.
6. For $A, B \in \mathbb{D}_g^{m\times n}$, the Cauchy–Binet formula (Chapter 4) yields
$$\mu(\alpha, A) \equiv \mu(\alpha, B) \quad \text{for all } \alpha \in Q_{k,m} \quad \text{if } A \sim_c B,$$
$$\nu(\beta, A) \equiv \nu(\beta, B) \quad \text{for all } \beta \in Q_{k,n} \quad \text{if } A \sim_r B,$$
$$\delta_k(A) \equiv \delta_k(B) \quad \text{if } A \sim B,$$
for $k = 1, \ldots, \min(m, n)$.
7. Any elementary matrix is a simple matrix, but not conversely.
8. The elementary row and column operations can be carried out by multiplications by A by suitable elementary matrices from the left and the right, respectively.
9. The simple row and column operations are carried out by multiplications by A by suitable simple matrices U from the left and right, respectively.
10. Let $\mathbb{D}_b$ be a Bezout domain, $A \in \mathbb{D}_b^{m\times n}$, rank $A = r$. Then A is row equivalent to $B = [b_{ij}] \in \mathbb{D}_b^{m\times n}$, in a Hermite normal form, which satisfies the following conditions.

    Let b_{in_i} be the first nonzero entry in the i-th row for $i = 1, \ldots, r$. Then $1 \leq n_1 < n_2 < \cdots < n_r \leq n$ are uniquely determined and the elements b_{in_i}, $i = 1, \ldots, r$ are uniquely determined, up to units, by the conditions
$$\nu((n_1, \ldots, n_i), A) = b_{1n_1} \cdots b_{in_i}, \quad i = 1, \ldots, r,$$
$$\nu(\alpha, A) = 0, \ \alpha \in Q_{i,n_i-1}, \quad i = 1, \ldots, r.$$

    The elements b_{jn_i}, $j = 1, \ldots, i-1$ are then successively uniquely determined up to the addition of arbitrary multiples of b_{in_i}. The remaining elements b_{ik} are now uniquely determined. The $\mathbb{D}$-invertible matrix Q, such that $B = QA$, can be given by a finite product of simple matrices.

    If b_{in_i} in the Hermite normal form is invertible, we assume the normalization conditions $b_{in_i} = 1$ and $b_{jn_i} = 0$ for $i < j$.
11. For Euclidean domains, we assume normalization conditions either $b_{jn_i} = 0$ or $d(b_{jn_i}) < d(b_{in_i})$ for $j < i$. Then for any $A \in \mathbb{D}_e^{m\times n}$, in a Hermite normal form $B = QA$, $Q \in \mathbf{GL}_m(\mathbb{D}_e)$ Q is a product of a finite elementary matrices.

12. $U \in \mathbf{GL}(n, \mathbb{D}_e)$ is a finite product of elementary $\mathbb{D}$-invertible matrices.
13. For $\mathbb{Z}$, we assume the normalization $b_{in_i} \geq 1$ and $0 \leq b_{jn_i} < b_{in_i}$ for $j < i$. For $F[x]$, we assume that b_{in_i} is a monic polynomial and $\deg b_{jn_i} < \deg b_{in_i}$ for $j < i$. Then for $\mathbb{D}_e = \mathbb{Z}, F[x]$, any $A \in \mathbb{D}_e^{m\times n}$ has a unique Hermite normal form.
14. $A, B \in \mathbb{D}_b$ are row equivalent if and only if A and B are row equivalent to the same Hermite normal form.
15. $A \in F^{m\times n}$ can be brought to its unique Hermite normal form, called the *reduced row echelon form* (RREF),

$$b_{in_i} = 1, \quad b_{jn_i} = 0, \quad j = 1, \ldots, i-1, \quad i = 1, \ldots, r = \text{rank } A,$$

by a finite number of elementary row operations. Hence, $A, B \in F^{m\times n}$ are row equivalent if and only if $r = \text{rank } A = \text{rank } B$ and they have the same RREF. (See Chapter 1.)
16. For $A \in \mathbb{D}_g^{m\times n}$ and $1 \leq p < q \leq \min(m, n)$, $\delta_p(A)|\delta_q(A)$.
17. For $A \in \mathbb{D}_g^{m\times n}$, $i_{j-1}(A)|i_j(A)$ for $j = 2, \ldots, \text{rank } A$.
18. Any $0 \neq A \in \mathbb{D}_{ed}^{m\times n}$ is equivalent to its Smith normal form $B = \text{diag}\,(i_1(A), \ldots, i_r(A), 0, \ldots, 0)$, where $r = \text{rank } A$ and $i_1(A), \ldots, i_r(A)$ are the invariants factors of A.
19. $A, B \in \mathbb{D}_{ed}^{m\times n}$ are equivalent if and only if A and B have the same rank and the same invariant factors.

Examples:

1. Let $A = \begin{bmatrix} 1 & a \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & b \\ 0 & 0 \end{bmatrix} \in \mathbb{D}^{2\times 2}$ be two Hermite normal forms. It is straightforward to show that $A \sim_r B$ if and only if $a = b$. Assume that $\mathbb{D}$ is a BD and let $a \neq 0$. Then rank $A = 1, \nu((1), A) = 1, \{\{\nu((2), A)\}\} = \{\{a\}\}, \nu((1,2), A) = 0$. If $\mathbb{D}$ has other units than 1, it follows that $\nu(\beta, A)$ for all $\beta \in Q_{k,2}, k = 1, 2$ do not determine the row equivalence class of A.
2. Let $A = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \in \mathbb{D}_b^{2\times 2}$. Then there exists $u, v \in \mathbb{D}_b$ such that $ua + vb = (a, b) = \nu((1), A)$. If $(a, b) \neq 0$, then $1 = (u, v)$. If $a = b = 0$, choose $u = 1, v = 0$. Hence, there exists $x, y \in \mathbb{D}_b$ such that $yu - xv = 1$. Thus $V = \begin{bmatrix} u & v \\ x & y \end{bmatrix} \in \mathbf{GL}(2, \mathbb{D}_b)$ and $VA = \begin{bmatrix} (a,b) & c' \\ b' & d' \end{bmatrix}$. Clearly $b' = xa + yb = (a, b)e$. Hence, $\begin{bmatrix} 1 & 0 \\ -e & 1 \end{bmatrix} VA = \begin{bmatrix} (a,b) & c' \\ 0 & f \end{bmatrix}$ is a Hermite normal form of A. This construction is easily extended to obtain a Hermite normal form for any $A \in \mathbb{D}_b^{m\times n}$, using simple row operations.
3. Let $A \in \mathbb{D}_e^{2\times 2}$ as in the previous example. Assume that $ab \neq 0$. Change the two rows of A if needed to assume that $d(a) \leq d(b)$. Let $a_1 = b, a_2 = a$ and do the first step of Euclid's algorithm: $a_1 = t_1 a_2 + a_3$, where $a_3 = 0$ or $d(a_3) < d(a_2)$. Let $V = \begin{bmatrix} 1 & 0 \\ -t_1 & 1 \end{bmatrix} \in \mathbf{GL}(2, \mathbb{D}_e)$ be an elementary matrix. Then $A_1 = VA = \begin{bmatrix} a_2 & * \\ a_3 & * \end{bmatrix}$. If $a_3 = 0$, then A_1 has a Hermite normal form. If $a_3 \neq 0$, continue as above. Since Euclid's algorithm terminates after a finite number of steps, it follows that A can be put into Hermite normal form by a finite number of elementary row operations. This statement holds similarly in the case $ab = 0$. This construction is easily extended to obtain a Hermite normal form for any $A \in \mathbb{D}_e^{m\times n}$ using elementary row operations.
4. Assume that $\mathbb{D}$ is a BD and let $A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \in \mathbb{D}^{2\times 2}$. Note that $\delta_1(A) = (a, b, c)$. If A is equivalent to a Smith normal form then there exists $V, U \in \mathbf{GL}(2, D)$ such that $VAU = \begin{bmatrix} (a,b,c) & * \\ * & * \end{bmatrix}$.

Assume that $V = \begin{bmatrix} p & q \\ \tilde{q} & \tilde{p} \end{bmatrix}, U = \begin{bmatrix} x & \tilde{y} \\ y & \tilde{x} \end{bmatrix}$. Then there exist $p, q, x, y \in \mathbb{D}$ such that $(px)a + (py)b + (qy)c = (a, b, c)$. Thus, if each $A \in \mathbb{D}^{2\times 2}$ is equivalent to Smith normal form in $\mathbb{D}^{2\times 2}$, then it follows that $\mathbb{D}$ is an EDD.

Conversely, suppose that $\mathbb{D}$ is an EDD. Then $\mathbb{D}$ is also a BD. Let $A \in \mathbb{D}^{2\times 2}$. First, bring A to an upper triangular Hermite normal form using simple row operations: $A_1 = WA = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}, W \in$ **GL**$(2, \mathbb{D})$. Note that $\delta_1(A) = \delta_1(A_1) = (a, b, c)$. Since $\mathbb{D}$ is an EDD, there exist $p, q, x, y \in \mathbb{D}$ such that $(px)a + (py)b + (qy)c = (a, b, c)$. If $(a, b, c) \neq (0, 0, 0)$, then $(p, q) = (x, y) = 1$. Otherwise $A = A_1 = 0$ and we are done. Hence, there exist $\tilde{p}, \tilde{q}, \tilde{x}, \tilde{y}$ such that $p\tilde{p} - q\tilde{q} = x\tilde{x} - y\tilde{y} = 1$. Let $V = \begin{bmatrix} p & q \\ \tilde{q} & \tilde{p} \end{bmatrix}, U = \begin{bmatrix} x & \tilde{y} \\ y & \tilde{x} \end{bmatrix}$. Thus, $G = VA_1U = \begin{bmatrix} \delta_1(A) & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$. Since $\delta_1(G) = \delta_1(A)$, we deduce that $\delta_1(A)$ divides g_{12} and g_{21}. Apply appropriate elementary row and column operations to deduce that A is equivalent to a diagonal matrix $C = \text{diag}\,(i_1(A), d_2)$. As $\delta_2(C) = i_1(A)d_2 = \delta_2(A)$, we see that C is has Smith normal form.

These arguments are easily extended to obtain a Smith normal form for any $A \in \mathbb{D}_{ed}^{m\times n}$, using simple row and column operations.

5. The converse of Fact 6 is false, as can be seen by considering $A = \begin{bmatrix} 2 & 2 \\ x & x \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \in \mathbb{Z}[x]^{2\times 2}$. $\mu(\{1\}, A) = \mu(\{1\}, B) = 1$, $\mu(\{2\}, A) = \mu(\{2\}, B) = 1$, $\mu(\{1, 2\}, A) = \mu(\{1, 2\}, B) = 0$, but there does not exist a $\mathbb{D}$-invertible P such that $PA = B$.
6. Let $\mathbb{D}$ be an integral domain and assume that $p(x) = x^m + \sum_{i=1}^{m} a_i x^{m-i} \in \mathbb{D}[x]$ is a monic polynomial of degree $m \geq 2$. Let $C(p) \in \mathbb{D}^{m\times m}$ be the companion matrix (see Chapter 4). Then $\det\,(xI_m - C(p)) = p(x)$. Assume that $\mathbb{D}[x]$ is a GCDD. Let $C(p)(x) = xI_m - C(p) \in \mathbb{D}[x]^{m\times m}$. By deleting the last column and first row, we obtain a triangular $(m-1)\times(m-1)$ submatrix with -1s on the diagonal, so it follows that $\delta_i(C(p)(x)) = 1$ for $i = 1, \ldots, m-1$. Hence, the invariant factors of $C(p)(x)$ are $i_1(C(p)(x)) = \ldots = i_{m-1}(C(p)(x)) = 1$ and $i_m(C(p)(x)) = p(x)$. If $\mathbb{D}$ is a field, then $C(p)(x)$ is equivalent over $\mathbb{D}[x]$ to $\text{diag}\,(1, \ldots, 1, p(x))$.

23.3 Linear Equations over Bezout Domains

Definitions:

M is a $\mathbb{D}$-**module** if **M** is an additive group with respect to the operation + and **M** admits a multiplication by a **scalar** $a \in \mathbb{D}$, i.e., there exists a mapping $\mathbb{D} \times \mathbf{M} \to \mathbf{M}$ which satisfies the standard distribution properties and $1\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in \mathbf{M}$ (the latter requirement sometimes results in the module being called unital). (For a field F, a module **M** is a vector space over F.)

M is **finitely generated** if there exist $\mathbf{v}_1, \ldots, \mathbf{v}_n \in \mathbf{M}$ so that every $\mathbf{v} \in \mathbf{M}$ is a **linear combination** of $\mathbf{v}_1, \ldots, \mathbf{v}_n$, over $\mathbb{D}$, i.e., $\mathbf{v} = a_1\mathbf{v}_1 + \ldots + a_n\mathbf{v}_n$ for some $a_1, \ldots, a_n \in \mathbb{D}$. $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is a **basis** of **M** if every **v** can be written as a unique linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_n$. $\dim \mathbf{M} = n$ means that **M** has a basis of n elements.

$\mathbf{N} \subseteq \mathbf{M}$ is a $\mathbb{D}$-**submodule** of **M** if **N** is closed under the addition and multiplication by scalars.

$\mathbb{D}^n (= \mathbb{D}^{n\times 1})$ is a $\mathbb{D}$-module. It has a **standard** basis $\mathbf{e}_i$ for $i = 1, \ldots, n$, where $\mathbf{e}_i$ is the i-th column of the identity matrix I_n.

For any $A \in \mathbb{D}^{m\times n}$, the **range** of A, denoted by range(A), is the set of all linear combinations of the columns of A.

The **kernel** of A, denoted by $\ker(A)$, is the set of all solutions to the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

Consider a system of m linear equations in n unknowns:

$\sum_{j=1}^{n} a_{ij} x_j = b_j, \quad i = 1, \ldots, m$, where $a_{ij}, b_i \in \mathbb{D}$ for $i = 1, \ldots, m, \ j = 1, \ldots, n$. In matrix notation this system is $A\mathbf{x} = \mathbf{b}$, where $A = [a_{ij}] \in \mathbb{D}^{m\times n}$, $\mathbf{x} = [x_1, \ldots, x_n]^T \in \mathbb{D}^n$, $\mathbf{b} = [b_1, \ldots, b_m]^T \in \mathbb{D}^m$. A and $[A, \mathbf{b}] \in \mathbb{D}^{m\times(n+1)}$ are called the **coefficient** matrix and the **augmented** matrix, respectively.

Let $A \in \mathrm{H}_0^{m\times n}$. Then $A = A(z) = [a_{ij}(z)]_{i=j=1}^{m,n}$ and $A(z)$ has the McLaurin expansion $A(z) = \sum_{k=0}^{\infty} A_k z^k$, where $A_k \in \mathbb{C}^{m\times n}, k = 0, \ldots$. Here, each $a_{ij}(z)$ has convergent McLaurin series for $|z| < R(A)$ for some $R(A) > 0$.

The invariant factors of A are called the **local invariant polynomials** of A, which are normalized to be of the form $i_k(A) = z^{i_k}$ for $0 \le i_1 \le i_2 \le \ldots \le i_r$, where $r = \text{rank } A$.

The integer i_r is the **index** of A and is denoted by $\eta = \eta(A)$. For a nonnegative integer p, denote by $\mathcal{K}_p = \mathcal{K}_p(A)$ the number of local invariant polynomials of A whose degree is equal to p.

Facts:

For modules, see [ZS58], [DF04], and [McD84]. The solvability of linear systems over EDD can be traced to [Hel43] and [Kap49]. The results for BD can be found in [Fri81] and [Frixx]. The results for H_0 are given in [Fri80]. The general theory of solvability of the systems of equations over commutative rings is discussed in [McD84, Exc. I.G.7–I.G.8]. (See Chapter 1 for information about solvability over fields.)

1. The system $A\mathbf{x} = \mathbf{b}$ is solvable over a Bezout domain $\mathbb{D}_b$ if and only if $r = \text{rank } A = \text{rank } [A, \mathbf{b}]$ and $\delta_r(A) = \delta_r([A, \mathbf{b}])$, which is equivalent to the statement that A and $[A, \mathbf{b}]$ have the same set of invariant factors, up to invertible elements. For a field F, this result reduces to the equality $\text{rank } A = \text{rank } [A, \mathbf{b}]$.
2. For $A \in \mathbb{D}_b^{m\times n}$, range A and ker A are modules in $\mathbb{D}_b^m$ and $\mathbb{D}_b^n$ having finite bases with rank A and null A elements, respectively. Moreover, the basis of ker A can be completed to a basis of $\mathbb{D}_b^n$.
3. For $A, B \in \mathrm{H}_0^{m\times n}$, let $C(z) = A(z) + z^{k+1}B(z)$, where k is a nonnegative integer. Then A and C have the same local invariant polynomials up to degree k. Moreover, if k is equal to the index of A, and A and C have the same rank, then A is equivalent to C.
4. Consider a system of linear equations over H_0 $A(z)\mathbf{u}(z) = \mathbf{b}(z)$, where $A(z) \in \mathrm{H}_0^{m\times n}$ and $\mathbf{b}(z) = \sum_{k=0}^{\infty} \mathbf{b}_k z^k \in \mathrm{H}_0^m, \quad \mathbf{b}_k \in \mathbb{C}^m, \ k = 0, \ldots$. Look for the power series solution $\mathbf{u}(z) = \sum_{k=0}^{\infty} \mathbf{u}_k z^k$, where $\mathbf{u}_k \in \mathbb{C}^n, \ k = 0, \ldots$. Then $\sum_{j=0}^{k} A_{k-j}\mathbf{u}_j = \mathbf{b}_k$, for $k = 0, \ldots$. This system is solvable for $k = 0, \ldots, q \in \mathbb{Z}_+$ if and only if $A(z)$ and $[A(z), \mathbf{b}(z)]$ have the same local invariant polynomials up to degree q.
5. Suppose that $A(z)\mathbf{u}(z) = \mathbf{b}(z)$ is solvable over H_0. Let $q = \eta(A)$ and suppose that $\mathbf{u}_0, \ldots, \mathbf{u}_q$ satisfies the system of equations, given in the previous fact, for $k = 0, \ldots, q$. Then there exists a solution $\mathbf{u}(z) \in \mathrm{H}_0^n$ satisfying $\mathbf{u}(0) = \mathbf{u}_0$.
6. Let $q \in \mathbb{Z}_+$ and $\mathbf{W}_q \subset \mathbb{C}^n$ be the subspace of all vectors $\mathbf{w}_0$ such that $\mathbf{w}_0, \ldots, \mathbf{w}_q$ is a solution to the homogenous system $\sum_{j=0}^{k} A_{k-j}\mathbf{w}_j = 0$, for $k = 0, \ldots, q$. Then $\dim \mathbf{W}_q = n - \sum_{j=0}^{q} \mathcal{K}_j(A)$. In particular, for $\eta = \eta(A)$ and any $\mathbf{w}_0 \in \mathbf{W}_\eta$ there exists $\mathbf{w}(z) \in \mathrm{H}_0^n$ such that $A(z)\mathbf{w}(z) = \mathbf{0}$, $\mathbf{w}(0) = \mathbf{w}_0$.

23.4 Strict Equivalence of Pencils

Definitions:

A matrix $A(x) \in \mathbb{D}[x]^{m\times n}$ is a **pencil** if $A(x) = A_0 + xA_1$, $A_0, A_1 \in \mathbb{D}^{m\times n}$.

A pencil $A(x)$ is **regular** if $m = n$ and $\det A(x)$ is not the zero polynomial. Otherwise $A(x)$ is a **singular** pencil.

Associate with a pencil $A(x) = A_0 + xA_1 \in \mathbb{D}[x]^{m\times n}$ the **homogeneous** pencil $A(x_0, x_1) = x_0A_0 + x_1A_1 \in \mathbb{D}[x_0, x_1]^{m\times n}$.

Two pencils $A(x), B(x) \in \mathbb{D}[x]^{m\times n}$ are **strictly** equivalent, denoted by $A(x) \overset{s}{\sim} B(x)$, if $B(x) = QA(x)P$ for some $P \in \mathbf{GL}_n(\mathbb{D})$, $Q \in \mathbf{GL}_m(\mathbb{D})$. Similarly, two homogeneous pencils $A(x_0, x_1), B(x_0, x_1)$

$\in \mathbb{D}[x_0, x_1]^{m\times n}$ are **strictly** equivalent, denoted by $A(x_0, x_1)\overset{s}{\sim}B(x_0, x_1)$, if $B(x_0, x_1) = QA(x_0, x_1)P$ for some $P \in \mathbf{GL}_n(\mathbb{D}),\ Q \in \mathbf{GL}_m(\mathbb{D})$.

For a UFD $\mathbb{D}_u$ let $\delta_k(x_0, x_1),\ i_k(x_0, x_1)$ be the invariant determinants and factors of $A(x_0, x_1)$, respectively, for $k = 1, \ldots, \text{rank}\ A(x_0, x_1)$. They are called **homogeneous** determinants and the invariant **homogeneous** polynomials (factors), respectively, of $A(x_0, x_1)$. (Sometimes, $\delta_k(x_0, x_1),\ i_k(x_0, x_1), k = 1, \ldots, \text{rank}\ A(x_0, x_1)$ are called the homogeneous determinants and the invariant homogeneous polynomials $A(x)$.)

Let $A(x) \in F[x]^{m\times n}$ and consider the module $\mathbf{M} \subset F[x]^n$ of all solutions of $A(x)\mathbf{w}(x) = 0$. The set of all solutions $\mathbf{w}(x)$ is an $F[x]$-module $\mathbf{M}$ with a finite basis $\mathbf{w}_1(x), \ldots, \mathbf{w}_s(x)$, where $s = n - \text{rank}\ A(x)$. Choose a basis $\mathbf{w}_1(x), \ldots, \mathbf{w}_s(x)$ in $\mathbf{M}$ such that $\mathbf{w}_k(x) \in \mathbf{M}$ has the lowest degree among all $\mathbf{w}(x) \in \mathbf{M}$, which are linearly independent over $F[x]$ of $\mathbf{w}_1, \ldots, \mathbf{w}_{k-1}(x)$ for $k = 1, \ldots, s$. Then the **column indices** $\alpha_1 \le \alpha_2 \le \ldots \le \alpha_s$ of $A(x)$ are given as $\alpha_k = \deg \mathbf{w}_k(x), k = 1, \ldots, s$. The **row indices** $0 \le \beta_1 \le \beta_2 \le \ldots \le \beta_t,\ t = m - \text{rank}\ A(x)$, of $A(x)$, are the column indices of $A(x)^{\mathrm{T}}$.

Facts:

The notion of strict equivalence of $n \times n$ regular pencils over the fields goes back to K. Weierstrass [Wei67]. The notion of strict similarity of $m \times n$ matrices over the fields is due to L. Kronecker [Kro90]. Most of the details can be found in [Gan59]. Some special results are proven in [Frixx]. For information about matrix pencils over fields see Section 43.1.

1. Let $A_0, A_1, B_0, B_1 \in \mathbb{D}^{m\times n}$. Then $A_0 + xA_1\overset{s}{\sim}B_0 + xB_0 \iff x_0A_0 + x_1A_1\overset{s}{\sim}B_0x_0 + B_1x_1$.
2. Let $A_0, A_1 \in \mathbb{D}_u$. Then the invariant determinants and the invariant polynomials $\delta_k(x_0, x_1)$, $i_k(x_0, x_1), k = 1, \ldots, \text{rank}\ x_0A_0 + x_1A_1$, of $x_0A_0 + x_1A_1$ are homogeneous polynomials. Moreover, if $\delta_k(x)$ and $i_k(x)$ are the invariant determinants and factors of the pencil $A_0 + xA_1$ for $k = 1, \ldots, \text{rank}\ A_0+xA_1$, then $\delta_k(x) = \delta_k(1, x),\ i_k(x) = i_k(1, x)$, for $k = 1, \ldots, \text{rank}\ A_0+xA_1$.
3. [Wei67] Let $A_0 + xA_1 \in F[x]^{n\times n}$ be a regular pencil. Then a pencil $B_0 + xB_1 \in F[x]^{n\times n}$ is strictly equivalent to $A_0 + xA_1$ if and only if $A_0 + xA_1$ and $B_0 + xB_1$ have the same invariant polynomials over $F[x]$.
4. [Frixx] Let $A_0 + xA_1, B_0 + xB_1 \in \mathbb{D}[x]^{n\times n}$. Assume that $A_1, B_1 \in \mathbf{GL}_n(\mathbb{D})$. Then $A_0 + xA_1\overset{s}{\sim}B_0 + xB_1 \iff A_0 + xA_1 \sim B_0 + xB_1$.
5. [Gan59] The column (row) indices are independent of a particular allowed choice of a basis $\mathbf{w}_1(x), \ldots, \mathbf{w}_s(x)$.
6. For singular pencils the invariant homogeneous polynomials alone do not determine the class of strictly equivalent pencils.
7. [Kro90], [Gan59] The pencils $A(x), B(x) \in F[x]^{m\times n}$ are strictly equivalent if and only if they have the same invariant homogeneous polynomials and the same row and column indices.

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24

Similarity of Families of Matrices

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This chapter uses the notations, definitions, and facts given in Chapter 23. The aim of this chapter is to acquaint the reader with two difficult problems in matrix theory:

1. Similarity of matrices over integral domains, which are not fields.
2. Simultaneous similarity of tuples of matrices over $\mathbb{C}$.

Problem *1* is notoriously difficult. We show that for the local ring H_0 this problem reduces to a Problem *2* for certain kind of matrices. We then discuss certain special cases of Problem *2* as simultaneous similarity of tuples of matrices to upper triangular and diagonal matrices. The L-property of pairs of matrices, which is discussed next, is closely related to simultaneous similarity of pair of matrices to a diagonal pair. The rest of the chapter is devoted to a "solution" of the Problem *2*, by the author, in terms of basic notions of algebraic geometry.

24.1 Similarity of Matrices

The classical result of K. Weierstrass [Wei67] states that the similarity class of $A \in F^{n\times n}$ is determined by the invariant factors of $-A + xI_n$ over $F[x]$. (See Chapter 6 and Chapter 23.) For a given $A, B \in F^{n\times n}$, one can easily determine if A and B are similar, by considering *only* the ranks of three specific matrices associated with A, B [GB77]. It is well known that it is a difficult problem to determine if $A, B \in \mathbb{D}^{n\times n}$ are $\mathbb{D}$-similar for most integral domains that are not fields. The emphasis of this chapter is the similarity over the local field H_0. The subject of similarity of matrices over H_0 arises naturally in theory linear differential equations having singularity with respect to a parameter. It was studied by Wasow in [Was63], [Was77], and [Was78].

Definitions:

For $E \in \mathbb{D}^{m\times n}, G \in \mathbb{D}^{p\times q}$ extend the definition of the **tensor** or **Kronecker** product $E \otimes G \in \mathbb{D}^{mp\times nq}$ of E with G to the domain $\mathbb{D}$ in the obvious way. (See Section 10.4.)

$A, B \in \mathbb{D}^{m\times m}$ are called **similar**, denoted by $A \approx B$ if $B = QAQ^{-1}$, for some $Q \in \mathrm{GL}_m(\mathbb{D})$.

Let $A, B \in \mathrm{H}(\Omega)^{n\times n}$. Then A and B are called **analytically similar**, denoted as $A \overset{a}{\approx} B$, if A and B are similar over $\mathrm{H}(\Omega)$.

A and B are called **locally similar** if for any $\zeta \in \Omega$, the restrictions A_ζ, B_ζ of A, B to the local rings H_ζ, respectively, are similar over H_ζ.

A, B are called **point-wise** similar if $A(\zeta), B(\zeta)$ are similar matrices in $\mathbb{C}^{n\times n}$ for each $\zeta \in \Omega$.

A, B are called **rationally** similar, denoted as $A \overset{r}{\approx} B$, if A, B are similar over the quotient field $\mathcal{M}(\Omega)$ of $\mathrm{H}(\Omega)$.

Let $A, B \in \mathrm{H}_0^{n\times n}$:

$$A(x) = \sum_{k=0}^{\infty} A_k x^k, \quad |x| < R(A), \qquad B(x) = \sum_{k=0}^{\infty} B_k x^k, \quad |x| < R(B).$$

Then $\eta(A, B)$ and $\mathcal{K}_p(A, B)$ are the index and the number of local invariant polynomials of degree p of the matrix $I_n \otimes A(x) - B(x)^{\mathrm{T}} \otimes I_n$, respectively, for $p = 0, 1, \ldots$.

$\lambda(x)$ is called an **algebraic** function if there exists a monic polynomial $p(\lambda, x) = \lambda^n + \sum_{i=1}^n q_i(x)\lambda^{n-i} \in (\mathbb{C}[x])[\lambda]$ of λ-degree $n \geq 1$ such that $p(\lambda(x), x) = 0$ identically. Then $\lambda(x)$ is a multivalued function on $\mathbb{C}$, which has n **branches**. At each point $\zeta \in \mathbb{C}$ each branch $\lambda_j(x)$ of $\lambda(x)$ has **Puiseaux** expansion: $\lambda_j(x) = \sum_{i=0}^{\infty} b_{ji}(\zeta)(x - \zeta)^{\frac{i}{m}}$, which converges for $|x - \zeta| < R(\zeta)$, and some integer m depending on $p(x)$. $\sum_{i=0}^{m} b_{ji}(\zeta)(x - \zeta)^{\frac{i}{m}}$ is called the **linear part** of $\lambda_j(x)$ at ζ. Two distinct branches $\lambda_j(x)$ and $\lambda_k(x)$ are called **tangent** at $\zeta \in \mathbb{C}$ if the linear parts of $\lambda_j(x)$ and $\lambda_k(x)$ coincide at ζ. Each branch $\lambda_j(x)$ has Puiseaux expansion around ∞: $\lambda_j(x) = x^l \sum_{i=0}^{\infty} c_{ji} x^{-\frac{i}{m}}$, which converges for $|x| > R$. Here, l is the smallest nonnegative integer such that $c_{j0} \neq 0$ at least for some branch λ_j. $x^l \sum_{i=0}^{m} c_{ji} x^{-\frac{i}{m}}$ is called the **principal part** of $\lambda_j(x)$ at ∞. Two distinct branches $\lambda_j(x)$ and $\lambda_k(x)$ are called **tangent** at ∞ if the principal parts of $\lambda_j(x)$ and $\lambda_k(x)$ coincide at ∞.

Facts:

The standard results on the tensor products can be found in Chapter 10 or Chapter 13 or in [MM64]. Most of the results of this section related to the analytic similarity over H_0 are taken from [Fri80].

1. The similarity relation is an equivalence relation on $\mathbb{D}^{m\times m}$.
2. $A \approx B \iff A(x) = -A + xI \overset{s}{\sim} B(x) = -B + xI$.
3. Let $A, B \in F^{n\times n}$. Then A and B are similar if and only if the pencils $-A + xI$ and $-B + xI$ have the same invariant polynomials over $F[x]$.
4. If $E = [e_{ij}] \in \mathbb{D}^{m\times n}, G \in \mathbb{D}^{p\times q}$, then $E \otimes G$ can be viewed as the $m \times n$ block matrix $[e_{ij}G]_{i,j=1}^{m,n}$. Alternatively, $E \otimes G$ can be identified with the linear transformation
$$L(E, G) : \mathbb{D}^{q\times n} \to \mathbb{D}^{p\times m}, \quad X \to GXE^{\mathrm{T}}.$$
5. $(E \otimes G)(U \otimes V) = EU \otimes GV$ whenever the products EU and GV are defined. Also $(E \otimes G)^{\mathrm{T}} = E^{\mathrm{T}} \otimes G^{\mathrm{T}}$ (cf. §2.5.4)).
6. For $A, B \in \mathbb{D}^{n\times n}$, if A is similar to B, then
$$I_n \otimes A - A^{\mathrm{T}} \otimes I_n \sim I_n \otimes A - B^{\mathrm{T}} \otimes I_n \sim I_n \otimes B - B^{\mathrm{T}} \otimes I_n.$$
7. [Gur80] There are examples over Euclidean domains for which the reverse of the implication in Fact 6 does not hold.
8. [GB77] For $\mathbb{D} = F$, the reverse of the implication in Fact 6 holds.
9. [Fri80] Let $A \in F^{m\times m},\ B \in F^{n\times n}$. Then
$$\mathrm{null}\,(I_n \otimes A - B^{\mathrm{T}} \otimes I_m) \leq \frac{1}{2}(\mathrm{null}\,(I_m \otimes A - A^{\mathrm{T}} \otimes I_m) + \mathrm{null}\,(I_n \otimes B - B^{\mathrm{T}} \otimes I_n)).$$
Equality holds if and only if $m = n$ and A and B are similar.

10. Let $A \in F^{n\times n}$ and assume that $p_1(x), \ldots, p_k(x) \in F[x]$ are the nontrivial normalized invariant polynomials of $-A+xI$, where $p_1(x)|p_2(x)|\ldots|p_k(x)$ and $p_1(x)p_2(x)\ldots p_k(x) = \det(xI - A)$. Then $A \approx C(p_1) \oplus C(p_2) \oplus \ldots \oplus C(p_k)$ and $C(p_1) \oplus C(p_2) \oplus \ldots \oplus C(p_k)$ is called the *rational canonical form* of A (cf. Chapter 6.6).
11. For $A, B \in \mathrm{H}(\Omega)^{n\times n}$, analytic similarity implies local similarity, local similarity implies point-wise similarity, and point-wise similarity implies rational similarity.
12. For $n = 1$, all the four concepts in Fact 11 are equivalent. For $n \geq 2$, local similarity, point-wise similarity, and rational similarity, are distinct (see Example 2).
13. The equivalence of the three matrices in Fact 6 over $\mathrm{H}(\Omega)$ implies the point-wise similarity of A and B.
14. Let $A, B \in \mathrm{H}_0^{n\times n}$. Then A and B are analytically similar over H_0 if and only if A and B are rationally similar over H_0 and there exists $\eta(A, A) + 1$ matrices $T_0, \ldots, T_\eta \in \mathbb{C}^{n\times n}$ ($\eta = \eta(A, A)$), such that $\det T_0 \neq 0$ and
$$\sum_{i=0}^{k} A_i T_{k-i} - T_{k-i} B_i = 0, \quad k = 0, \ldots, \eta(A, A).$$
15. Suppose that the characteristic polynomial of $A(x)$ splits over H_0:
$$\det(\lambda I - A(x)) = \prod_{i=1}^{n}(\lambda - \lambda_i(x)), \quad \lambda_i(x) \in \mathrm{H}_0,\ i = 1, \ldots, n.$$
Then $A(x)$ is analytically similar to
$$\begin{aligned} &C(x) = \oplus_{i=1}^{\ell} C_i(x), \quad C_i(x) \in \mathrm{H}_0^{n_i\times n_i}, \\ &(\alpha_i I_{n_i} - C_i(0))^{n_i} = 0,\ \alpha_i = \lambda_{n_i}(0),\ \alpha_i \neq \alpha_j \quad \text{for} \quad i \neq j,\ i, j = 1, \ldots, \ell. \end{aligned}$$
16. Assume that the characteristic polynomial of $A(x) \in \mathrm{H}_0$ splits in H_0.Then $A(x)$ is analytically similar to a block diagonal matrix $C(x)$ of the form Fact 15 such that each $C_i(x)$ is an upper triangular matrix whose off-diagonal entries are polynomials in x. Moreover, the degree of each polynomial entry above the diagonal in the matrix $C_i(x)$ does not exceed $\eta(C_i, C_i)$ for $i = 1, \ldots, \ell$.
17. Let $P(x)$ and $Q(x)$ be matrices of the form
$$\begin{aligned} &P(x) = \oplus_{i=1}^{p} P_i(x),\ P_i(x) \in \mathrm{H}_0^{m_i\times m_i}, \\ &(\alpha_i I_{m_i} - P_i(0))^{m_i} = 0,\ \alpha_i \neq \alpha_j \quad \text{for} \quad i \neq j, i, j = 1, \ldots, p, \\ &Q(x) = \oplus_{j=1}^{q} Q_j(x),\ Q_j(x) \in \mathrm{H}_0^{n_j\times n_j}, \\ &(\beta_j I_{n_j} - Q_j(0))^{n_j} = 0,\ \beta_i \neq \beta_j \quad \text{for} \quad i \neq j,\ i, j = 1, \ldots, q. \end{aligned}$$
Assume furthermore that
$$\alpha_i = \beta_i,\ i = 1, \ldots, t,\ \alpha_j \neq \beta_j,\ i = t+1, \ldots, p,\ j = t+1, \ldots, q,\ 0 \leq t \leq \min(p, q).$$
Then the nonconstant local invariant polynomials of $I \otimes P(x) - Q(x)^{\mathrm{T}} \otimes I$ are the nonconstant local invariant polynomials of $I \otimes P_i(x) - Q_i(x)^{\mathrm{T}} \otimes I$ for $i = 1, \ldots, t$:
$$\mathcal{K}_p(P, Q) = \sum_{i=1}^{t} \mathcal{K}_p(P_i, Q_i), \quad p = 1, \ldots, .$$
In particular, if $C(x)$ is of the form in Fact 15, then
$$\eta(C, C) = \max_{1\leq i\leq \ell} \eta(C_i, C_i).$$

18. $A(x)\overset{a}{\approx}B(x) \iff A(y^m)\overset{a}{\approx}B(y^m)$ for any $2 \le m \in \mathbb{N}$.
19. [GR65] (*Weierstrass preparation theorem*) For any monic polynomial $p(\lambda, x) = \lambda^n + \sum_{i=1}^n a_i(x)\lambda^{n-i} \in \mathrm{H}_0[\lambda]$ there exists $m \in \mathbb{N}$ such that $p(\lambda, y^m)$ splits over H_0.
20. For a given rational canonical form $A(x) \in \mathrm{H}_0^{2\times 2}$ there are at most a countable number of analytic similarity classes. (See Example 3.)
21. For a given rational canonical form $A(x) \in \mathrm{H}_0^{n\times n}$, where $n \ge 3$, there may exist a family of distinct similarity classes corresponding to a finite dimensional variety. (See Example 4.)
22. Let $A(x) \in H_0^{n\times n}$ and assume that the characteristic polynomial of $A(x)$ splits in H_0 as in Fact 15. Let $B(x) = \mathrm{diag}\,(\lambda_1(x), \dots, \lambda_n(x))$. Then $A(x)$ and $B(x)$ are not analytically similar if and only if there exists a nonnegative integer p such that

$$\mathcal{K}_p(A, A) + \mathcal{K}_p(B, B) < 2\mathcal{K}_p(A, B),$$
$$\mathcal{K}_j(A, A) + \mathcal{K}_j(B, B) = 2\mathcal{K}_j(A, B), \ \ j = 0, \dots, p-1, \quad \text{if } p \ge 1.$$

In particular, $A(x)\overset{a}{\approx}B(x)$ if and only if the three matrices given in Fact 6 are equivalent over H_0.
23. [Fri78] Let $A(x) \in \mathbb{C}[x]^{n\times n}$. Then each eigenvalue $\lambda(x)$ of $A(x)$ is an algebraic function. Assume that $A(\zeta)$ is diagonalizable for some $\zeta \in \mathbb{C}$. Then the linear part of each branch of $\lambda_j(x)$ is linear at ζ, i.e., is of the form $\alpha + \beta x$ for some $\alpha, \beta \in \mathbb{C}$.
24. Let $A(x) \in \mathbb{C}[x]^{n\times n}$ be of the form $A(x) = \sum_{k=0}^{\ell} A_k x^k$, where $A_k \in \mathbb{C}^{n\times n}$ for $k = 0, \dots, \ell$ and $\ell \ge 1$, $A_\ell \ne 0$. Then one of the following conditions imply that $A(x) = S(x)B(x)S^{-1}(x)$, where $S(x) \in \mathrm{GL}(n, \mathbb{C}[x])$ and $B(x) \in \mathbb{C}[x]^{n\times n}$ is a diagonal matrix of the form $\oplus_{i=1}^m \lambda_i(x) I_{k_i}$, where $k_1, \dots, k_m \ge 1$. Furthermore, $\lambda_1(x), \dots, \lambda_m(x)$ are m distinct polynomials satisfying the following conditions:

    (a) $\deg \lambda_1 = \ell \ge \deg \lambda_i(x), i = 2, \dots, m-1$.

    (b) The polynomial $\lambda_i(x) - \lambda_j(x)$ has only simple roots in $\mathbb{C}$ for $i \ne j$. ($\lambda_i(\zeta) = \lambda_j(\zeta) \Rightarrow \lambda_i'(\zeta) \ne \lambda_j'(\zeta)$).

    i. The characteristic polynomial of $A(x)$ splits in $\mathbb{C}[x]$, i.e., all the eigenvalues of $A(x)$ are polynomials. $A(x)$ is point-wise diagonalizable in $\mathbb{C}$ and no two distinct eigenvalues are tangent at any $\zeta \in \mathbb{C}$.

    ii. $A(x)$ is point-wise diagonalizable in $\mathbb{C}$ and A_ℓ is diagonalizable. No two distinct eigenvalues are tangent at any point $\zeta \in \mathbb{C} \cup \{\infty\}$. Then $A(x)$ is strictly similar to $B(x)$, i.e., $S(x)$ can be chosen in $\mathrm{GL}(n, \mathbb{C})$. Furthermore, $\lambda_1(x), \dots, \lambda_m(x)$ satisfy the additional condition:

    (c) For $i \ne j$, either $\frac{d^\ell \lambda_i}{d^\ell x}(0) \ne \frac{d^\ell \lambda_j}{d^\ell x}(0)$ or $\frac{d^\ell \lambda_i}{d^\ell x}(0) = \frac{d^\ell \lambda_j}{d^\ell x}(0)$ and $\frac{d^{\ell-1} \lambda_i}{d^{\ell=1} x}(0) \ne \frac{d^{\ell-1} \lambda_j}{d^{\ell-1} x}(0)$.

Examples:

1. Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}, \ B = \begin{bmatrix} 1 & 1 \\ 0 & 5 \end{bmatrix} \in \mathbb{Z}^{2\times 2}.$$

Then $A(x)$ and $B(x)$ have the same invariant polynomials over $\mathbb{Z}[x]$ and A and B are not similar over $\mathbb{Z}$.

2. Let

$$A(z) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad D(z) = \begin{bmatrix} z & 0 \\ 0 & 1 \end{bmatrix}.$$

Then $zA(z) = D(z)A(z)D(z)^{-1}$, i.e., $A(z), zA(z)$ are rationally similar. Clearly $A(z)$ and $zA(z)$ are not point-wise similar for any Ω containing 0. Now $zA(z), z^2A(z)$ are point-wise similar in $\mathbb{C}$, but they are not locally similar on H_0.

3. Let $A(x) \in \mathrm{H}_0^{2\times 2}$ and assume that $\det(\lambda I - A(x)) = (\lambda - \lambda_1(x))(\lambda - \lambda_2(x))$. Then $A(x)$ is analytically similar either to a diagonal matrix or to

$$B(x) = \begin{bmatrix} \lambda_1(x) & x^k \\ 0 & \lambda_2(x) \end{bmatrix}, \quad k = 0, \ldots, p \ (p \geq 0).$$

Furthermore, if $A(x) \overset{a}{\approx} B(x)$, then $\eta(A, A) = k$.

4. Let $A(x) \in \mathrm{H}_0^{3\times 3}$. Assume that

$$A(x) \overset{r}{\approx} C(p), \quad p(\lambda, x) = \lambda(\lambda - x^{2m})(\lambda - x^{4m}), \quad m \geq 1.$$

Then $A(x)$ is analytically similar to a matrix

$$B(x,a) = \begin{bmatrix} 0 & x^{k_1} & a(x) \\ 0 & x^{2m} & x^{k_2} \\ 0 & 0 & x^{4m} \end{bmatrix}, \quad 0 \leq k_1, k_2 \leq \infty \ (x^{\infty} = 0),$$

where $a(x)$ is a polynomial of degree $4m - 1$ at most. Furthermore, $B(x,a) \overset{a}{\approx} B(x,b)$ if and only if

(a) If $a(0) \neq 1$, then $b - a$ is divisible by x^m.

(b) If $a(0) = 1$ and $\frac{d^i a}{dx^i} = 0$, $i = 1, \ldots, k-1$, $\frac{d^k a}{dx^k} \neq 0$ for $1 \leq k < m$, then $b - a$ is divisible by x^{m+k}.

(c) If $a(0) = 1$ and $\frac{d^i a}{dx^i} = 0$, $i = 1, \ldots, m$, then $b - a$ is divisible by x^{2m}.

Then for $k_1 = k_2 = m$ and $a(0) \in \mathbb{C}\backslash\{1\}$, we can assume that $a(x)$ is a polynomial of degree less than m. Furthermore, the similarity classes of $A(x)$ are uniquely determined by such $a(x)$. These similarity classes are parameterized by $\mathbb{C}\backslash\{1\} \times \mathbb{C}^{m-1}$ (the Taylor coefficients of $a(x)$).

24.2 Simultaneous Similarity of Matrices

In this section, we introduce the notion of simultaneous similarity of matrices over a domain $\mathbb{D}$. The problem of simultaneous similarity of matrices over a field F, i.e., to describe the similarity class of a given m (≥ 2) tuple of matrices or to decide when a given two tuples of matrices are simultaneously similar, is in general a hard problem, which will be discussed in the next sections. There are some cases where this problem has a relatively simple solution. As shown below, the problem of analytic similarity of $A(x), B(x) \in \mathrm{H}_0^{n\times n}$ reduces to the problem of simultaneously similarity of certain 2-tuples of matrices.

Definitions:

For $A_0, \ldots, A_l \in \mathbb{D}^{n\times n}$ denote by $\mathcal{A}(A_0, \ldots, A_l) \subset \mathbb{D}^{n\times n}$ the minimal algebra in $\mathbb{D}^{n\times n}$ containing I_n and $A_0, \ldots, A_l$. Thus, every matrix $G \in \mathcal{A}(A_0, \ldots, A_l)$ is a noncommutative polynomial in $A_0, \ldots, A_l$.

For $l \geq 1$, $(A_0, A_1, \ldots, A_\ell), (B_0, \ldots, B_\ell) \in (\mathbb{D}^{n\times n})^{\ell+1}$ are called **simultaneously** similar, denoted by $(A_0, A_1, \ldots, A_\ell) \approx (B_0, \ldots, B_\ell)$, if there exists $P \in \mathrm{GL}(n, \mathbb{D})$ such that $B_i = P A_i P^{-1}, i = 0, \ldots, \ell$, i.e., $(B_0, B_1, \ldots, B_\ell) = P(A_0, A_1, \ldots, A_\ell)P^{-1}$.

Associate with $(A_0, A_1, \ldots, A_\ell), (B_0, \ldots, B_\ell) \in (\mathbb{D}^{n\times n})^{\ell+1}$ the matrix polynomials $A(x) = \sum_{i=0}^{\ell} A_i x^i$, $B(x) = \sum_{i=0}^{\ell} B_i x^i \in \mathbb{D}[x]^{n\times n}$. $A(x)$ and $B(x)$ are called **strictly** similar ($A \overset{s}{\approx} B$) if there exists $P \in \mathrm{GL}(n, \mathbb{D})$ such that $B(x) = P A(x) P^{-1}$.

Facts:

1. $A \overset{s}{\approx} B \iff (A_0, A_1, \ldots, A_\ell) \approx (B_0, \ldots, B_\ell)$.
2. $(A_0, \ldots, A_\ell) \in (\mathbb{C}^{n\times n})^{\ell+1}$ is simultaneously similar to a diagonal tuple $(B_0, \ldots, B_\ell) \in (\mathbb{C}^{n\times n})^{\ell+1}$, i.e., each B_i is a diagonal matrix if and only if $A_0, \ldots, A_\ell$ are $\ell + 1$ commuting diagonalizable matrices: $A_i A_j = A_j A_i$ for $i, j = 0, \ldots, \ell$.

3. If $A_0, \ldots, A_\ell \in \mathbb{C}^{n\times n}$ commute, then $(A_0, \ldots, A_\ell)$ is simultaneously similar to an upper triangular tuple $(B_0, \ldots, B_\ell)$.
4. Let $l \in \mathbb{N}$, $(A_0, \ldots, A_l), (B_0, \ldots, B_l) \in (\mathbb{C}^{n\times n})^{l+1}$, and $U = [U_{ij}]_{i,j=1}^{l+1}, V = [V_{ij}]_{i,j=1}^{l+1}, W = [W_{ij}]_{i,j=1}^{l+1} \in \mathbb{C}^{n(l+1)\times n(l+1)}$, $U_{ij}, V_{ij}, W_{ij} \in \mathbb{C}^{n\times n}, i, j = 1, \ldots, l+1$ be block upper triangular matrices with the following block entries:

$$U_{ij} = A_{j-i},\ V_{ij} = B_{j-i},\ W_{ij} = \delta_{(i+1)j} I_n, \quad i = 1, \ldots, l+1, j = i, \ldots, l+1.$$

   Then the system in Fact 14 of section 24.1 is solvable with $T_0 \in \mathrm{GL}(n, \mathbb{C})$ if and only for $l = \kappa(A, A)$ the pairs (U, W) and (V, W) are simultaneously similar.
5. For $A_0, \ldots, A_\ell \in (\mathbb{C}^{n\times n})^{\ell+1}$ TFAE:
   - $(A_0, \ldots, A_\ell)$ is simultaneously similar to an upper triangular tuple $(B_0, \ldots, B_\ell) \in (\mathbb{C}^{n\times n})^{\ell+1}$.
   - For any $0 \le i < j \le \ell$ and $M \in \mathcal{A}(A_0, \ldots, A_\ell)$, the matrix $(A_i A_j - A_j A_i)M$ is nilpotent.
6. Let $\mathcal{X}_0 = \mathcal{A}(A_0, \ldots, A_\ell) \subseteq F^{n\times n}$ and define recursively

$$\mathcal{X}_k = \sum_{0\le i<j\le \ell} (A_i A_j - A_j A_i)\mathcal{X}_{k-1} \subseteq F^{n\times n}, \quad k = 1, \ldots.$$

   Then $(A_0, \ldots, A_\ell)$ is simultaneously similar to an upper triangular tuple if and only if the following two conditions hold:
   - $A_i \mathcal{X}_k \subseteq \mathcal{X}_k, \quad i = 0, \ldots, \ell,\ k = 0, \ldots.$
   - There exists $q \ge 1$ such that $\mathcal{X}_q = \{0\}$ and $\mathcal{X}_k$ is a strict subspace of $\mathcal{X}_{k-1}$ for $k = 1, \ldots, q$.

Examples:

1. This example illustrates the construction of the matrices U and W in Fact 4. Let $A_0 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $A_1 = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$, and $A_2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. Then

$$U = \begin{bmatrix} 1 & 2 & 5 & 6 & 1 & -1 \\ 3 & 4 & 7 & 8 & -1 & 1 \\ 0 & 0 & 1 & 2 & 5 & 6 \\ 0 & 0 & 3 & 4 & 7 & 8 \\ 0 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 3 & 4 \end{bmatrix} \quad \text{and} \quad W = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

24.3 Property L

Property L was introduced and studied in [MT52] and [MT55]. In this section, we consider only square pencils $A(x) = A_0 + A_1 x \in \mathbb{C}[x]^{n\times n}$, $A(x_0, x_1) \in \mathbb{C}[x_0, x_1]^{n\times n}$, where $A_1 \ne 0$.

Definitions:

A pencil $A(x) \in \mathbb{C}[x]^{n\times n}$ has **property** L if all the eigenvalues of $A(x_0, x_1)$ are linear functions. That is, $\lambda_i(x_0, x_1) = \alpha_i x_0 + \beta_i x_1$ is an eigenvalue of $A(x_0, x_1)$ of multiplicity n_i for $i = 1, \ldots, m$, where

$$n = \sum_{i=1}^{m} n_i, \quad (\alpha_i, \beta_i) \ne (\alpha_j, \beta_j), \quad \text{for} \quad 1 \le i < j \le m.$$

A pencil $A(x) = A_0 + A_1 x$ is **Hermitian** if A^0, A^1 are Hermitian.

Facts:

Most of the results of this section can be found in [MF80], [Fri81], and [Frixx].

1. For a pencil $A(x) = A_0 + xA_1 \in \mathbb{C}[x]^{n\times n}$ TFAE:
   - $A(x)$ has property L.
   - The eigenvalues of $A(x)$ are polynomials of degree 1 at most.
   - The characteristic polynomial of $A(x)$ splits into linear factors over $\mathbb{C}[x]$.
   - There is an ordering of the eigenvalues of A_0 and A_1, $\alpha_1, \ldots, \alpha_n$ and $\beta_1, \ldots, \beta_n$, respectively, such that the eigenvalues of $A_0x_0 + A_1x_1$ are $\alpha_1x_0 + \beta_1x_1, \ldots, \alpha_nx_0 + \beta_nx_1$.
2. A pencil in $A(x)$ has property L if one of the following conditions hold:
   - $A(x)$ is similar over $\mathbb{C}(x)$ to an upper triangular matrix $U(x) \in \mathbb{C}(x)^{n\times n}$.
   - $A(x)$ is strictly similar to an upper triangular pencil $U(x) = U_0 + U_1x$.
   - $A(x)$ is similar over $\mathbb{C}[x]$ to a diagonal matrix $B(x) \in \mathbb{C}[x]^{n\times n}$.
   - $A(x)$ is strictly similar to diagonal pencil.
3. If a pencil $A(x_0, x_1)$ has property L, then any two distinct eigenvalues are not tangent at any point of $\mathbb{C} \cup \infty$.
4. Assume that $A(x)$ is point-wise diagonalizable on $\mathbb{C}$. Then $A(x)$ has property L. Furthermore, $A(x)$ is similar over $\mathbb{C}[x]$ to a diagonal pencil $B(x) = B_0 + B_1x$. Suppose furthermore that A_1 is diagonalizable, i.e., $A(x_0, x_1)$ is point-wise diagonalizable on $\mathbb{C}^2$. Then $A(x)$ is strictly similar to a diagonal pencil $B(x)$, i.e., A_0 and A_1 are commuting diagonalizable matrices.
5. Let $A(x) = A_0 + A_1x \in \mathbb{C}[x]^{n\times n}$ such that A_1 and A_2 are diagonalizable and $A_0A_1 \neq A_1A_0$. Then exactly one of the following conditions hold:
   - $A(x)$ is not diagonalizable exactly at the points $\zeta_1, \ldots, \zeta_p$, where $1 \le p \le n(n-1)$.
   - For $n \ge 3$, $A(x) \in \mathbb{C}[x]^{n\times n}$ is diagonalizable exactly at the points $\zeta_1 = 0, \ldots, \zeta_q$ for some $q \ge 1$. (*We do not know if this condition is satisfied for some pencil.*)
6. Let $A(x) = A_0 + A_1x$ be a Hermitian pencil satisfying $A_0A_1 \neq A_1A_0$. Then there exists $2q$ distinct complex points $\zeta_1, \overline{\zeta}_1 \ldots, \zeta_q, \overline{\zeta}_q \in \mathbb{C}\backslash\mathbb{R}$, $1 \le q \le \frac{n(n-1)}{2}$ such that $A(x)$ is not diagonalizable if and only if $x \in \{\zeta_1, \overline{\zeta}_1, \ldots, \zeta_q, \overline{\zeta}_q\}$.

Examples:

1. This example illustrates the case $n = 2$ of Fact 5. Let

$$A_0 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{and} \quad A_1 = \begin{bmatrix} 1 & 3 \\ -3 & 1 \end{bmatrix}, \quad \text{so} \quad A(x) = \begin{bmatrix} x+1 & 3x \\ -3x & x+2 \end{bmatrix}.$$

For $\zeta \in \mathbb{C}$, the only possible way $A(\zeta)$ can fail to be diagonalizable is if $A(\zeta)$ has repeated eigenvalues. The eigenvalues of $A(\zeta)$ are $\frac{1}{2}\left(2\zeta - \sqrt{1-36\zeta^2} + 3\right)$ and $\frac{1}{2}\left(2\zeta + \sqrt{1-36\zeta^2} + 3\right)$, so the only values of ζ at which it is possible that $A(\zeta)$ is not diagonalizeable are $\zeta = \pm\frac{1}{6}$, and in fact $A(\pm\frac{1}{6})$ is not diagonalizable.

24.4 Simultaneous Similarity Classification I

This section outlines the setting for the classification of conjugacy classes of $l+1$ tuples $(A_0, A_1, \ldots, A_l) \in (\mathbb{C}^{n\times n})^{l+1}$ under the simultaneous similarity. This classification depends on certain standard notions in algebraic geometry that are explained briefly in this section. A detailed solution to the classification of conjugacy classes of $l+1$ tuples is outlined in the next section.

Definitions:

$\mathcal{X} \subset \mathbb{C}^N$ is called an **affine algebraic variety** (called here a **variety**) if it is the zero set of a finite number of polynomial equations in $\mathbb{C}^N$.

$\mathcal{X}$ is irreducible if $\mathcal{X}$ does not decompose in a nontrivial way to a union of two varieties.

If $\mathcal{X}$ is a finite nontrivial union of irreducible varieties, these irreducible varieties are called the **irreducible** components of $\mathcal{X}$.

$x \in \mathcal{X}$ is called a **regular** (**smooth**) point of irreducible $\mathcal{X}$ if in the neighborhood of this point $\mathcal{X}$ is a complex manifold of a fixed dimension d, which is called the **dimension** of $\mathcal{X}$ and is denoted by dim $\mathcal{X}$. $\emptyset$ is an irreducible variety of dimension -1.

For a reducible variety $\mathcal{Y} \subset \mathbb{C}^N$, the **dimension** of $\mathcal{Y}$, denoted by dim $\mathcal{Y}$, is the maximum dimension of its irreducible components.

The set of singular (nonsmooth) points of $\mathcal{X}$ is denoted by $\mathcal{X}_s$.

A set $\mathcal{Z}$ is a **quasi-irreducible** variety if there exists a nonempty irreducible variety $\mathcal{X}$ and a strict subvariety $\mathcal{Y} \subset \mathcal{X}$ such that $\mathcal{Z} = \mathcal{X}\backslash\mathcal{Y}$. The dimension of $\mathcal{Z}$, denoted by dim $\mathcal{Z}$, is defined to be equal to the dimension of $\mathcal{X}$.

A quasi-irreducible variety $\mathcal{Z}$ is **regular** if $\mathcal{Z} \subset \mathcal{X}\backslash\mathcal{X}_s$.

A **stratification** of $\mathbb{C}^N$ is a decomposition of $\mathbb{C}^N$ to a finite disjoint union of $\mathcal{X}_1, \ldots, \mathcal{X}_p$ of regular quasi-irreducible varieties such that Cl $(\mathcal{X}_i)\backslash\mathcal{X}_i = \cup_{j\in\mathcal{A}_i}\mathcal{X}_j$ for some $\mathcal{A}_i \subset \{1, \ldots, p\}$ for $i = 1, \ldots, p$. (Cl $(\mathcal{X}_i) = \mathcal{X}_i \iff \mathcal{A}_i = \emptyset$.)

Denote by $\mathbb{C}[\mathbb{C}^N]$ the ring of polynomial in N variables with coefficients in $\mathbb{C}$.

Denote by $\mathcal{W}_{n,l+1,r+1}$ the finite dimensional vector space of multilinear polynomials in $(l+1)n^2$ variables of degree at most $r+1$. That is, the degree of each variable in any polynomial is at most 1. $N(n,l,r) :=$ dim $\mathcal{W}_{n,l+1,r+1}$. $\mathcal{W}_{n,l+1,r+1}$ has a standard basis $\mathbf{e}_1, \ldots, \mathbf{e}_{N(n,l,r)}$ in $\mathcal{W}_{n,l+1,r+1}$ consisting of monomials in $(l+1)n^2$ variables of degree $r+1$ at most, arranged in a lexicographical order.

Let $\mathcal{X} \subset \mathbb{C}^N$ be a quasi-irreducible variety. Denote by $\mathbb{C}[\mathcal{X}]$ the restriction of all polynomials $f(x) \in \mathbb{C}[\mathbb{C}^N]$ to $\mathcal{X}$, where $f, g \in \mathbb{C}[\mathbb{C}^N]$ are identified if $f - g$ vanishes on $\mathcal{X}$. Let $\mathbb{C}(\mathcal{X})$ denote the quotient field of $\mathbb{C}[\mathcal{X}]$.

A rational function $h \in \mathbb{C}(\mathcal{X})$ is **regular** if h is defined everywhere in $\mathcal{X}$. A regular rational function on $\mathcal{X}$ is an analytic function.

Denote by $\mathbf{A}$ the $l+1$ tuple $(A_0, \ldots, A_l) \in (\mathbb{C}^{n\times n})^{l+1}$. The group GL$(n,\mathbb{C})$ acts by conjugation on $(\mathbb{C}^{n\times n})^{l+1}$: $T\mathbf{A}T^{-1} = (TA_0T^{-1}, \ldots TA_lT^{-1})$ for any $\mathbf{A} \in (\mathbb{C}^{n\times n})^{l+1}$ and $T \in$ GL$(n,\mathbb{C})$.

Let orb $(\mathbf{A}) := \{T\mathbf{A}T^{-1} :\ T \in \text{GL}(n,\mathbb{C})\}$ be the **orbit** of $\mathbf{A}$ (under the action of GL$(n,\mathbb{C})$).

Let $\mathcal{X} \subset (\mathbb{C}^{n\times n})^{l+1}$ be a quasi-irreducible variety. $\mathcal{X}$ is called **invariant** (under the action of GL$(n,\mathbb{C})$) if $T\mathcal{X}T^{-1} = \mathcal{X}$ for all $T \in$ GL$(n,\mathbb{C})$.

Assume that $\mathcal{X}$ is an invariant quasi-irreducible variety. A rational function $h \in \mathbb{C}(\mathcal{X})$ is called **invariant** if h is the same value on any two points of a given orbit in $\mathcal{X}$, where h is defined. Denote by $\mathbb{C}[\mathcal{X}]^{\text{inv}} \subseteq \mathbb{C}[\mathcal{X}]$ and $\mathbb{C}(\mathcal{X})^{\text{inv}} \subseteq \mathbb{C}(\mathcal{X})$ the subdomain of invariant polynomials and subfield of invariant functions, respectively.

Facts:

For general background, consult for example [Sha77]. More specific details are given in [Fri83], [Fri85], and [Fri86].

1. An intersection of a finite or infinite number of varieties is a variety, which can be an empty set.
2. A finite union of varieties in $\mathbb{C}^N$ is a variety.
3. Every variety $\mathcal{X}$ is a finite nontrivial union of irreducible varieties.
4. Let $\mathcal{X} \subset \mathbb{C}^N$ be an irreducible variety. Then $\mathcal{X}$ is path-wise connected.
5. $\mathcal{X}_s$ is a proper subvariety of the variety $\mathcal{X}$ and dim $\mathcal{X}_s <$ dim $\mathcal{X}$.
6. dim $\mathbb{C}^N = N$ and $(\mathbb{C}^N)_s = \emptyset$. For any $\mathbf{z} \in \mathbb{C}^N$, the set $\{\mathbf{z}\}$ is an irreducible variety of dimension 0.
7. A quasi-irreducible variety $\mathcal{Z} = \mathcal{X}\backslash\mathcal{Y}$ is path-wise connected and its closure, denoted by Cl $(\mathcal{Z})$, is equal to $\mathcal{X}$. Cl $(\mathcal{Z})\backslash\mathcal{Z}$ is a variety of dimension strictly less than the dimension of $\mathcal{Z}$.

8. The set of all regular points of an irreducible variety $\mathcal{X}$, denoted by $\mathcal{X}_r := \mathcal{X}\backslash\mathcal{X}_s$, is a quasi-irreducible variety. Moreover, $\mathcal{X}_r$ is a path-wise connected complex manifold of complex dimension dim $\mathcal{X}$.
9. $N(n,l,r) := \dim \mathcal{W}_{n,l+1,r+1} = \sum_{i=0}^{r+1} \binom{(l+1)n^2}{i}$.
10. For an irreducible $\mathcal{X}$, $\mathbb{C}[\mathcal{X}]$ is an integral domain.
11. For a quasi-irreducible $\mathcal{X}$, $\mathbb{C}[\mathcal{X}], \mathbb{C}(\mathcal{X})$ can be identified with $\mathbb{C}[\mathrm{Cl}\ (\mathcal{X})], \mathbb{C}(\mathrm{Cl}\ (\mathcal{X}))$, respectively.
12. For $\mathbf{A} \in (\mathbb{C}^{n\times n})^{l+1}$, orb $(\mathbf{A})$ is a quasi-irreducible variety in $(\mathbb{C}^{n\times n})^{l+1}$.
13. Let $\mathcal{X} \subset (\mathbb{C}^{n\times n})^{l+1}$ be a quasi-irreducible variety. $\mathcal{X}$ is invariant if $\mathbf{A} \in \mathcal{X} \iff$ orb $(\mathbf{A}) \subseteq \mathcal{X}$.
14. Let $\mathcal{X}$ be an invariant quasi-irreducible variety. The quotient field of $\mathbb{C}[\mathcal{X}]^{\mathrm{inv}}$ is a subfield of $\mathbb{C}(\mathcal{X})^{\mathrm{inv}}$, and in some interesting cases the quotient field of $\mathbb{C}[\mathcal{X}]^{\mathrm{inv}}$ is a strict subfield of $\mathbb{C}(\mathcal{X})^{\mathrm{inv}}$.
15. Assume that $\mathcal{X} \subset (\mathbb{C}^{n\times n})^{l+1}$ is an invariant quasi-irreducible variety. Then $\mathbb{C}[\mathcal{X}]^{\mathrm{inv}}$ and $\mathbb{C}(\mathcal{X})^{\mathrm{inv}}$ are finitely invariant generated. That is, there exists $f_1, \ldots, f_i \in \mathbb{C}[\mathcal{X}]^{\mathrm{inv}}$ and $g_1, \ldots, g_j \in \mathbb{C}(\mathcal{X})^{\mathrm{inv}}$ such that any polynomial in $\mathbb{C}[\mathcal{X}]^{\mathrm{inv}}$ is a polynomial in $f_1, \ldots, f_i$, and any rational function in $\mathbb{C}(\mathcal{X})^{\mathrm{inv}}$ is a rational function in $g_1, \ldots, g_j$.
16. (**Classification Theorem**) Let $n \geq 2$ and $l \geq 0$ be fixed integers. Then there exists a stratification $\cup_{i=1}^{p}\mathcal{X}_i$ of $(\mathbb{C}^{n\times n})^{l+1}$ with the following properties. For each $\mathcal{X}_i$ there exist m_i regular rational functions $g_{1,i}, \ldots, g_{m_i,i} \in \mathbb{C}(\mathcal{X}_i)^{\mathrm{inv}}$ such that the values of $g_{j,i}$ for $j = 1, \ldots, m_i$ on any orbit in $\mathcal{X}_i$ determines this orbit uniquely.
    The rational functions $g_{1,i}, \ldots, g_{m_i,i}$ are the generators of $\mathbb{C}(\mathcal{X}_i)^{\mathrm{inv}}$ for $i = 1, \ldots, p$.

Examples:

1. Let $\mathcal{S}$ be an irreducible variety of scalar matrices $\mathcal{S} := \{A \in \mathbb{C}^{2\times 2} : A = \frac{\mathrm{tr}\ A}{2} I_2\}$ and $\mathcal{X} := \mathbb{C}^{2\times 2}\backslash\mathcal{S}$ be a quasi-irreducible variety. Then dim $\mathcal{X} = 4$, dim $\mathcal{S} = 1$, and $\mathbb{C}^{2\times 2} = \mathcal{X} \cup \mathcal{S}$ is a stratification of $\mathbb{C}^{2\times 2}$.
2. Let $\mathcal{U} \subset (\mathbb{C}^{2\times 2})^2$ be the set of all pairs $(A, B) \in (\mathbb{C}^{2\times 2})^2$, which are simultaneously similar to a pair of upper triangular matrices. Then $\mathcal{U}$ is a variety given by the zero of the following polynomial:

    $$\mathcal{U} := \{(A, B) \in (\mathbb{C}^{2\times 2})^2 : (2\,\mathrm{tr}\ A^2 - (\mathrm{tr}\ A)^2)(2\,\mathrm{tr}\ B^2 - (\mathrm{tr}\ B)^2) - (2\,\mathrm{tr}\ AB - \mathrm{tr}\ A\,\mathrm{tr}\ B)^2 = 0\}.$$

    Let $\mathcal{C} \subset \mathcal{U}$ be the variety of commuting matrices:

    $$\mathcal{C} := \{(A, B) \in (\mathbb{C}^{2\times 2})^2 : AB - BA = 0\}.$$

    Let $\mathcal{V}$ be the variety given by the zeros of the following three polynomials:

    $$\mathcal{V} := \{(A, B) \in (\mathbb{C}^{2\times 2})^2 : 2\,\mathrm{tr}\ A^2 - (\mathrm{tr}\ A)^2 = 2\,\mathrm{tr}\ B^2 - (\mathrm{tr}\ B)^2 = 2\,\mathrm{tr}\ AB - \mathrm{tr}\ A\,\mathrm{tr}\ B = 0\}.$$

    Then $\mathcal{V}$ is the variety of all pairs (A, B), which are simultaneously similar to a pair of the form $\left(\begin{bmatrix} \lambda & \alpha \\ 0 & \lambda \end{bmatrix}, \begin{bmatrix} \mu & \beta \\ 0 & \mu \end{bmatrix}\right)$. Hence, $\mathcal{V} \subset \mathcal{C}$. Let $\mathcal{W} := \{A \in (\mathbb{C}^{2\times 2}) : 2\,\mathrm{tr}\ A^2 - (\mathrm{tr}\ A)^2 = 0\}$ and $\mathcal{S} \subset \mathcal{W}$ be defined as in the previous example. Define the following quasi-irreducible varieties in $(\mathbb{C}^{2\times 2})^2$:

    $$\mathcal{X}_1 := (\mathbb{C}^{2\times 2})^2\backslash\mathcal{U},\ \mathcal{X}_2 := \mathcal{U}\backslash\mathcal{C},\ \mathcal{X}_3 = \mathcal{C}\backslash\mathcal{V},\ \mathcal{X}_4 := \mathcal{V}\backslash(\mathcal{S} \times \mathcal{W} \cup \mathcal{W} \times \mathcal{S}),$$
    $$\mathcal{X}_5 := \mathcal{S} \times (\mathcal{W}\backslash S),\ \mathcal{X}_6 := (\mathcal{W}\backslash\mathcal{S}) \times \mathcal{S},\ \mathcal{X}_7 = \mathcal{S} \times \mathcal{S}.$$

    Then

    $$\dim \mathcal{X}_1 = 8,\ \dim \mathcal{X}_2 = 7,\ \dim \mathcal{X}_3 = 6,\ \dim \mathcal{X}_4 = 5,$$
    $$\dim \mathcal{X}_5 = \dim \mathcal{X}_6 = 4,\ \dim \mathcal{X}_7 = 2,$$

    and $\cup_{i=1}^{7}\mathcal{X}_i$ is a stratification of $(\mathbb{C}^{2\times 2})^2$.
3. In the classical case of similarity classes in $\mathbb{C}^{n\times n}$, i.e., $l = 0$, it is possible to choose a fixed set of polynomial invariant functions as $g_j(A) = \mathrm{tr}\ (A^j)$ for $j = 1, \ldots, n$. However, we still have to stratify $\mathbb{C}^{n\times n}$ to $\cup_{i=1}^{p}\mathcal{X}_i$, where each $A \in \mathcal{X}_i$ has some specific Jordan structures.

4. Consider the stratification $\mathbb{C}^{2\times 2} = \mathcal{X} \cup \mathcal{S}$ as in Example 1. Clearly $\mathcal{X}$ and $\mathcal{S}$ are invariant under the action of $\mathrm{GL}(2, \mathbb{C})$. The invariant functions $\mathrm{tr}\ A, \mathrm{tr}\ A^2$ determine uniquely orb (A) on $\mathcal{X}$. The Jordan canonical for of any A in $\mathcal{X}$ is either consists of two distinct Jordan blocks of order 1 or one Jordan block of order 2. The invariant function $\mathrm{tr}\ A$ determines orb (A) for any $A \in \mathcal{S}$. It is possible to refine the stratification of $\mathbb{C}^{2\times 2}$ to three invariant components $\mathbb{C}^{2\times 2}\backslash\mathcal{W}, \mathcal{W}\backslash\mathcal{S}, \mathcal{S}$, where $\mathcal{W}$ is defined in Example 2. Each component contains only matrices with one kind of Jordan block. On the first component, $\mathrm{tr}\ A, \mathrm{tr}\ A^2$ determine the orbit, and on the second and third component, $\mathrm{tr}\ A$ determines the orbit.
5. To see the fundamental difference between similarity ($l = 0$) and simultaneous similarity $l \geq 1$, it is suffice to consider Example 2. Observe first that the stratification of $(\mathbb{C}^{2\times 2})^2 = \cup_{i=1}^{7} \mathcal{X}_i$ is invariant under the action of $\mathrm{GL}(2, \mathbb{C})$. On $\mathcal{X}_1$ the five invariant polynomials $\mathrm{tr}\ A$, $\mathrm{tr}\ A^2$, $\mathrm{tr}\ B$, $\mathrm{tr}\ B^2$, $\mathrm{tr}\ AB$, which are algebraically independent, determine uniquely any orbit in $\mathcal{X}_1$.

   Let $(A = [a_{ij}], B = [b_{ij}]) \in \mathcal{X}_2$. Then A and B have a unique one-dimensional common eigenspace corresponding to the eigenvalues λ_1, μ_1 of A, B, respectively. Assume that $a_{12}b_{21} - a_{21}b_{12} \neq 0$. Define

$$\lambda_1 = \alpha(A, B) := \frac{(b_{11} - b_{22})a_{12}a_{21} + a_{22}a_{12}b_{21} - a_{11}a_{21}b_{12}}{a_{12}b_{21} - a_{21}b_{12}},$$
$$\mu_1 = \alpha(B, A).$$

   Then $\mathrm{tr}\ A$, $\mathrm{tr}\ B$, $\alpha(A, B)$, $\alpha(B, A)$ are regular, algebraically independent, rational invariant functions on $\mathcal{X}_2$, whose values determine orb (A, B). Cl (orb (A, B)) contains an orbit generated by two diagonal matrices diag (λ_1, λ_2) and diag (μ_1, μ_2). Hence, $\mathbb{C}[\mathcal{X}_2]^{\mathrm{inv}}$ is generated by the five invariant polynomials $\mathrm{tr}\ A, \mathrm{tr}\ A^2, \mathrm{tr}\ B, \mathrm{tr}\ B^2, \mathrm{tr}\ AB$, which are algebraically dependent. Their values coincide exactly on two distinct orbits in $\mathcal{X}_2$. On $\mathcal{X}_3$ the above invariant polynomials separate the orbits. Any $(A = [a_{ij}], B = [b_{ij}]) \in \mathcal{X}_4$ is simultaneously similar a unique pair of the form $\left(\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}, \begin{bmatrix} \mu & t \\ 0 & \mu \end{bmatrix}\right)$. Then $t = \gamma(A, B) := \frac{b_{12}}{a_{12}}$. Thus, $\mathrm{tr}\ A, \mathrm{tr}\ B, \gamma(A, B)$ are three algebraically independent regular rational invariant functions on $\mathcal{X}_4$, whose values determine a unique orbit in $\mathcal{X}_4$. Clearly $(\lambda I_2, \mu I_2) \in$ Cl (X_4). Then $\mathbb{C}[\mathcal{X}_4]^{\mathrm{inv}}$ is generated by $\mathrm{tr}\ A, \mathrm{tr}\ B$. The values of $\mathrm{tr}\ A = 2\lambda, \mathrm{tr}\ B = 2\mu$ correspond to a complex line of orbits in $\mathcal{X}_4$. Hence, the classification problem of simultaneous similarity classes in $\mathcal{X}_4$ or $\mathcal{V}$ is a *wild* problem.

   On $\mathcal{X}_5$, $\mathcal{X}_6$, $\mathcal{X}_7$, the algebraically independent functions $\mathrm{tr}\ A, \mathrm{tr}\ B$ determine the orbit in each of the stratum.

24.5 Simultaneous Similarity Classification II

In this section, we give an invariant stratification of $(\mathbb{C}^{n\times n})^{l+1}$, for $l \geq 1$, under the action of $\mathrm{GL}(n, \mathbb{C})$ and describe a set of invariant regular rational functions on each stratum, which separate the orbits up to a finite number. We assume the nontrivial case $n > 1$. It is conjectured that the continuous invariants of the given orbit determine uniquely the orbit on each stratum given in the Classification Theorem.

Classification of simultaneous similarity classes of matrices is a known *wild* problem [GP69]. For another approach to classification of simultaneous similarity classes of matrices using *Belitskii* reduction see [Ser00]. See other applications of these techniques to classifications of linear systems [Fri85] and to canonical forms [Fri86].

Definitions:

For $\mathbf{A} = (A_0, \ldots, A_l), \mathbf{B} = (B_0, \ldots, B_l) \in (\mathbb{C}^{n\times n})^{l+1}$ let $L(\mathbf{B}, \mathbf{A}) : \mathbb{C}^{n\times n} \to (\mathbb{C}^{n\times n})^{l+1}$ be the linear operator given by $U \mapsto (B_0U - UA_0, \ldots, B_lU - UA_l)$. Then $L(\mathbf{B}, \mathbf{A})$ is represented by the $(l+1)n^2 \times n^2$ matrix $(I_n \otimes B_0^{\mathrm{T}} - A_0 \otimes I_n, \ldots, I_n \otimes B_l^{\mathrm{T}} - A_l \otimes I_n)^{\mathrm{T}}$, where $U \mapsto (I_n \otimes B_0^{\mathrm{T}} - A_0 \otimes I_n, \ldots, I_n \otimes B_l^{\mathrm{T}} - A_l \otimes I_n)^{\mathrm{T}} U$. Let $L(\mathbf{A}) := L(\mathbf{A}, \mathbf{A})$. The dimension of orb $(\mathbf{A})$ is denoted by dim orb $(\mathbf{A})$.

Let $\mathcal{S}_n := \{A \in \mathbb{C}^{n\times n} : A = \frac{\operatorname{tr} A}{n} I_n\}$ be the variety of scalar matrices. Let

$$\mathcal{M}_{n,l+1,r} := \{\mathbf{A} \in (\mathbb{C}^{n\times n})^{l+1} : \operatorname{rank}\ L(\mathbf{A}) = r\}, \quad r = 0, 1, \ldots, n^2 - 1.$$

Facts:

Most of the results in this section are given in [Fri83].

1. For $\mathbf{A} = (A_0, \ldots, A_l), \in (\mathbb{C}^{n\times n})^{l+1}$, dim orb $(\mathbf{A})$ is equal to the rank of $L(\mathbf{A})$.
2. Since any $U \in \mathcal{S}_n$ commutes with any $B \in \mathbb{C}^{n\times n}$ it follows that $\ker L(\mathbf{A}) \supset \mathcal{S}_n$. Hence, rank $L(\mathbf{A}) \le n^2 - 1$.
3. $\mathcal{M}_{n,l+1,n^2-1}$ is a invariant quasi-irreducible variety of dimension $(l+1)n^2$, i.e., Cl $(\mathcal{M}_{n,l+1,n^2-1}) = (\mathbb{C}^{n\times n})^{l+1}$. The sets $\mathcal{M}_{n,l+1,r}, r = n^2 - 2, \ldots, 0$ have the decomposition to invariant quasi-irreducible varieties, each of dimension strictly less than $(l+1)n^2$.
4. Let $r \in [0, n^2 - 1]$, $\mathbf{A} \in \mathcal{M}_{n,l+1,r}$, and $\mathbf{B} = T\mathbf{A}T^{-1}$. Then $L(\mathbf{B}, \mathbf{A}) = \operatorname{diag}(I_n \otimes T, \ldots, I_n \otimes T)L(\mathbf{A})(I_n \otimes T^{-1})$, rank $L(\mathbf{B}, \mathbf{A}) = r$ and $\det L(\mathbf{B}, \mathbf{A})[\alpha, \beta] = 0$ for any $\alpha \in \mathrm{Q}_{r+1,(l+1)n^2}, \beta \in \mathrm{Q}_{r+1,n^2}$. (See Chapter 23.2)
5. Let $\mathbf{X} = (X_0, \ldots, X_l) \in (\mathbb{C}^{n\times n})^{l+1}$ with the indeterminate entries $X_k = [x_{k,ij}]$ for $k = 0, \ldots, l$. Each $\det L(\mathbf{X}, \mathbf{A})[\alpha, \beta], \alpha \in \mathrm{Q}_{r+1,(l+1)n^2}, \beta \in \mathrm{Q}_{r+1,n^2}$ is a vector in $\mathcal{W}_{n,l+1,r+1}$, i.e., it is a multilinear polynomial in $(l+1)n^2$ variables of degree $r+1$ at most. We identify $\det L(\mathbf{X}, \mathbf{A})[\alpha, \beta], \alpha \in \mathrm{Q}_{r+1,(l+1)n^2}, \beta \in \mathrm{Q}_{r+1,n^2}$ with the **row** vector $\mathbf{a}(\mathbf{A}, \alpha, \beta) \in \mathbb{C}^{N(n,l,r)}$ given by its coefficients in the basis $\mathbf{e}_1, \ldots, \mathbf{e}_{N(n,l,r)}$. The number of these vectors is $M(n,l,r) := \binom{(l+1)n^2}{r+1}\binom{n^2}{r+1}$. Let $R(\mathbf{A}) \in \mathbb{C}^{M(n,l,r)N(n,l,r)\times M(n,l,r)N(n,l,r)}$ be the matrix with the rows $\mathbf{a}(A, \alpha, \beta)$, where the pairs $(\alpha, \beta) \in \mathrm{Q}_{r+1,(l+1)n^2} \times \mathrm{Q}_{r+1,n^2}$ are listed in a lexicographical order.
6. All points on the orb $(\mathbf{A})$ satisfy the following polynomial equations in $\mathbb{C}[(\mathbb{C}^{n\times n})^{l+1}]$:

$$\begin{aligned}&\det L(\mathbf{X}, \mathbf{A})[\alpha, \beta] = 0,\\ &\text{for all } \alpha \in \mathrm{Q}_{r+1,(l+1)n^2}, \beta \in \mathrm{Q}_{r+1,n^2}.\end{aligned} \tag{24.1}$$

   Thus, the matrix $R(\mathbf{A})$ determines the above variety.
7. If $\mathbf{B} = T\mathbf{A}T^{-1}$, then $R(\mathbf{A})$ is row equivalent to $R(\mathbf{B})$. To each orb $(\mathbf{A})$ we can associate a unique reduced row echelon form $F(\mathbf{A}) \in \mathbb{C}^{M(n,l,r)N(n,l,r)\times M(n,l,r)N(n,l,r)}$ of $R(\mathbf{A})$. $\varrho(\mathbf{A}) :=$ rank $R(\mathbf{A})$ is the number of linearly independent polynomials given in (24.1). Let $\mathcal{I}(\mathbf{A}) = \{(1, j_1), \ldots, (\varrho(\mathbf{A}), j_{\varrho(\mathbf{A})})\} \subset \{1, \ldots, \varrho(\mathbf{A})\} \times \{1, \ldots, N(n,l,r)\}$ be the location of the pivots in the $M(n,l,r) \times N(n,l,r)$ matrix $F(\mathbf{A}) = [f_{ij}(\mathbf{A})]$. That is, $1 \le j_1 < \ldots < j_{\varrho(\mathbf{A})} \le N(n,l,r)$, $f_{ij_i}(\mathbf{A}) = 1$ for $i = 1, \ldots, \varrho(\mathbf{A})$ and $f_{ij} = 0$ unless $j \ge i$ and $i \in [1, \varrho(\mathbf{A})]$. The nontrivial entries $f_{ij}(\mathbf{A})$ for $j > i$ are rational functions in the entries of the $l+1$ tuple $\mathbf{A}$. Thus, $F(\mathbf{B}) = F(\mathbf{A})$ for $\mathbf{B} \in$ orb $(\mathbf{A})$. The numbers $r(\mathbf{A}) :=$ rank $L(\mathbf{A}), \varrho(\mathbf{A})$ and the set $\mathcal{I}(\mathbf{A})$ are called the **discrete** invariants of orb $(\mathbf{A})$. The rational functions $f_{ij}(\mathbf{A}), i = 1, \ldots, \varrho(\mathbf{A}), j = i+1, \ldots, N(n,l,r)$ are called the **continuous** invariants of orb $(\mathbf{A})$.
8. (*Classification Theorem for Simultaneous Similarity*) Let $l \ge 1, n \ge 2$ be integers. Fix an integer $r \in [0, n^2 - 1]$ and let $M(n,l,r), N(n,l,r)$ be the integers defined as above. Let $0 \le \varrho \le \min(M(n,l,r), N(n,l,r))$ and the set $\mathcal{I} = \{(1, j_1), \ldots, (\varrho, j_\varrho) \subset \{1, \ldots, \varrho\} \times \{1, \ldots, N(n,l,r)\}, 1 \le j_1 < \ldots < j_\varrho \le N(n,l,r)$ be given. Let $\mathcal{M}_{n,l+1.r}(\varrho, \mathcal{I})$ be the set of all $\mathbf{A} \in (\mathbb{C}^{n\times n})^{l+1}$ such that rank $L(\mathbf{A}) = r$, $\varrho(\mathbf{A}) = \varrho$, and $\mathcal{I}(\mathbf{A}) = \mathcal{I}$. Then $\mathcal{M}_{n,l+1.r}(\varrho, \mathcal{I})$ is invariant quasi-irreducible variety under the action of $\mathrm{GL}(n, \mathbb{C})$. Suppose that $\mathcal{M}_{n,l+1.r}(\varrho, \mathcal{I}) \ne \emptyset$. Recall that for each $\mathbf{A} \in \mathcal{M}_{n,l+1.r}(\varrho, \mathcal{I})$ the continuous invariants of $\mathbf{A}$, which correspond to the entries $f_{ij}(\mathbf{A}),\ i = 1, \ldots, \varrho,\ j = i+1, \ldots, N(n,l,r)$ of the reduced row echelon form of $R(\mathbf{A})$, are regular rational invariant functions on $\mathcal{M}_{n,l+1.r}(\varrho, \mathcal{I})$. Then the values of the continuous invariants determine a finite number of orbits in $\mathcal{M}_{n,l+1.r}(\varrho, \mathcal{I})$.

   The quasi-irreducible variety $\mathcal{M}_{n,l+1.r}(\varrho, \mathcal{I})$ decomposes uniquely as a finite union of invariant regular quasi-irreducible varieties. The union of all these decompositions of $\mathcal{M}_{n,l+1.r}(\varrho, \mathcal{I})$ for all possible values r, ϱ, and the sets $\mathcal{I}$ gives rise to an invariant stratification of $(\mathbb{C}^{n\times n})^{l+1}$.

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25

Max-Plus Algebra

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Max-plus algebra has been discovered more or less independently by several schools, in relation with various mathematical fields. This chapter is limited to finite dimensional linear algebra. For more information, the reader may consult the books [CG79], [Zim81], [CKR84], [BCOQ92], [KM97], [GM02], and [HOvdW06]. The collections of articles [MS92], [Gun98], and [LM05] give a good idea of current developments.

25.1 Preliminaries

Definitions:

The **max-plus semiring** $\mathbb{R}_{\max}$ is the set $\mathbb{R} \cup \{-\infty\}$, equipped with the **addition** $(a, b) \mapsto \max(a, b)$ and the **multiplication** $(a, b) \mapsto a + b$. The identity element for the addition, **zero**, is $-\infty$, and the identity element for the multiplication, **unit**, is 0. To illuminate the linear algebraic nature of the results, the generic notations $+\!\!\!+$, $\sum$, $\times\!\!\!\times$ (or concatenation), $\mathbb{0}$ and $\mathbb{1}$ are used for the addition, the sum, the multiplication, the zero, and the unit of $\mathbb{R}_{\max}$, respectively, so that when a, b belong to $\mathbb{R}_{\max}$, $a +\!\!\!+ b$ will mean $\max(a, b)$, $a \times\!\!\!\times b$ or ab will mean the usual sum $a + b$. We use blackboard (double struck) fonts to denote the max-plus operations (compare "$+\!\!\!+$" with "$+$").

The **min-plus semiring** $\mathbb{R}_{\min}$ is the set $\mathbb{R} \cup \{+\infty\}$ equipped with the addition $(a, b) \mapsto \min(a, b)$ and the multiplication $(a, b) \mapsto a + b$. The zero is $+\infty$, the unit 0. The name **tropical** is now also used essentially as a synonym of min-plus. Properly speaking, it refers to the **tropical semiring**, which is the subsemiring of $\mathbb{R}_{\min}$ consisting of the elements in $\mathbb{N} \cup \{+\infty\}$.

The **completed max-plus semiring** $\overline{\mathbb{R}}_{\max}$ is the set $\mathbb{R} \cup \{\pm\infty\}$ equipped with the addition $(a, b) \mapsto \max(a, b)$ and the multiplication $(a, b) \mapsto a+b$, with the convention that $-\infty+(+\infty) = +\infty+(-\infty) = -\infty$. The **completed min-plus semiring**, $\overline{\mathbb{R}}_{\min}$, is defined in a dual way.

Many classical algebraic definitions have max-plus analogues. For instance, $\mathbb{R}^n_{\max}$ is the set of n-dimensional **vectors** and $\mathbb{R}^{n\times p}_{\max}$ is the set of $n \times p$ **matrices** with entries in $\mathbb{R}_{\max}$. They are equipped with the vector and matrix operations, defined and denoted in the usual way. The $n \times p$ **zero matrix**, $\mathbf{0}_{np}$ or $\mathbf{0}$, has all its entries equal to $\mathbb{0}$. The $n \times n$ **identity matrix**, I_n or I, has diagonal entries equal to $\mathbb{1}$, and nondiagonal entries equal to $\mathbb{0}$. Given a matrix $A = (A_{ij}) \in \mathbb{R}_{\max}^{n \times p}$, we denote by $A_{i \cdot}$ and $A_{\cdot j}$ the i-th row and the j-th column of A. We also denote by A the linear map $\mathbb{R}_{\max}^{p} \to \mathbb{R}_{\max}^{n}$ sending a vector x to Ax. **Semimodules** and **subsemimodules** over the semiring $\mathbb{R}_{\max}$ are defined as the analogues of modules and submodules over rings. A subset F of a semimodule M over $\mathbb{R}_{\max}$ **spans** M, or is a **spanning family** of M, if every element $\mathbf{x}$ of M can be expressed as a finite linear combination of the elements of F, meaning that $\mathbf{x} = \sum_{\mathbf{f} \in F} \lambda_{\mathbf{f}} . \mathbf{f}$, where $(\lambda_{\mathbf{f}})_{\mathbf{f} \in F}$ is a family of elements of $\mathbb{R}_{\max}$ such that $\lambda_{\mathbf{f}} = \mathbb{0}$ for all but finitely many $\mathbf{f} \in F$. A semimodule is **finitely generated** if it has a finite spanning family.

The sets $\mathbb{R}_{\max}$ and $\overline{\mathbb{R}}_{\max}$ are ordered by the usual order of $\mathbb{R} \cup \{\pm\infty\}$. Vectors and matrices over $\overline{\mathbb{R}}_{\max}$ are ordered with the product ordering. The supremum and the infimum operations are denoted by $\vee$ and $\wedge$, respectively. Moreover, the sum of the elements of an arbitrary set X of scalars, vectors, or matrices with entries in $\overline{\mathbb{R}}_{\max}$ is by definition the supremum of X.

If $A \in \overline{\mathbb{R}}_{\max}^{n \times n}$, the **Kleene star** of A is the matrix $A^\star = I \oplus A \oplus A^2 \oplus \cdots$.

The **digraph** $\Gamma(A)$ associated to an $n \times n$ matrix A with entries in $\overline{\mathbb{R}}_{\max}$ consists of the vertices $1, \ldots, n$, with an arc from vertex i to vertex j when $A_{ij} \neq \mathbb{0}$. The **weight** of a walk W given by $(i_1, i_2), \ldots, (i_{k-1}, i_k)$ is $|W|_A := A_{i_1 i_2} \cdots A_{i_{k-1} i_k}$, and its **length** is $|W| := k - 1$. The matrix A is **irreducible** if $\Gamma(A)$ is strongly connected.

Facts:

1. When $A \in \overline{\mathbb{R}}_{\max}^{n \times n}$, the weight of a walk $W = ((i_1, i_2), \ldots, (i_{k-1}, i_k))$ in $\Gamma(A)$ is given by the usual sum $|W|_A = A_{i_1 i_2} + \cdots + A_{i_{k-1} i_k}$, and $A^\star_{ij}$ gives the maximal weight $|W|_A$ of a walk from vertex i to vertex j. One can also define the matrix $A^\star$ when $A \in \overline{\mathbb{R}}_{\min}^{n \times n}$. Then, $A^\star_{ij}$ is the minimal weight of a walk from vertex i to vertex j. Computing $A^\star$ is the same as the all pairs' shortest path problem.
2. [CG79], [BCOQ92, Th. 3.20] If $A \in \overline{\mathbb{R}}_{\max}^{n \times n}$ and the weights of the cycles of $\Gamma(A)$ do not exceed $\mathbb{1}$, then $A^\star = I \oplus A \oplus \cdots \oplus A^{n-1}$.
3. [BCOQ92, Th. 4.75 and Rk. 80] If $A \in \overline{\mathbb{R}}_{\max}^{n \times n}$ and $\mathbf{b} \in \overline{\mathbb{R}}_{\max}^{n}$, then the smallest $\mathbf{x} \in \overline{\mathbb{R}}_{\max}^{n}$ such that $\mathbf{x} = A\mathbf{x} \oplus \mathbf{b}$ coincides with the smallest $\mathbf{x} \in \overline{\mathbb{R}}_{\max}^{n}$ such that $\mathbf{x} \geq A\mathbf{x} \oplus \mathbf{b}$, and it is given by $A^\star \mathbf{b}$.
4. [BCOQ92, Th. 3.17] When $A \in \mathbb{R}_{\max}^{n \times n}$, $\mathbf{b} \in \mathbb{R}_{\max}^{n}$, and when all the cycles of $\Gamma(A)$ have a weight strictly less than $\mathbb{1}$, then $A^\star \mathbf{b}$ is the unique solution $\mathbf{x} \in \mathbb{R}_{\max}^{n}$ of $\mathbf{x} = A\mathbf{x} \oplus \mathbf{b}$.
5. Let $A \in \mathbb{R}_{\max}^{n \times n}$ and $\mathbf{b} \in \mathbb{R}_{\max}^{n}$. Construct the sequence:

$$\mathbf{x}_0 = \mathbf{b},\ \mathbf{x}_1 = A\mathbf{x}_0 \oplus \mathbf{b},\ \mathbf{x}_2 = A\mathbf{x}_1 \oplus \mathbf{b}, \ldots .$$

   The sequence $\mathbf{x}_k$ is nondecreasing. If all the cycles of $\Gamma(A)$ have a weight less than or equal to $\mathbb{1}$, then, $\mathbf{x}_{n-1} = \mathbf{x}_n = \cdots = A^\star \mathbf{b}$. Otherwise, $\mathbf{x}_{n-1} \neq \mathbf{x}_n$. Computing the sequence $\mathbf{x}_k$ to determine $A^\star \mathbf{b}$ is a special instance of label correcting shortest path algorithm [GP88].
6. [BCOQ92, Lemma 4.101] For all $a \in \overline{\mathbb{R}}_{\max}^{n \times n}$, $b \in \overline{\mathbb{R}}_{\max}^{n \times p}$, $c \in \overline{\mathbb{R}}_{\max}^{p \times n}$, and $d \in \overline{\mathbb{R}}_{\max}^{p \times p}$, we have

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^\star = \begin{bmatrix} a^\star \oplus a^\star b (c a^\star b \oplus d)^\star c a^\star & a^\star b (c a^\star b \oplus d)^\star \\ (c a^\star b \oplus d)^\star c a^\star & (c a^\star b \oplus d)^\star \end{bmatrix} .$$

   This fact and the next one are special instances of well-known results of language theory [Eil74], concerning unambiguous rational identities. Both are valid in more general semirings.
7. [MY60] Let $A \in \overline{\mathbb{R}}_{\max}^{n \times n}$. Construct the sequence of matrices $A^{(0)}, \ldots, A^{(n)}$ such that $A^{(0)} = A$ and

$$A^{(k)}_{ij} = A^{(k-1)}_{ij} \oplus A^{(k-1)}_{ik} (A^{(k-1)}_{kk})^\star A^{(k-1)}_{kj},$$

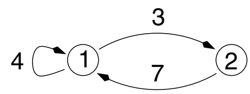
   for $i, j = 1, \ldots, n$ and $k = 1, \ldots, n$. Then, $A^{(n)} = A \oplus A^2 \oplus \cdots$.

Example:

1. Consider the matrix

$$A = \begin{bmatrix} 4 & 3 \\ 7 & -\infty \end{bmatrix}.$$

The digraph $\Gamma(A)$ is



We have

$$A^2 = \begin{bmatrix} 10 & 7 \\ 11 & 10 \end{bmatrix}.$$

For instance, $A^2_{11} = A_{1\cdot}A_{\cdot 1} = [4\ 3][4\ 7]^T = \max(4+4, 3+7) = 10$. This gives the maximal weight of a walk of length 2 from vertex 1 to vertex 1, which is attained by the walk $(1,2),(2,1)$. Since there is one cycle with positive weight in $\Gamma(A)$ (for instance, the cycle $(1,1)$ has weight 4), and since A is irreducible, the matrix $A^\star$ has all its entries equal to $+\infty$. To get a Kleene star with finite entries, consider the matrix

$$C = (-5)A = \begin{bmatrix} -1 & -2 \\ 2 & -\infty \end{bmatrix}.$$

The only cycles in $\Gamma(A)$ are $(1,1)$ and $(1,2),(2,1)$ (up to a cyclic conjugacy). They have weights -1 and 0. Applying Fact 2, we get

$$C^\star = I \oplus C = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}.$$

Applications:

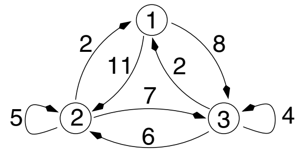
1. *Dynamic programming.* Consider a deterministic Markov decision process with a set of states $\{1, \dots, n\}$ in which one player can move from state i to state j, receiving a payoff of $A_{ij} \in \mathbb{R} \cup \{-\infty\}$. To every state i, associate an initial payoff $\mathbf{c}_i \in \mathbb{R} \cup \{-\infty\}$ and a terminal payoff $\mathbf{b}_i \in \mathbb{R} \cup \{-\infty\}$. The value in horizon k is by definition the maximum of the sums of the payoffs (including the initial and terminal payoffs) corresponding to all the trajectories consisting exactly of k moves. It is given by $\mathbf{c}A^k\mathbf{b}$, where the product and the power are understood in the max-plus sense. The special case where the initial state is equal to some given $m \in \{1, \dots, n\}$ (and where there is no initial payoff) can be modeled by taking $\mathbf{c} := \mathbf{e}_m$, the m-th max-plus basis vector (whose entries are all equal to $\mathbb{0}$, except the m-th entry, which is equal to $\mathbb{1}$). The case where the final state is fixed can be represented in a dual way. Deterministic Markov decision problems (which are the same as shortest path problems) are ubiquitous in operations research, mathematical economics, and optimal control.
2. [BCOQ92] *Discrete event systems.* Consider a system in which certain repetitive events, denoted by $1, \dots, n$, occur. To every event i is associated a dater function $x_i : \mathbb{Z} \to \mathbb{R}$, where $x_i(k)$ represents the date of the k-th occurrence of event i. Precedence constraints between the repetitive events are given by a set of arcs $E \subset \{1, \dots, n\}^2$, equipped with two valuations $\nu : E \to \mathbb{N}$ and $\tau : E \to \mathbb{R}$. If $(i,j) \in E$, the k-th execution of event i cannot occur earlier than τ_{ij} time units before the $(k - \nu_{ij})$-th execution of event j, so that $x_i(k) \geq \max_{j:\,(i,j)\in E} \tau_{ij} + x_j(k - \nu_{ij})$. This can be rewritten, using the max-plus notation, as

$$\mathbf{x}(k) \geq A_0\mathbf{x}(k) \oplus \cdots \oplus A_{\bar{\nu}}\mathbf{x}(k - \bar{\nu}),$$

where $\bar{\nu} := \max_{(i,j)\in E} \nu_{ij}$ and $\mathbf{x}(k) \in \mathbb{R}^n_{\max}$ is the vector with entries $x_i(k)$. Often, the dates $x_i(k)$ are only defined for positive k, then appropriate initial conditions must be incorporated in the model. One is particularly interested in the earliest dynamics, which, by Fact 3, is given by $\mathbf{x}(k) = A_0^\star A_1 \mathbf{x}(k-1) \oplus \cdots \oplus A_0^\star A_{\bar{\nu}} \mathbf{x}(k-\bar{\nu})$. The class of systems following dynamics of these forms is known in the Petri net literature as **timed event graphs**. It is used to model certain manufacturing systems [CDQV85], or transportation or communication networks [BCOQ92], [HOvdW06].

3. N. Bacaër [Bac03] observed that max-plus algebra appears in a familiar problem, crop rotation. Suppose n different crops can be cultivated every year. Assume for simplicity that the income of the year is a deterministic function, $(i, j) \mapsto A_{ij}$, depending only on the crop i of the preceding year, and of the crop j of the current year (a slightly more complex model in which the income of the year depends on the crops of the two preceding years is needed to explain the historical variations of crop rotations [Bac03]). The income of a sequence $i_1, \ldots, i_k$ of crops can be written as $\mathbf{c}_{i_1} A_{i_1 i_2} \cdots A_{i_{k-1} i_k}$, where $\mathbf{c}_{i_1}$ is the income of the first year. The maximal income in k years is given by $\mathbf{c}A^{k-1}\mathbf{b}$, where $\mathbf{b} = (\mathbb{1}, \ldots, \mathbb{1})$. We next show an example.

$$A = \begin{bmatrix} -\infty & 11 & 8 \\ 2 & 5 & 7 \\ 2 & 6 & 4 \end{bmatrix}$$



Here, vertices 1, 2, and 3 represent fallow (no crop), wheat, and oats, respectively. (We put no arc from 1 to 1, setting $A_{11} = -\infty$, to disallow two successive years of fallow.) The numerical values have no pretension to realism; however, the income of a year of wheat is 11 after a year of fallow, this is greater than after a year of cereal (5 or 6, depending on whether wheat or oats was cultivated). An initial vector coherent with these data may be $\mathbf{c} = [-\infty \;\; 11 \;\; 8]$, meaning that the income of the first year is the same as the income after a year of fallow. We have $\mathbf{c}A\mathbf{b} = 18$, meaning that the optimal income in 2 years is 18. This corresponds to the optimal walk $(2, 3)$, indicating that wheat and oats should be successively cultivated during these 2 years.

25.2 The Maximal Cycle Mean

Definitions:

1. The **maximal cycle mean**, $\rho_{\max}(A)$, of a matrix $A \in \mathbb{R}^{n\times n}_{\max}$, is the maximum of the weight-to-length ratio over all cycles c of $\Gamma(A)$, that is,

$$\rho_{\max}(A) = \max_{c \text{ cycle of } \Gamma(A)} \frac{|c|_A}{|c|} = \max_{k\geq 1} \max_{i_1,\ldots,i_k} \frac{A_{i_1 i_2} + \cdots + A_{i_k i_1}}{k}. \tag{25.1}$$

2. Denote by $\mathbb{R}^{n\times n}_+$ the set of real $n \times n$ matrices with nonnegative entries. For $A \in \mathbb{R}^{n\times n}_+$ and $p > 0$, $A^{(p)}$ is by definition the matrix such that $(A^{(p)})_{ij} = (A_{ij})^p$, and

$$\rho_p(A) := (\rho(A^{(p)}))^{1/p},$$

where ρ denotes the (usual) spectral radius. We also define $\rho_\infty(A) = \lim_{p\to+\infty} \rho_p(A)$.

Facts:

1. [CG79], [Gau92, Ch. IV], [BSvdD95] *Max-plus Collatz–Wielandt formula, I.* Let $A \in \mathbb{R}^{n\times n}_{\max}$ and $\lambda \in \mathbb{R}$. The following assertions are equivalent: (i) There exists $\mathbf{u} \in \mathbb{R}^n$ such that $A\mathbf{u} \leq \lambda\mathbf{u}$; (ii) $\rho_{\max}(A) \leq \lambda$. It follows that

$$\rho_{\max}(A) = \inf_{\mathbf{u}\in\mathbb{R}^n} \max_{1\leq i\leq n} (A\mathbf{u})_i \mathbin{/\!\!/} \mathbf{u}_i$$

(the product $A\mathbf{u}$ and the division by $\mathbf{u}_i$ should be understood in the max-plus sense). If $\rho_{\max}(A) > \mathbb{0}$, then this infimum is attained by some $\mathbf{u} \in \mathbb{R}^n$. If in addition A is irreducible, then Assertion (i) is equivalent to the following: (i') there exists $\mathbf{u} \in \mathbb{R}^n_{\max} \setminus \{\mathbf{0}\}$ such that $A\mathbf{u} \leq \lambda\mathbf{u}$.

2. [Gau92, Ch. IV], [BSvdD95] *Max-plus Collatz–Wielandt formula, II.* Let $\lambda \in \mathbb{R}_{\max}$. The following assertions are equivalent: (i) There exists $\mathbf{u} \in \mathbb{R}^n_{\max} \setminus \{\mathbf{0}\}$ such that $A\mathbf{u} \geq \lambda\mathbf{u}$; (ii) $\rho_{\max}(A) \geq \lambda$. It follows that

$$\rho_{\max}(A) = \max_{\mathbf{u} \in \mathbb{R}^n_{\max} \setminus \{\mathbf{0}\}} \min_{\substack{1 \leq i \leq n \\ \mathbf{u}_i \neq \mathbb{0}}} (A\mathbf{u})_i \;/\!\!/\; \mathbf{u}_i .$$

3. [Fri86] For $A \in \mathbb{R}_+^{n \times n}$, we have $\rho_\infty(A) = \exp(\rho_{\max}(\log(A)))$, where log is interpreted entrywise.
4. [KO85] For all $A \in \mathbb{R}_+^{n \times n}$, and $1 \leq q \leq p \leq \infty$, we have $\rho_p(A) \leq \rho_q(A)$.
5. For all $A, B \in \mathbb{R}_+^{n \times n}$, we have

$$\rho(A \circ B) \leq \rho_p(A)\rho_q(B) \quad \text{for all} \quad p, q \in [1, \infty] \quad \text{such that} \quad \frac{1}{p} + \frac{1}{q} = 1.$$

This follows from the classical Kingman's inequality [Kin61], which states that the map $\log \circ \rho \circ \exp$ is convex (exp is interpreted entrywise). We have in particular $\rho(A \circ B) \leq \rho_\infty(A)\rho(B)$.

6. [Fri86] For all $A \in \mathbb{R}_+^{n \times n}$, we have

$$\rho_\infty(A) \leq \rho(A) \leq \rho_\infty(A)\rho(\hat{A}) \leq \rho_\infty(A)n,$$

where $\hat{A}$ is the pattern matrix of A, that is, $\hat{A}_{ij} = 1$ if $A_{ij} \neq 0$ and $\hat{A}_{ij} = 0$ if $A_{ij} = 0$.

7. [Bap98], [EvdD99] For all $A \in \mathbb{R}_+^{n \times n}$, we have $\lim_{k \to \infty} (\rho_\infty(A^k))^{1/k} = \rho(A)$.
8. [CG79] *Computing $\rho_{\max}(A)$ by linear programming.* For $A \in \mathbb{R}^{n \times n}_{\max}$, $\rho_{\max}(A)$ is the value of the linear program

$$\inf \lambda \text{ s.t. } \exists \mathbf{u} \in \mathbb{R}^n, \quad \forall (i, j) \in E, \quad A_{ij} + \mathbf{u}_j \leq \lambda + \mathbf{u}_i,$$

where $E = \{(i, j) \mid 1 \leq i, j \leq n,\ A_{ij} \neq \mathbb{0}\}$ is the set of arcs of $\Gamma(A)$.

9. *Dual linear program to compute $\rho_{\max}(A)$.* Let $\mathcal{C}$ denote the set of nonnegative vectors $x = (x_{ij})_{(i,j) \in E}$ such that

$$\forall 1 \leq i \leq n, \quad \sum_{1 \leq k \leq n,\ (k,i) \in E} x_{ki} = \sum_{1 \leq j \leq n, (i,j) \in E} x_{ij}, \quad \text{and} \quad \sum_{(i,j) \in E} x_{ij} = 1.$$

To every cycle c of $\Gamma(A)$ corresponds bijectively the extreme point of the polytope $\mathcal{C}$ that is given by $x_{ij} = 1/|c|$ if (i, j) belongs to c, and $x_{ij} = 0$ otherwise. Moreover, $\rho_{\max}(A) = \sup\{\sum_{(i,j) \in E} A_{ij} x_{ij} \mid x \in \mathcal{C}\}$.

10. [Kar78] *Karp's formula.* If $A \in \mathbb{R}^{n \times n}_{\max}$ is irreducible, then, for all $1 \leq i \leq n$,

$$\rho_{\max}(A) = \max_{\substack{1 \leq j \leq n \\ A^n_{ij} \neq \mathbb{0}}} \min_{1 \leq k \leq n} \frac{(A^n)_{ij} - (A^{n-k})_{ij}}{k}. \tag{25.2}$$

To evaluate the right-hand side expression, compute the sequence $\mathbf{u}^0 = \mathbf{e}_i, \mathbf{u}^1 = \mathbf{u}^0 A, \mathbf{u}^n = \mathbf{u}^{n-1} A$, so that $\mathbf{u}^k = A^k_{i\cdot}$ for all $0 \leq k \leq n$. This takes a time $O(nm)$, where m is the number of arcs of $\Gamma(A)$. One can avoid storing the vectors $\mathbf{u}^0, \ldots, \mathbf{u}^n$, at the price of recomputing the sequence $\mathbf{u}^0, \ldots, \mathbf{u}^{n-1}$ once $\mathbf{u}^n$ is known. The time and space complexity of Karp's algorithm are $O(nm)$ and $O(n)$, respectively. The policy iteration algorithm of [CTCG+98] seems experimentally more efficient than Karp's algorithm. Other algorithms are given in particular in [CGL96], [BO93], and [EvdD99]. A comparison of maximal cycle mean algorithms appears in [DGI98]. When the entries of A take only two finite values, the maximal cycle mean of A can be computed in linear time [CGB95]. The Karp and policy iteration algorithms, as well as the general max-plus operations (full and sparse matrix products, matrix residuation, etc.) are implemented in the **Maxplus toolbox** of **Scilab**, freely available in the contributed section of the Web site www.scilab.org.

Example:

1. For the matrix A in Application 3 of section 25.1, we have $\rho_{\max}(A) = \max(5, 4, (2 + 11)/2, (2 + 8)/2, (7+6)/2, (11+7+2)/3, (8+6+2)/3) = 20/3$, which gives the maximal reward per year. This is attained by the cycle $(1, 2), (2, 3), (3, 1)$, corresponding to the rotation of crops: fallow, wheat, oats.

25.3 The Max-Plus Eigenproblem

The results of this section and of the next one constitute max-plus spectral theory. Early and fundamental contributions are due to Cuninghame–Green (see [CG79]), Vorobyev [Vor67], Romanovskiĭ [Rom67], Gondran and Minoux [GM77], and Cohen, Dubois, Quadrat, and Viot [CDQV83]. General presentations are included in [CG79], [BCOQ92], and [GM02]. The infinite dimensional max-plus spectral theory (which is not covered here) has been developed particularly after Maslov, in relation with Hamilton–Jacobi partial differential equations; see [MS92] and [KM97]. See also [MPN02], [AGW05], and [Fat06] for recent developments.

In this section and the next two, A denotes a matrix in $\mathbb{R}_{\max}^{n\times n}$.

Definitions:

An **eigenvector** of A is a vector $\mathbf{u} \in \mathbb{R}^n_{\max} \setminus \{\mathbb{0}\}$ such that $A\mathbf{u} = \lambda\mathbf{u}$, for some scalar $\lambda \in \mathbb{R}_{\max}$, which is called the **(geometric) eigenvalue** corresponding to $\mathbf{u}$. With the notation of classical algebra, the equation $A\mathbf{u} = \lambda\mathbf{u}$ can be rewritten as

$$\max_{1\le j\le n} A_{ij} + \mathbf{u}_j = \lambda + \mathbf{u}_i, \quad \forall 1 \le i \le n.$$

If λ is an eigenvalue of A, the set of vectors $\mathbf{u} \in \mathbb{R}^n_{\max}$ such that $A\mathbf{u} = \lambda\mathbf{u}$ is the **eigenspace** of A for the eigenvalue λ.

The **saturation digraph** with respect to $\mathbf{u} \in \mathbb{R}^n_{\max}$, $\mathrm{Sat}(A, \mathbf{u})$, is the digraph with vertices $1, \dots, n$ and an arc from vertex i to vertex j when $A_{ij}\mathbf{u}_j = (A\mathbf{u})_i$.

A cycle $c = (i_1, i_2), \dots, (i_k, i_1)$ that attains the maximum in (25.1) is called **critical**. The **critical digraph** is the union of the critical cycles. The **critical vertices** are the vertices of the critical digraph.

The **normalized matrix** is $\tilde{A} = \rho_{\max}(A)^{-1}A$ (when $\rho_{\max}(A) \neq \mathbb{0}$).

For a digraph Γ, vertex i **has access** to a vertex j if there is a walk from i to j in Γ. The **(access equivalent) classes** of Γ are the equivalence classes of the set of its vertices for the relation "i has access to j and j has access to i." A class C **has access** to a class C' if some vertex of C has access to some vertex of C'. A class is **final** if it has access only to itself.

The **classes** of a matrix A are the classes of $\Gamma(A)$, and the **critical classes** of A are the classes of the critical digraph of A. A class C of A is **basic** if $\rho_{\max}(A[C, C]) = \rho_{\max}(A)$.

Facts:

The proof of most of the following facts can be found in particular in [CG79] or [BCOQ92, Sec. 3.7]; we give specific references when needed.

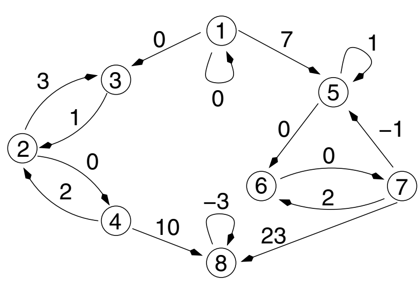
1. For any matrix A, $\rho_{\max}(A)$ is an eigenvalue of A, and any eigenvalue of A is less than or equal to $\rho_{\max}(A)$.
2. An eigenvalue of A associated with an eigenvector in $\mathbb{R}^n$ must be equal to $\rho_{\max}(A)$.
3. [ES75] *Max-plus diagonal scaling.* Assume that $\mathbf{u} \in \mathbb{R}^n$ is an eigenvector of A. Then the matrix B such that $B_{ij} = \mathbf{u}_i^{-1}A_{ij}\mathbf{u}_j$ has all its entries less than or equal to $\rho_{\max}(A)$, and the maximum of every of its rows is equal to $\rho_{\max}(A)$.
4. If A is irreducible, then $\rho_{\max}(A) > \mathbb{0}$ and it is the only eigenvalue of A. *From now on, we assume that $\Gamma(A)$ has at least one cycle, so that $\rho_{\max}(A) > \mathbb{0}$.*

5. For all critical vertices i of A, the column $\tilde{A}^\star_{\cdot i}$ is an eigenvector of A for the eigenvalue $\rho_{\max}(A)$. Moreover, if i and j belong to the same critical class of A, then $\tilde{A}^\star_{\cdot i} = \tilde{A}^\star_{\cdot j}\tilde{A}^\star_{ji}$.
6. *Eigenspace for the eigenvalue* $\rho_{\max}(A)$. Let $C_1, \dots, C_s$ denote the critical classes of A, and let us choose arbitrarily one vertex $i_t \in C_t$, for every $t = 1, \dots, s$. Then, the columns $\tilde{A}^\star_{\cdot,i_t}$, $t = 1, \dots, s$ span the eigenspace of A for the eigenvalue $\rho_{\max}(A)$. Moreover, any spanning family of this eigenspace contains some scalar multiple of every column $\tilde{A}^\star_{\cdot,i_t}$, $t = 1, \dots, s$.
7. Let C denote the set of critical vertices, and let $T = \{1, \dots, n\} \setminus C$. The following facts are proved in a more general setting in [AG03, Th. 3.4], with the exception of (b), which follows from Fact 4 of Section 25.1.
   (a) The restriction $\mathbf{v} \mapsto \mathbf{v}[C]$ is an isomorphism from the eigenspace of A for the eigenvalue $\rho_{\max}(A)$ to the eigenspace of $A[C, C]$ for the same eigenvalue.
   (b) An eigenvector $\mathbf{u}$ for the eigenvalue $\rho_{\max}(A)$ is determined from its restriction $\mathbf{u}[C]$ by $\mathbf{u}[T] = (\tilde{A}[T, T])^\star \tilde{A}[T, C]\mathbf{u}[C]$.
   (c) Moreover, $\rho_{\max}(A)$ is the only eigenvalue of $A[C, C]$ and the eigenspace of $A[C, C]$ is stable by infimum and by convex combination in the usual sense.
8. *Complementary slackness.* If $\mathbf{u} \in \mathbb{R}^n_{\max}$ is such that $A\mathbf{u} \le \rho_{\max}(A)\mathbf{u}$, then $(A\mathbf{u})_i = \rho_{\max}(A)\mathbf{u}_i$, for all critical vertices i.
9. *Critical digraph vs. saturation digraph.* Let $\mathbf{u} \in \mathbb{R}^n$ be such that $A\mathbf{u} \le \rho_{\max}(A)\mathbf{u}$. Then, the union of the cycles of $\mathrm{Sat}(A, \mathbf{u})$ is equal to the critical digraph of A.
10. [CQD90], [Gau92, Ch. IV], [BSvdD95] *Spectrum of reducible matrices.* A scalar $\lambda \ne \mathbb{0}$ is an eigenvalue of A if and only if there is at least one class C of A such that $\rho_{\max}(A[C, C]) = \lambda$ and $\rho_{\max}(A[C, C]) \ge \rho_{\max}(A[C', C'])$ for all classes C' that have access to C.
11. [CQD90], [BSvdD95] The matrix A has an eigenvector in $\mathbb{R}^n$ if and only if all its final classes are basic.
12. [Gau92, Ch. IV] *Eigenspace for an eigenvalue* λ. Let $C^1, \dots, C^m$ denote all the classes C of A such that $\rho_{\max}(A[C, C]) = \lambda$ and $\rho_{\max}(A[C', C']) \le \lambda$ for all classes C' that have access to C. For every $1 \le k \le m$, let $C^k_1, \dots, C^k_{s_k}$ denote the critical classes of the matrix $A[C^k, C^k]$. For all $1 \le k \le m$ and $1 \le t \le s_k$, let us choose arbitrarily an element $j_{k,t}$ in C^k_t. Then, the family of columns $(\lambda^{-1}A)^\star_{\cdot,j_{k,t}}$, indexed by all these k and t, spans the eigenspace of A for the eigenvalue λ, and any spanning family of this eigenspace contains a scalar multiple of every $(\lambda^{-1}A)^\star_{\cdot,j_{k,t}}$.
13. *Computing the eigenvectors.* Observe first that any vertex j that attains the maximum in Karp's formula (25.2) is critical. To compute one eigenvector for the eigenvalue $\rho_{\max}(A)$, it suffices to compute $\tilde{A}^\star_{\cdot j}$ for some critical vertex j. This is equivalent to a single source shortest path problem, which can be solved in $O(nm)$ time and $O(n)$ space. Alternatively, one may use the *policy iteration algorithm* of [CTCG$^+$98] or the improvement in [EvdD99] of the *power algorithm* [BO93]. Once a particular eigenvector is known, the critical digraph can be computed from Fact 9 in $O(m)$ additional time.

Examples:

1. For the matrix A in Application 3 of section 25.1, the only critical cycle is $(1, 2), (2, 3), (3, 1)$ (up to a circular permutation of vertices). The critical digraph consists of the vertices and arcs of this cycle. By Fact 6, any eigenvector $\mathbf{u}$ of A is proportional to $\tilde{A}^\star_{\cdot 1} = [0\ {-13/3}\ {-14/3}]^T$ (or equivalently, to $\tilde{A}^\star_{\cdot 2}$ or $\tilde{A}^\star_{\cdot 3}$). Observe that an eigenvector yields a relative price information between the different states.
2. Consider the matrix and its associated digraph:

$$A = \left[\begin{array}{c|ccc|ccc|c}
0 & \cdot & 0 & \cdot & 7 & \cdot & \cdot & \cdot \\
\hline
\cdot & \cdot & 3 & 0 & \cdot & \cdot & \cdot & \cdot \\
\cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & 2 & \cdot & \cdot & \cdot & \cdot & \cdot & 10 \\
\hline
\cdot & \cdot & \cdot & \cdot & 1 & 0 & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 0 & \cdot \\
\cdot & \cdot & \cdot & \cdot & -1 & 2 & \cdot & 23 \\
\hline
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & -3
\end{array}\right]$$



(We use $\cdot$ to represent the element $-\infty$.) The classes of A are $C^1 = \{1\}$, $C^2 = \{2,3,4\}$, $C^3 = \{5,6,7\}$, and $C^4 = \{8\}$. We have $\rho_{\max}(A) = \rho_{\max}(A[C^2,C^2]) = 2$, $\rho_{\max}(A[C^1,C^1]) = 0$, $\rho_{\max}(A[C^3,C^3]) = 1$, and $\rho_{\max}(A[C^4,C^4]) = -3$. The critical digraph is reduced to the critical cycle $(2,3)(3,2)$. By Fact 6, any eigenvector for the eigenvalue $\rho_{\max}(A)$ is proportional to $\tilde{A}^{\star}_{\cdot 2} = [-3\ 0\ -1\ 0\ -\infty\ -\infty\ -\infty\ -\infty]^T$. By Fact 10, the other eigenvalues of A are 0 and 1. By Fact 12, any eigenvector for the eigenvalue 0 is proportional to $A^{\star}_{\cdot 1} = \mathbf{e}_1$. Observe that the critical classes of $A[C^3,C^3]$ are $C^3_1 = \{5\}$ and $C^3_2 = \{6,7\}$. Therefore, by Fact 12, any eigenvector for the eigenvalue 1 is a max-plus linear combination of $(1^{-1}A)^{\star}_{\cdot 5} = [6\ -\infty\ -\infty\ -\infty\ 0\ -3\ -2\ -\infty]^T$ and $(1^{-1}A)^{\star}_{\cdot 6} = [5\ -\infty\ -\infty\ -\infty\ -1\ 0\ 1\ -\infty]^T$. The eigenvalues of A^T are 2, 1, and -3. So A and A^T have only two eigenvalues in common.

25.4 Asymptotics of Matrix Powers

Definitions:

A sequence $s_0, s_1, \ldots$ of elements of $\mathbb{R}_{\max}$ is **recognizable** if there exists a positive integer p, vectors $\mathbf{b} \in \mathbb{R}_{\max}^{p\times 1}$ and $\mathbf{c} \in \mathbb{R}_{\max}^{1\times p}$, and a matrix $M \in \mathbb{R}_{\max}^{p\times p}$ such that $s_k = \mathbf{c}M^k\mathbf{b}$, for all nonnegative integers k.

A sequence $s_0, s_1, \ldots$ of elements of $\mathbb{R}_{\max}$ is **ultimately geometric** with **rate** $\lambda \in \mathbb{R}_{\max}$ if $s_{k+1} = \lambda s_k$ for k large enough.

The **merge** of q sequences $s^1, \ldots, s^q$ is the sequence s such that $s_{kq+i-1} = s^i_k$, for all $k \geq 0$ and $1 \leq i \leq q$.

Facts:

1. [Gun94], [CTGG99] If every row of the matrix A has at least one entry different from $\mathbb{0}$, then, for all $1 \leq i \leq n$ and $\mathbf{u} \in \mathbb{R}^n$, the limit
$$\chi_i(A) = \lim_{k\to\infty} (A^k\mathbf{u})_i^{1/k}$$
exists and is independent of the choice of $\mathbf{u}$. The vector $\chi(A) = (\chi_i(A))_{1\leq i\leq n} \in \mathbb{R}^n$ is called the **cycle-time** of A. It is given by
$$\chi_i(A) = \max\{\rho_{\max}(A[C,C]) \mid C \text{ is a class of } A \text{ to which } i \text{ has access}\}.$$
In particular, if A is irreducible, then $\chi_i(A) = \rho_{\max}(A)$ for all $i = 1, \ldots, n$.
2. The following constitutes the cyclicity theorem, due to Cohen, Dubois, Quadrat, and Viot [CDQV83]. See [BCOQ92] and [AGW05] for more accessible accounts.
   (a) If A is irreducible, there exists a positive integer γ such that $A^{k+\gamma} = \rho_{\max}(A)^{\gamma}A^k$ for k large enough. The minimal value of γ is called the **cyclicity** of A.
   (b) Assume again that A is irreducible. Let $C_1, \ldots, C_s$ be the critical classes of A, and for $i = 1, \ldots, s$, let γ_i denote the g.c.d. (greatest common divisor) of the lengths of the critical cycles of A belonging to C_i. Then, the cyclicity γ of A is the l.c.m. (least common multiple) of $\gamma_1, \ldots, \gamma_s$.
   (c) Assume that $\rho_{\max}(A) \neq \mathbb{0}$. The **spectral projector** of A is the matrix $P := \lim_{k\to\infty} \tilde{A}^k\tilde{A}^{\star} = \lim_{k\to\infty} \tilde{A}^k \oplus \tilde{A}^{k+1} \oplus \cdots$. It is given by $P = \bigoplus_{i\in C} \tilde{A}^{\star}_{\cdot i}\tilde{A}^{\star}_{i\cdot}$, where C denotes the set of critical vertices of A. When A is irreducible, the limit is attained in finite time. If, in addition, A has cyclicity one, then $A^k = \rho_{\max}(A)^k P$ for k large enough.
3. Assume that A is irreducible, and let m denote the number of arcs of its critical digraph. Then, the cyclicity of A can be computed in $O(m)$ time from the critical digraph of A, using the algorithm of Denardo [Den77].

4. The smallest integer k such that $A^{k+\gamma} = \rho_{\max}(A)^{\gamma} A^k$ is called the **coupling time**. It is estimated in [HA99], [BG01], [AGW05] (assuming again that A is irreducible).
5. [AGW05, Th. 7.5] *Turnpike theorem.* Define a walk of $\Gamma(A)$ to be optimal if it has a maximal weight amongst all walks with the same ends and length. If A is irreducible, then the number of noncritical vertices of an optimal walk (counted with multiplicities) is bounded by a constant depending only on A.
6. [Mol88], [Gau94], [KB94], [DeS00] A sequence of elements of $\mathbb{R}_{\max}$ is recognizable if and only if it is a merge of ultimately geometric sequences. In particular, for all $1 \leq i, j \leq n$, the sequence $(A^k)_{ij}$ is a merge of ultimately geometric sequences.
7. [Sim78], [Has90], [Sim94], [Gau96] One can decide whether a finitely generated semigroup S of matrices with effective entries in $\mathbb{R}_{\max}$ is finite. One can also decide whether the set of entries in a given position of the matrices of S is finite (limitedness problem). However [Kro94], whether this set contains a given entry is undecidable (even when the entries of the matrices belong to $\mathbb{Z} \cup \{-\infty\}$).

Example:

1. For the matrix A in Application 3 of section 25.1, the cyclicity is 3, and the spectral projector is

$$P = \tilde{A}^{\star}_{\cdot 1} \tilde{A}^{\star}_{1 \cdot} = \begin{bmatrix} 0 \\ -13/3 \\ -14/3 \end{bmatrix} \begin{bmatrix} 0 & 13/3 & 14/3 \end{bmatrix}^T = \begin{bmatrix} 0 & 13/3 & 14/3 \\ -13/3 & 0 & 1/3 \\ -14/3 & -1/3 & 0 \end{bmatrix}.$$

2. For the matrix A in Example 2 of Section 25.3, the cycle-time is $\chi(A) = [2\ 2\ 2\ 2\ 1\ 1\ 1\ -3]^T$. The cyclicity of $A[C^2, C^2]$ is 2 because there is only one critical cycle, which has length 2. Let $B := A[C^3, C^3]$. The critical digraph of B has two strongly connected components consisting, respectively, of the cycles $(5,5)$ and $(6,7), (7,6)$. So B has cyclicity l.c.m. $(1,2) = 2$. The sequence $s_k := (A^k)_{18}$ is such that $s_{k+2} = s_k + 4$, for $k \geq 24$, with $s_{24} = s_{25} = 51$. Hence, s_k is the merge of two ultimately geometric sequences, both with rate 4. To get an example where different rates appear, replace the entries A_{11} and A_{88} of A by $-\infty$. Then, the same sequence s_k is such that $s_{k+2} = s_k + 4$, for all even $k \geq 24$, and $s_{k+2} = s_k + 2$, for all odd $k \geq 5$, with $s_5 = 31$ and $s_{24} = 51$.

25.5 The Max-Plus Permanent

Definitions:

The (max-plus) **permanent** of A is per $A = \sum_{\sigma \in S_n} A_{1\sigma(1)} \cdots A_{n\sigma(n)}$, or with the usual notation of classical algebra, per $A = \max_{\sigma \in S_n} A_{1\sigma(1)} + \cdots + A_{n\sigma(n)}$, which is the value of the optimal assignment problem with weights A_{ij}.

A **max-plus polynomial function** P is a map $\mathbb{R}_{\max} \to \mathbb{R}_{\max}$ of the form $P(x) = \sum_{i=0}^{n} p_i x^i$ with $p_i \in \mathbb{R}_{\max}$, $i = 0, \ldots, n$. If $p_n \neq \mathbb{0}$, P is of **degree** n.

The **roots** of a nonzero max-plus polynomial function P are the points of nondifferentiability of P, together with the point $\mathbb{0}$ when the derivative of P near $-\infty$ is positive. The **multiplicity** of a root α of P is defined as the variation of the derivative of P at the point α, $P'(\alpha^+) - P'(\alpha^-)$, when $\alpha \neq \mathbb{0}$, and as its derivative near $-\infty$, $P'(\mathbb{0}^+)$, when $\alpha = \mathbb{0}$.

The (max-plus) **characteristic polynomial function** of A is the polynomial function P_A given by $P_A(x) = \text{per}(A + xI)$ for $x \in \mathbb{R}_{\max}$. The **algebraic eigenvalues** of A are the roots of P_A.

Facts:

1. [CGM80] Any nonzero max-plus polynomial function P can be factored uniquely as $P(x) = a(x + \alpha_1) \cdots (x + \alpha_n)$, where $a \in \mathbb{R}$, n is the degree of P, and the α_i are the roots of P, counted with multiplicities.

2. [CG83], [ABG04, Th. 4.6 and 4.7]. The greatest algebraic eigenvalue of A is equal to $\rho_{\max}(A)$. Its multiplicity is less than or equal to the number of critical vertices of A, with equality if and only if the critical vertices can be covered by disjoint critical cycles.
3. Any geometric eigenvalue of A is an algebraic eigenvalue of A. (This can be deduced from Fact 2 of this section, and Fact 10 of section 25.3.)
4. [Yoe61] If $A \geq I$ and per $A = \mathbb{1}$, then $A^{\star}_{ij} = \text{per } A(j,i)$, for all $1 \leq i, j \leq n$.
5. [But00] Assume that all the entries of A are different from $\mathbb{0}$. The following are equivalent: (i) there is a vector $b \in \mathbb{R}^n$ that has a unique preimage by A; (ii) there is only one permutation σ such that $|\sigma|_A := A_{1\sigma(1)} \cdots A_{n\sigma(n)} = \text{per } A$. Further characterizations can be found in [But00] and [DSS05].
6. [Bap95] *Alexandroff inequality over* $\mathbb{R}_{\max}$. Construct the matrix B with columns $A_{\cdot 1}, A_{\cdot 1}, A_{\cdot 3}, \ldots,$ $A_{\cdot n}$ and the matrix C with columns $A_{\cdot 2}, A_{\cdot 2}, A_{\cdot 3}, \ldots, A_{\cdot n}$. Then $(\text{per } A)^2 \geq (\text{per } B)(\text{per } C)$, or with the notation of classical algebra, $2 \times \text{per } A \geq \text{per } B + \text{per } C$.
7. [BB03] The max-plus characteristic polynomial function of A can be computed by solving $O(n)$ optimal assignment problems.

Example:

1. For the matrix A in Example 2 of section 25.3, the characteristic polynomial of A is the product of the characteristic polynomials of the matrices $A[C^i, C^i]$, for $i = 1, \ldots, 4$. Thus, $P_A(x) = (x \oplus 0)(x \oplus 2)^2 x(x \oplus 1)^3(x \oplus (-3)))$, and so, the algebraic eigenvalues of A are $-\infty, -3, 0, 1,$ and 2, with respective multiplicities 1, 1, 1, 3, and 2.

25.6 Linear Inequalities and Projections

Definitions:

If $A \in \overline{\mathbb{R}}_{\max}^{n\times p}$, the **range** of A, denoted range A, is $\{A\mathbf{x} \mid \mathbf{x} \in \overline{\mathbb{R}}_{\max}^{p}\} \subset \overline{\mathbb{R}}_{\max}^{n}$. The **kernel** of A, denoted ker A, is the set of **equivalence classes modulo** A, which are the classes for the equivalence relation "$\mathbf{x} \sim \mathbf{y}$ if $A\mathbf{x} = A\mathbf{y}$."

The **support** of a vector $\mathbf{b} \in \overline{\mathbb{R}}_{\max}^{n}$ is $\text{supp } \mathbf{b} := \{i \in \{1, \ldots, n\} \mid \mathbf{b}_i \neq \mathbb{0}\}$.

The **orthogonal congruence** of a subset U of $\overline{\mathbb{R}}_{\max}^{n}$ is $U^{\perp} := \{(\mathbf{x}, \mathbf{y}) \in \overline{\mathbb{R}}_{\max}^{n} \times \overline{\mathbb{R}}_{\max}^{n} \mid \mathbf{u}\cdot\mathbf{x} = \mathbf{u}\cdot\mathbf{y}\ \forall \mathbf{u} \in U\}$, where "$\cdot$" denotes the max-plus scalar product. The **orthogonal space** of a subset C of $\overline{\mathbb{R}}_{\max}^{n} \times \overline{\mathbb{R}}_{\max}^{n}$ is $C^{\top} := \{\mathbf{u} \in \overline{\mathbb{R}}_{\max}^{n} \mid \mathbf{u} \cdot \mathbf{x} = \mathbf{u} \cdot \mathbf{y}\ \forall (\mathbf{x}, \mathbf{y}) \in C\}$.

Facts:

1. For all $a,\ b \in \overline{\mathbb{R}}_{\max}$, the maximal $c \in \overline{\mathbb{R}}_{\max}$ such that $ac \leq b$, denoted by $a \setminus b$ (or b / a), is given by $a \setminus b = b - a$ if $(a, b) \notin \{(-\infty, -\infty), (+\infty, +\infty)\}$, and $a \setminus b = +\infty$ otherwise.
2. [BCOQ92, Eq. 4.82] If $A \in \overline{\mathbb{R}}_{\max}^{n\times p}$ and $B \in \overline{\mathbb{R}}_{\max}^{n\times q}$, then the inequation $AX \leq B$ has a maximal solution $X \in \overline{\mathbb{R}}_{\max}^{p\times q}$ given by the matrix $A \setminus B$ defined by $(A \setminus B)_{ij} = \bigwedge_{1\leq k\leq n} A_{ki} \setminus B_{kj}$. Similarly, for $A \in \overline{\mathbb{R}}_{\max}^{n\times p}$ and $C \in \overline{\mathbb{R}}_{\max}^{r\times p}$, the maximal solution $C / A \in \overline{\mathbb{R}}_{\max}^{r\times n}$ of $XA \leq C$ exists and is given by $(C / A)_{ij} = \bigwedge_{1\leq k\leq p} C_{ik} / A_{jk}$.
3. The equation $AX = B$ has a solution if and only if $A(A \setminus B) = B$.
4. For $A \in \overline{\mathbb{R}}_{\max}^{n\times p}$, the map $A^{\sharp}:\ \mathbf{y} \in \overline{\mathbb{R}}_{\min}^{n} \to A \setminus \mathbf{y} \in \overline{\mathbb{R}}_{\min}^{p}$ is linear. It is represented by the matrix $-A^T$.
5. [BCOQ92, Table 4.1] For matrices A, B, C with entries in $\overline{\mathbb{R}}_{\max}$ and with appropriate dimensions, we have

$$A(A \setminus (AB)) = AB, \quad A \setminus (A(A \setminus B)) = A \setminus B,$$
$$(A \oplus B) \setminus C = (A \setminus C) \wedge (B \setminus C), \quad A \setminus (B \wedge C) = (A \setminus B) \wedge (A \setminus C),$$
$$(AB) \setminus C = B \setminus (A \setminus C), \quad A \setminus (B / C) = (A \setminus B) / C.$$

The first five identities have dual versions, with $/\!\!/$ instead of $\backslash\!\!\backslash$. Due to the last identity, we shall write $A \backslash\!\!\backslash B /\!\!/ C$ instead of $A \backslash\!\!\backslash (B /\!\!/ C)$.

6. [CGQ97] Let $A \in \overline{\mathbb{R}}_{\max}^{n\times p}$, $B \in \overline{\mathbb{R}}_{\max}^{n\times q}$ and $C \in \overline{\mathbb{R}}_{\max}^{r\times p}$. We have range $A \subset$ range $B \iff A = B(B \backslash\!\!\backslash A)$, and $\ker A \subset \ker C \iff C = (C /\!\!/ A)A$.
7. [CGQ96] Let $A \in \overline{\mathbb{R}}_{\max}^{n\times p}$. The map $\Pi_A := A \circ A^\sharp$ is a projector on the range of A, meaning that $(\Pi_A)^2 = \Pi_A$ and range $\Pi_A =$ range A. Moreover, $\Pi_A(x)$ is the greatest element of the range of A, which is less than or equal to x. Similarly, the map $\Pi^A := A^\sharp \circ A$ is a projector on the range of $A^\sharp$, and $\Pi^A(x)$ is the smallest element of the range of $A^\sharp$ that is greater than or equal to x. Finally, every equivalence class modulo A meets the range of $A^\sharp$ at a unique point.
8. [CGQ04], [DS04] For any $A \in \overline{\mathbb{R}}_{\max}^{n\times p}$, the map $x \mapsto A(-x)$ is a bijection from range (A^T) to range (A), with inverse map $x \mapsto A^T(-x)$.
9. [CGQ96], [CGQ97] *Projection onto a range parallel to a kernel.* Let $B \in \overline{\mathbb{R}}_{\max}^{n\times p}$ and $C \in \overline{\mathbb{R}}_{\max}^{q\times n}$. For all $x \in \overline{\mathbb{R}}_{\max}^{n}$, there is a greatest ξ on the range of B such that $C\xi \leq Cx$. It is given by $\Pi_B^C(x)$, where $\Pi_B^C := \Pi_B \circ \Pi^C$. We have $(\Pi_B^C)^2 = \Pi_B^C$. Assume now that every equivalence class modulo C meets the range of B at a unique point. This is the case if and only if range $(CB) =$ range C and $\ker(CB) = \ker B$. Then $\Pi_B^C(x)$ is the unique element of the range of B, which is equivalent to x modulo C, the map Π_B^C is a linear projector on the range of B, and it is represented by the matrix $(B /\!\!/ (CB))C$, which is equal to $B((CB) \backslash\!\!\backslash C)$.
10. [CGQ97] *Regular matrices.* Let $A \in \overline{\mathbb{R}}_{\max}^{n\times p}$. The following assertions are equivalent: (i) there is a linear projector from $\overline{\mathbb{R}}_{\max}^{n}$ to range A; (ii) $A = AXA$ for some $X \in \overline{\mathbb{R}}_{\max}^{p\times n}$; (iii) $A = A(A \backslash\!\!\backslash A /\!\!/ A)A$.
11. [Vor67], [Zim76, Ch. 3] (See also [But94], [AGK05].) *Vorobyev–Zimmermann covering theorem.* Assume that $A \in \mathbb{R}_{\max}^{n\times p}$ and $\mathbf{b} \in \overline{\mathbb{R}}_{\max}^{n}$. For $j \in \{1, \ldots, p\}$, let
$$S_j = \{i \in \{1, \ldots, n\} \mid A_{ij} \neq \mathbb{0} \text{ and } A_{ij} \backslash\!\!\backslash \mathbf{b}_i = (A \backslash\!\!\backslash \mathbf{b})_j\}.$$
The equation $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\cup_{1\leq j\leq p} S_j \supset \operatorname{supp} \mathbf{b}$ or equivalently $\cup_{j\in \operatorname{supp}(A \backslash\!\!\backslash \mathbf{b})} S_j \supset \operatorname{supp} \mathbf{b}$. It has a unique solution if and only if $\cup_{j\in \operatorname{supp}(A \backslash\!\!\backslash \mathbf{b})} S_j \supset \operatorname{supp} \mathbf{b}$ and $\cup_{j\in J} S_j \not\supset \operatorname{supp} \mathbf{b}$ for all strict subsets J of $\operatorname{supp}(A \backslash\!\!\backslash \mathbf{b})$.
12. [Zim77], [SS92], [CGQ04], [CGQS05], [DS04] *Separation theorem.* Let $A \in \overline{\mathbb{R}}_{\max}^{n\times p}$ and $\mathbf{b} \in \overline{\mathbb{R}}_{\max}^{n}$. If $\mathbf{b} \notin$ range A, then there exists $\mathbf{c}, \mathbf{d} \in \overline{\mathbb{R}}_{\max}^{n}$ such that the **halfspace** $H := \{\mathbf{x} \in \overline{\mathbb{R}}_{\max}^{n} \mid \mathbf{c}\cdot\mathbf{x} \geq \mathbf{d}\cdot\mathbf{x}\}$ contains range A but not $\mathbf{b}$. We can take $\mathbf{c} = -\mathbf{b}$ and $\mathbf{d} = -\Pi_A(\mathbf{b})$. Moreover, when A and $\mathbf{b}$ have entries in $\mathbb{R}_{\max}$, $\mathbf{c}, \mathbf{d}$ can be chosen with entries in $\mathbb{R}_{\max}$.
13. [GP97] For any $A \in \overline{\mathbb{R}}_{\max}^{n\times p}$, we have $((\text{range } A)^\perp)^\top =$ range A.
14. [LMS01], [CGQ04] A linear form defined on a finitely generated subsemimodule of $\overline{\mathbb{R}}_{\max}^{n}$ can be extended to $\overline{\mathbb{R}}_{\max}^{n}$. This is a special case of a max-plus analogue of the Riesz representation theorem.
15. [BH84], [GP97] Let $A, B \in \overline{\mathbb{R}}_{\max}^{n\times p}$. The set of solutions $\mathbf{x} \in \overline{\mathbb{R}}_{\max}^{p}$ of $A\mathbf{x} = B\mathbf{x}$ is a finitely generated subsemimodule of $\overline{\mathbb{R}}_{\max}^{p}$.
16. [GP97], [Gau98] Let X, Y be finitely generated subsemimodules of $\overline{\mathbb{R}}_{\max}^{n}$, $A \in \overline{\mathbb{R}}_{\max}^{n\times p}$ and $B \in \overline{\mathbb{R}}_{\max}^{r\times n}$. Then $X \cap Y$, $X \mathbin{+\!\!\!+} Y := \{\mathbf{x} \mathbin{+\!\!\!+} \mathbf{y} \mid \mathbf{x} \in X,\ \mathbf{y} \in Y\}$, and $X - Y := \{\mathbf{z} \in \overline{\mathbb{R}}_{\max}^{n} \mid \exists \mathbf{x} \in X, \mathbf{y} \in Y,\ \mathbf{x} = \mathbf{y} \mathbin{+\!\!\!+} \mathbf{z}\}$ are finitely generated subsemimodules of $\overline{\mathbb{R}}_{\max}^{n}$. Also, $A^{-1}(X)$, $B(X)$, and $X^\perp$ are finitely generated subsemimodules of $\overline{\mathbb{R}}_{\max}^{p}$, $\overline{\mathbb{R}}_{\max}^{r}$, and $\overline{\mathbb{R}}_{\max}^{n} \times \overline{\mathbb{R}}_{\max}^{n}$, respectively. Similarly, if Z is a finitely generated subsemimodule of $\overline{\mathbb{R}}_{\max}^{n} \times \overline{\mathbb{R}}_{\max}^{n}$, then $Z^\top$ is a finitely generated subsemimodule of $\overline{\mathbb{R}}_{\max}^{n}$.
17. Facts 13 to 16 still hold if $\overline{\mathbb{R}}_{\max}$ is replaced by $\mathbb{R}_{\max}$.
18. When $A, B \in \mathbb{R}_{\max}^{n\times p}$, algorithms to find one solution of $A\mathbf{x} = B\mathbf{x}$ are given in [WB98] or [CGB03]. One can also use the general algorithm of [GG98] to compute a finite fixed point of a min-max function, together with the observation that $\mathbf{x}$ satisfies $A\mathbf{x} = B\mathbf{x}$ if and only if $\mathbf{x} = f(\mathbf{x})$, where $f(\mathbf{x}) = \mathbf{x} \wedge (A \backslash\!\!\backslash (B\mathbf{x})) \wedge (B \backslash\!\!\backslash (A\mathbf{x}))$.

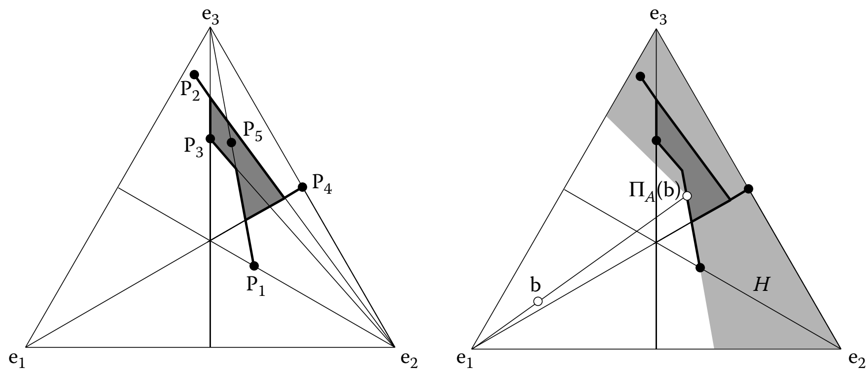


FIGURE 25.1 Projection of a point on a range.

Examples:

1. In order to illustrate Fact 11, consider

$$A = \begin{bmatrix} 0 & 0 & 0 & -\infty & 0.5 \\ 1 & -2 & 0 & 0 & 1.5 \\ 0 & 3 & 2 & 0 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 0 \\ 0.5 \end{bmatrix}. \tag{25.3}$$

Let $\bar{\mathbf{x}} := A \setminus\!\!\setminus \mathbf{b}$. We have $\bar{\mathbf{x}}_1 = \min(-0+3, -1+0, -0+0.5) = -1$, and so, $S_1 = \{2\}$ because the minimum is attained only by the second term. Similarly, $\bar{\mathbf{x}}_2 = -2.5$, $S_2 = \{3\}$, $\bar{\mathbf{x}}_3 = -1.5$, $S_3 = \{3\}$, $\bar{\mathbf{x}}_4 = 0$, $S_4 = \{2\}$, $\bar{\mathbf{x}}_5 = -2.5$, $S_5 = \{3\}$. Since $\cup_{1\le j\le 5} S_j = \{2,3\} \not\supset \operatorname{supp} \mathbf{b} = \{1,2,3\}$, Fact 11 shows that the equation $Ax = \mathbf{b}$ has no solution. This also follows from the fact that $\Pi_A(\mathbf{b}) = A(A \setminus\!\!\setminus \mathbf{b}) = [-1\ 0\ 0.5]^T < \mathbf{b}$.

2. The range of the previous matrix A is represented in Figure 25.1 (left). A nonzero vector $\mathbf{x} \in \mathbb{R}^3_{\max}$ is represented by the point that is the barycenter with weights $(\exp(\beta \mathbf{x}_i))_{1\le i\le 3}$ of the vertices of the simplex, where $\beta > 0$ is a fixed scaling parameter. Every vertex of the simplex represents one basis vector $\mathbf{e}_i$. Proportional vectors are represented by the same point. The i-th column of A, $A_{\cdot i}$, is represented by the point $\mathbf{p}_i$ on the figure. Observe that the broken segment from $\mathbf{p}_1$ to $\mathbf{p}_2$, which represents the semimodule generated by $A_{\cdot 1}$ and $A_{\cdot 2}$, contains $\mathbf{p}_5$. Indeed, $A_{\cdot 5} = 0.5 A_{\cdot 1} \oplus A_{\cdot 2}$. The range of A is represented by the closed region in dark grey and by the bold segments joining the points $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_4$ to it.

   We next compute a half-space separating the point b defined in (25.3) from range A. Recall that $\Pi_A(\mathbf{b}) = [-1\ 0\ 0.5]^T$. So, by Fact 12, a half-space containing range A and not $\mathbf{b}$ is $H := \{\mathbf{x} \in \overline{\mathbb{R}}^3_{\max} (-3)\mathbf{x}_1 \oplus \mathbf{x}_2 \oplus (-0.5)\mathbf{x}_3 \ge 1\mathbf{x}_1 \oplus \mathbf{x}_2 \oplus (-0.5)\mathbf{x}_3\}$. We also have $H \cap \mathbb{R}^3_{\max} = \{\mathbf{x} \in \mathbb{R}^3_{\max} \mid \mathbf{x}_2 \oplus (-0.5)\mathbf{x}_3 \ge 1\mathbf{x}_1\}$. The set of nonzero points of $H \cap \mathbb{R}^3_{\max}$ is represented by the light gray region in Figure 25.1 (right).

25.7 Max-Plus Linear Independence and Rank

Definitions:

If M is a subsemimodule of $\mathbb{R}^n_{\max}$, $\mathbf{u} \in M$ is an **extremal generator** of M, or $\mathbb{R}_{\max}\mathbf{u} := \{\lambda.\mathbf{u} \mid \lambda \in \mathbb{R}_{\max}\}$ is an **extreme ray** of M, if $\mathbf{u} \ne \mathbb{0}$ and if $\mathbf{u} = \mathbf{v} \oplus \mathbf{w}$ with $\mathbf{v}, \mathbf{w} \in M$ imply that $\mathbf{u} = \mathbf{v}$ or $\mathbf{u} = \mathbf{w}$.

A family $\mathbf{u}_1, \ldots, \mathbf{u}_r$ of vectors of $\mathbb{R}^n_{\max}$ is **linearly independent in the Gondran–Minoux sense** if for all disjoints subsets I and J of $\{1, \ldots, r\}$, and all $\lambda_i \in \mathbb{R}_{\max}$, $i \in I \cup J$, we have $\bigoplus_{i\in I} \lambda_i.\mathbf{u}_i \ne \bigoplus_{j\in J} \lambda_j.\mathbf{u}_j$, unless $\lambda_i = \mathbb{0}$ for all $i \in I \cup J$.

For $A \in \mathbb{R}_{\max}^{n\times n}$, we define

$$\det{}^+ A := \sum_{\sigma\in S_n^+} A_{1\sigma(1)}\cdots A_{n\sigma(n)}, \qquad \det{}^- A := \sum_{\sigma\in S_n^-} A_{1\sigma(1)}\cdots A_{n\sigma(n)},$$

where S_n^+ and S_n^- are, respectively, the sets of even and odd permutations of $\{1,\dots,n\}$. The **bideterminant** [GM84] of A is $(\det^+ A, \det^- A)$.

For $A \in \mathbb{R}_{\max}^{n\times p} \setminus \{\mathbf{0}\}$, we define

- The **row rank** (resp. the **column rank**) of A, denoted $\mathrm{rk}_{\mathrm{row}}(A)$ (resp. $\mathrm{rk}_{\mathrm{col}}(A)$), as the number of extreme rays of range A^T (resp. range A).
- The **Schein** rank of A as $\mathrm{rk}_{\mathrm{Sch}}(A) := \min\{r \geq 1 \mid A = BC, \text{ with } B \in \mathbb{R}_{\max}^{n\times r}, C \in \mathbb{R}_{\max}^{r\times p}\}$.
- The **strong rank** of A, denoted $\mathrm{rk}_{\mathrm{st}}(A)$, as the maximal $r \geq 1$ such that there exists an $r \times r$ submatrix B of A for which there is only one permutation σ such that $|\sigma|_B = \operatorname{per} B$.
- The row (resp. column) **Gondran–Minoux rank** of A, denoted $\mathrm{rk}_{\mathrm{GMr}}(A)$ (resp. $\mathrm{rk}_{\mathrm{GMc}}$), as the maximal $r \geq 1$ such that A has r linearly independent rows (resp. columns) in the Gondran–Minoux sense.
- The **symmetrized rank** of A, denoted $\mathrm{rk}_{\mathrm{sym}}(A)$, as the maximal $r \geq 1$ such that A has an $r \times r$ submatrix B such that $\det^+ B \neq \det^- B$.

(A new rank notion, **Kapranov rank**, which is not discussed here, has been recently studied [DSS05]. We also note that the Schein rank is called in this reference Barvinok rank.)

Facts:

1. [Hel88], [Mol88], [Wag91], [Gau98], [DS04] Let M be a finitely generated subsemimodule of $\mathbb{R}_{\max}^n$. A subset of vectors of M spans M if and only if it contains at least one nonzero element of every extreme ray of M.
2. [GM02] The columns of $A \in \mathbb{R}_{\max}^{n\times n}$ are linearly independent in the Gondran–Minoux sense if and only if $\det^+ A \neq \det^- A$.
3. [Plu90], [BCOQ92, Th. 3.78]. *Max-plus Cramer's formula.* Let $A \in \mathbb{R}_{\max}^{n\times n}$, and let $\mathbf{b}^-, \mathbf{b}^+ \in \mathbb{R}_{\max}^n$. Define the i-th positive Cramer's determinant by

$$D_i^+ := \det{}^+(A_{\cdot 1}\dots A_{\cdot,i-1}\mathbf{b}^+ A_{\cdot,i+1}\dots A_{\cdot n}) \oplus \det{}^-(A_{\cdot 1}\dots A_{\cdot,i-1}\mathbf{b}^- A_{\cdot,i+1}\dots A_{\cdot n}),$$

   and the i-th negative Cramer's determinant, D_i^-, by exchanging $\mathbf{b}^+$ and $\mathbf{b}^-$ in the definition of D_i^+. Assume that $\mathbf{x}^+, \mathbf{x}^- \in \mathbb{R}_{\max}^n$ have disjoint supports. Then $A\mathbf{x}^+ \oplus \mathbf{b}^- = A\mathbf{x}^- \oplus \mathbf{b}^+$ implies that

$$(\det{}^+ A)\mathbf{x}_i^+ \oplus (\det{}^- A)\mathbf{x}_i^- \oplus D_i^- = (\det{}^- A)\mathbf{x}_i^+ \oplus (\det{}^+ A)\mathbf{x}_i^- \oplus D_i^+ \quad \forall 1 \leq i \leq n. \tag{25.4}$$

   The converse implication holds, and the vectors $\mathbf{x}^+$ and $\mathbf{x}^-$ are uniquely determined by (25.4), if $\det^+ A \neq \det^- A$, and if $D_i^+ \neq D_i^-$ or $D_i^+ = D_i^- = \mathbb{0}$, for all $1 \leq i \leq n$. This result is formulated in a simpler way in [Plu90], [BCOQ92] using the **symmetrization** of the max-plus semiring, which leads to more general results. We note that the converse implication relies on the following semiring analogue of the classical adjugate identity: $A \operatorname{adj}^+ A \oplus \det^- A\, I = A \operatorname{adj}^- A \oplus \det^+ A\, I$, where $\operatorname{adj}^\pm A := (\det^\pm A(j,i))_{1\leq i,j\leq n}$. This identity, as well as analogues of many other determinantal identities, can be obtained using the general method of [RS84]. See, for instance, [GBCG98], where the derivation of the Binet–Cauchy identity is detailed.
4. For $A \in \mathbb{R}_{\max}^{n\times p}$, we have

$$\mathrm{rk}_{\mathrm{st}}(A) \leq \mathrm{rk}_{\mathrm{sym}}(A) \leq \begin{Bmatrix} \mathrm{rk}_{\mathrm{GMr}}(A) \\ \mathrm{rk}_{\mathrm{GMc}}(A) \end{Bmatrix} \leq \mathrm{rk}_{\mathrm{Sch}}(A) \leq \begin{cases} \mathrm{rk}_{\mathrm{row}}(A) \\ \mathrm{rk}_{\mathrm{col}}(A) \end{cases}.$$

The second inequality follows from Fact 2, the third one from Facts 2 and 3. The other inequalities are immediate. Moreover, all these inequalities become equalities if A is regular [CGQ06].

Examples:

1. The matrix A in Example 1 of section 25.6 has column rank 4: The extremal rays of range A are generated by the first four columns of A. All the other ranks of A are equal to 3.

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26
Matrices Leaving a Cone Invariant

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Generalizations of the Perron–Frobenius theory of nonnegative matrices to linear operators leaving a cone invariant were first developed for operators on a Banach space by Krein and Rutman [KR48], Karlin [Kar59], and Schaefer [Sfr66], although there are early examples in finite dimensions, e.g., [Sch65] and [Bir67]. In this chapter, we describe a generalization that is sometimes called the geometric spectral theory of nonnegative linear operators in finite dimensions, which emerged in the late 1980s. Motivated by a search for geometric analogs of results in the previously developed combinatorial spectral theory of (reducible) nonnegative matrices (for reviews see [Sch86] and [Her99]), this area is a study of the Perron–Frobenius theory of a nonnegative matrix and its generalizations from the cone-theoretic viewpoint. The treatment is linear-algebraic and cone-theoretic (geometric) with the facial and duality concepts and occasionally certain elementary analytic tools playing the dominant role. The theory is particularly rich when the underlying cone is polyhedral (finitely generated) and it reduces to the nonnegative matrix case when the cone is simplicial.

26.1 Perron—Frobenius Theorem for Cones

We work with cones in a real vector space, as "cone" is a real concept. To deal with cones in $\mathbb{C}^n$, we can identify the latter space with $\mathbb{R}^{2n}$. For a discussion on the connection between the real and complex case of the spectral theory, see [TS94, Sect. 8].

Definitions:

A **proper cone** K in a finite-dimensional real vector space V is a closed, pointed, full convex cone, *viz.*

- $K + K \subseteq K$, viz. $\mathbf{x}, \mathbf{y} \in K \Longrightarrow \mathbf{x} + \mathbf{y} \in K$.
- $\mathbb{R}^+ K \subseteq K$, viz. $\mathbf{x} \in K, \alpha \in \mathbb{R}^+ \Longrightarrow \alpha\mathbf{x} \in K$.
- K is closed in the usual topology of V.

- $K \cap (-K) = \{\mathbf{0}\}$, viz. $\mathbf{x}, -\mathbf{x} \in K \Longrightarrow \mathbf{x} = \mathbf{0}$.
- $\text{int}K \neq \emptyset$, where $\text{int}K$ is the interior of K.

Usually, the unqualified term **cone** is defined by the first two items in the above definition. However, in this chapter we call a proper cone simply a **cone**. We denote by K a cone in $\mathbb{R}^n$, $n \geq 2$.

The vector $\mathbf{x} \in \mathbb{R}^n$ is K**-nonnegative**, written $\mathbf{x} \geq^K \mathbf{0}$, if $\mathbf{x} \in K$.

The vector $\mathbf{x}$ is K**-semipositive**, written $\mathbf{x} \gneq^K \mathbf{0}$, if $\mathbf{x} \geq^K \mathbf{0}$ and $\mathbf{x} \neq \mathbf{0}$.

The vector $\mathbf{x}$ is K**-positive**, written $\mathbf{x} >^K \mathbf{0}$, if $\mathbf{x} \in \text{int } K$.

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, we write $\mathbf{x} \geq^K \mathbf{y}$ ($\mathbf{x} \gneq^K \mathbf{y}$, $\mathbf{x} >^K \mathbf{y}$) if $\mathbf{x} - \mathbf{y}$ is K-nonnegative (K-semipositive, K-positive).

The matrix $A \in \mathbb{R}^{n \times n}$ is K**-nonnegative**, written $A \geq^K \mathbf{0}$, if $AK \subseteq K$.

The matrix A is K**-semipositive**, written $A \gneq^K \mathbf{0}$, if $A \geq^K \mathbf{0}$ and $A \neq \mathbf{0}$.

The matrix A is K**-positive**, written $A >^K \mathbf{0}$, if $A(K \setminus \{\mathbf{0}\}) \subseteq \text{int } K$.

For $A, B \in \mathbb{R}^{n \times n}$, $A \geq^K B$ ($A \gneq^K B$, $A >^K B$) means $A - B \geq^K \mathbf{0}$ ($A - B \gneq^K \mathbf{0}$, $A - B >^K \mathbf{0}$).

A **face** F of a cone $K \subseteq \mathbb{R}^n$ is a subset of K, which is a cone in the linear span of F such that $\mathbf{x} \in F$, $\mathbf{x} \geq^K \mathbf{y} \geq^K \mathbf{0} \Longrightarrow \mathbf{y} \in F$.

(In this chapter, F will always denote a face rather than a field, since the only fields involved are $\mathbb{R}$ and $\mathbb{C}$.) Thus, F satisfies all definitions of a cone except that its interior may be empty.

A face F of K is a **trivial face** if $F = \{\mathbf{0}\}$ or $F = K$.

For a subset S of a cone K, the intersection of all faces of K including S is called the **face of** K **generated by** S and is denoted by $\Phi(S)$. If $S = \{\mathbf{x}\}$, then $\Phi(S)$ is written simply as $\Phi(\mathbf{x})$.

For faces F, G of K, their **meet** and **join** are given respectively by $F \wedge G = F \cap G$ and $F \vee G = \Phi(F \cup G)$.

A vector $\mathbf{x} \in K$ is an **extreme vector** if either $\mathbf{x}$ is the zero vector or $\mathbf{x}$ is nonzero and $\Phi(\mathbf{x}) = \{\lambda\mathbf{x} : \lambda \geq 0\}$; in the latter case, the face $\Phi(\mathbf{x})$ is called an **extreme ray**.

If P is K-nonnegative, then a face F of K is a P**-invariant face** if $PF \subseteq F$.

If P is K-nonnegative, then P is K**-irreducible** if the only P-invariant faces are the trivial faces.

If K is a cone in $\mathbb{R}^n$, then a cone, called the **dual cone** of K, is denoted and given by

$$K^* = \{\mathbf{y} \in \mathbb{R}^n : \mathbf{y}^T\mathbf{x} \geq 0 \text{ for all } \mathbf{x} \in K\}.$$

If A is an $n \times n$ complex matrix and $\mathbf{x}$ is a vector in $\mathbb{C}^n$, then the **local spectral radius of** A **at** $\mathbf{x}$ is denoted and given by $\rho_{\mathbf{x}}(A) = \limsup_{m\to\infty} \|A^m\mathbf{x}\|^{1/m}$, where $\|\cdot\|$ is any norm of $\mathbb{C}^n$. For $A \in \mathbb{C}^{n \times n}$, its spectral radius is denoted by $\rho(A)$ (or ρ) (cf. Section 4.3).

Facts:

Let K be a cone in $\mathbb{R}^n$.

1. The condition $\text{int}K \neq \emptyset$ in the definition of a cone is equivalent to $K - K = V$, *viz.*, for all $\mathbf{z} \in V$ there exist $\mathbf{x}, \mathbf{y} \in K$ such that $\mathbf{z} = \mathbf{x} - \mathbf{y}$.
2. A K-positive matrix is K-irreducible.
3. [Van68], [SV70] Let P be a K-nonnegative matrix. The following are equivalent:
   (a) P is K-irreducible.
   (b) $\sum_{i=0}^{n-1} P^i >^K \mathbf{0}$.
   (c) $(I + P)^{n-1} >^K \mathbf{0}$.
   (d) No eigenvector of P (for any eigenvalue) lies on the boundary of K.
4. (Generalization of Perron–Frobenius Theorem) [KR48], [BS75] Let P be a K-irreducible matrix with spectral radius ρ. Then
   (a) ρ is positive and is a simple eigenvalue of P.
   (b) There exists a (up to a scalar multiple) unique K-positive (right) eigenvector $\mathbf{u}$ of P corresponding to ρ.
   (c) $\mathbf{u}$ is the only K-semipositive eigenvector for P (for any eigenvalue).
   (d) $K \cap (\rho I - P)\mathbb{R}^n = \{\mathbf{0}\}$.

5. (Generalization of Perron–Frobenius Theorem) Let P be a K-nonnegative matrix with spectral radius ρ. Then

   (a) ρ is an eigenvalue of P.

   (b) There is a K-semipositive eigenvector of P corresponding to ρ.

6. If P, Q are K-nonnegative and $Q \stackrel{K}{\leq} P$, then $\rho(Q) \leq \rho(P)$. Further, if P is K-irreducible and $Q \stackrel{K}{\lneq} P$, then $\rho(Q) < \rho(P)$.
7. P is K-nonnegative (K-irreducible) if and only if P^T is K^*-nonnegative (K^*-irreducible).
8. If A is an $n \times n$ complex matrix and $\mathbf{x}$ is a vector in $\mathbf{C}^n$, then the local spectral radius $\rho_{\mathbf{x}}(A)$ of A at $\mathbf{x}$ is equal to the spectral radius of the restriction of A to the A-cyclic subspace generated by $\mathbf{x}$, i.e., $\text{span}\{A^i\mathbf{x} : i = 0, 1, \ldots\}$. If $\mathbf{x}$ is nonzero and $\mathbf{x} = \mathbf{x}_1 + \cdots + \mathbf{x}_k$ is the representation of $\mathbf{x}$ as a sum of generalized eigenvectors of A corresponding, respectively, to distinct eigenvalues $\lambda_1, \ldots, \lambda_k$, then $\rho_{\mathbf{x}}(A)$ is also equal to $\max_{1 \leq i \leq k} |\lambda_i|$.
9. Barker and Schneider [BS75] developed Perron–Frobenius theory in the setting of a (possibly infinite-dimensional) vector space over a fully ordered field without topology. They introduced the concepts of irreducibility and strong irreducibility, and show that these two concepts are equivalent if the underlying cone has ascending chain condition on faces. See [ERS95] for the role of real closed-ordered fields in this theory.

Examples:

1. The nonnegative orthant $(\mathbb{R}_0^+)^n$ in $\mathbb{R}^n$ is a cone. Then $\mathbf{x} \geq^K \mathbf{0}$ if and only if $\mathbf{x} \geq \mathbf{0}$, *viz.* the entries of $\mathbf{x}$ are nonnegative, and F is a face of $(\mathbb{R}_0^+)^n$ if and only if F is of the form F_J for some $J \subseteq \{1, \ldots, n\}$, where

$$F_J = \{\mathbf{x} \in (\mathbb{R}_0^+)^n \ : \ x_i = 0, \ i \notin J\}.$$

   Further, $P \geq^K \mathbf{0}$ ($P \gneq^K \mathbf{0}$, $P >^K \mathbf{0}$, P is K-irreducible) if and only if $P \geq \mathbf{0}$ ($P \gneq \mathbf{0}$, $P > \mathbf{0}$, P is irreducible) in the sense used for nonnegative matrices, cf. Chapter 9.
2. The nontrivial faces of the Lorentz (ice cream) cone K_n in $\mathbb{R}^n$, *viz.*

$$K_n = \{\mathbf{x} \in \mathbb{R}^n : (x_1^2 + \cdots + x_{n-1}^2)^{1/2} \leq x_n\},$$

   are precisely its extreme rays, each generated by a nonzero boundary vector, that is, one for which the equality holds above. The matrix

$$P = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

   is K_3-irreducible [BP79, p. 22].

26.2 Collatz—Wielandt Sets and Distinguished Eigenvalues

Collatz–Wielandt sets were apparently first defined in [BS75]. However, they are so-called because they are closely related to Wielandt's proof of the Perron–Frobenius theorem for irreducible nonnegative matrices, [Wie50], which employs an inequality found in Collatz [Col42]. See also [Sch96] for further remarks on Collatz–Wielandt sets and related max-min and min-max characterizations of the spectral radius of nonnegative matrices and their generalizations.

Definitions:

Let P be a K-nonnegative matrix.

The **Collatz–Wielandt sets** associated with P ([BS75], [TW89], [TS01], [TS03], and [Tam01]) are defined by

$$\begin{aligned}\Omega(P) &= \{\omega \geq 0 : \exists \mathbf{x} \in K\backslash\{\mathbf{0}\},\ P\mathbf{x} \geq^K \omega\mathbf{x}\}.\\ \Omega_1(P) &= \{\omega \geq 0 : \exists \mathbf{x} \in \operatorname{int} K,\ P\mathbf{x} \geq^K \omega\mathbf{x}\}.\\ \Sigma(P) &= \{\sigma \geq 0 : \exists \mathbf{x} \in K\backslash\{\mathbf{0}\},\ P\mathbf{x} \,{}^K\!\!\leq \sigma\mathbf{x}\}.\\ \Sigma_1(P) &= \{\sigma \geq 0 : \exists \mathbf{x} \in \operatorname{int} K,\ P\mathbf{x} \,{}^K\!\!\leq \sigma\mathbf{x}\}.\end{aligned}$$

For a K-nonnegative vector $\mathbf{x}$, the **lower and upper Collatz–Wielandt numbers** of $\mathbf{x}$ with respect to P are defined by

$$\begin{aligned}r_P(x) &= \sup\ \{\omega \geq 0 : P\mathbf{x} \geq^K \omega\mathbf{x}\},\\ R_P(x) &= \inf\ \{\sigma \geq 0 : P\mathbf{x} \,{}^K\!\!\leq \sigma\mathbf{x}\},\end{aligned}$$

where we write $R_P(\mathbf{x}) = \infty$ if no σ exists such that $P\mathbf{x} \,{}^K\!\!\leq \sigma\mathbf{x}$.

A (nonnegative) eigenvalue of P is a **distinguished eigenvalue** for K if it has an associated K-semipositive eigenvector.

The **Perron space** $N_\rho^\nu(P)$ (or N_ρ^ν) is the subspace consisting of all $\mathbf{u} \in \mathbb{R}^n$ such that $(P - \rho I)^k\mathbf{u} = \mathbf{0}$ for some positive integer k. (See Chapter 6.1 for a more general definition of $N_\lambda^\nu(A)$.)

If F is a P-invariant face of K, then the restriction of P to span$F$ is written as $P|F$. The spectral radius of $P|F$ is written as $\rho[F]$, and if λ is an eigenvalue of $P|F$, its index is written as $\nu_\lambda[F]$.

A cone K in $\mathbb{R}^n$ is **polyhedral** if it is the set of linear combinations with nonnegative coefficients of vectors taken from a finite subset of $\mathbb{R}^n$, and is **simplicial** if the finite subset is linearly independent.

Facts:

Let P be a K-nonnegative matrix.

1. [TW89] A real number λ is a distinguished eigenvalue of P for K if and only if $\lambda = \rho_{\mathbf{b}}(P)$ for some K-semipositive vector $\mathbf{b}$.
2. [Tam90] Consider the following conditions:
   (a) ρ is the only distinguished eigenvalue of P for K.
   (b) $\mathbf{x} \geq^K \mathbf{0}$ and $P\mathbf{x} \,{}^K\!\!\leq \rho\mathbf{x}$ imply that $P\mathbf{x} = \rho\mathbf{x}$.
   (c) The Perron space of P^T contains a K^*-positive vector.
   (d) $\rho \in \Omega_1(P^T)$.

   Conditions (a), (b), and (c) are always equivalent and are implied by condition (d). When K is polyhedral, condition (d) is also an equivalent condition.
3. [Tam90] The following conditions are equivalent:
   (a) $\rho(P)$ is the only distinguished eigenvalue of P for K and the index of $\rho(P)$ is one.
   (b) For any vector $\mathbf{x} \in \mathbb{R}^n$, $P\mathbf{x} \,{}^K\!\!\leq \rho(P)\mathbf{x}$ implies that $P\mathbf{x} = \rho(P)\mathbf{x}$.
   (c) $K \cap (\rho I - P)\mathbb{R}^n = \{\mathbf{0}\}$.
   (d) P^T has a K^*-positive eigenvector (corresponding to $\rho(P)$).
4. [TW89] The following statements all hold:
   (a) [BS75] If P is K-irreducible, then
   $\sup\ \Omega(P) = \sup\ \Omega_1(P) = \inf\ \Sigma(P) = \inf\ \Sigma_1(P) = \rho(P)$.
   (b) $\sup\ \Omega(P) = \inf\ \Sigma_1(P) = \rho(P)$.
   (c) $\inf\ \Sigma(P)$ is equal to the least distinguished eigenvalue of P for K.

(d) $\sup \Omega_1(P) = \inf \Sigma(P^T)$ and, hence, is equal to the least distinguished eigenvalue of P^T for K^*.

(e) $\sup \Omega(P) \in \Omega(P)$ and $\inf \Sigma(P) \in \Sigma(P)$.

(f) When K is polyhedral, we have $\sup \Omega_1(P) \in \Omega_1(P)$. For general cones, we may have $\sup \Omega_1(P) \notin \Omega_1(P)$.

(g) [Tam90] When K is polyhedral, $\rho(P) \in \Omega_1(P)$ if and only if $\rho(A)$ is the only distinguished eigenvalue of P^T for K^*.

(h) [TS03] $\rho(P) \in \Sigma_1(P)$ if and only if $\Phi((N_\rho^1(P) \cap K) \cup C) = K$, where C is the set $\{\mathbf{x} \in K : \rho_{\mathbf{x}}(P) < \rho(P)\}$ and $N_\rho^1(P)$ is the Perron eigenspace of P.

5. In the irreducible nonnegative matrix case, statement (b) of the preceding fact reduces to the well-known max-min and min-max characterizations of $\rho(P)$ due to Wielandt. Schaefer [Sfr84] generalized the result to irreducible compact operators in L^p-spaces and more recently Friedland [Fri90], [Fri91] also extended the characterizations in the settings of a Banach space or a C^*-algebra.
6. [TW89, Theorem 2.4(i)] For any $\mathbf{x} \geq^K \mathbf{0}, r_P(\mathbf{x}) \leq \rho_{\mathbf{x}}(P) \leq R_P(\mathbf{x})$. (This fact extends the well-known inequality $r_P(\mathbf{x}) \leq \rho(P) \leq R_P(\mathbf{x})$ in the nonnegative matrix case, due to Collatz [Col42] under the assumption that $\mathbf{x}$ is a positive vector and due to Wielandt [Wie50] under the assumption that P is irreducible and $\mathbf{x}$ is semipositive. For similar results concerning a nonnegative linear continuous operator in a Banach space, see [FN89].)
7. A discussion on estimating $\rho(P)$ or $\rho_x(P)$ by a convergent sequence of (lower or upper) Collatz–Wielandt numbers can be found in [TW89, Sect. 5] and [Tam01, Subsect. 3.1.4].
8. [GKT95, Corollary 3.2] If K is strictly convex (i.e., each boundary vector is extreme), then P has at most two distinguished eigenvalues. This fact supports the statement that the spectral theory of nonnegative linear operators depends on the geometry of the underlying cone.

26.3 The Peripheral Spectrum, the Core, and the Perron–Schaefer Condition

In addition to using Collatz–Wielandt sets to study Perron–Frobenius theory, we may also approach this theory by considering the core (whose definition will be given below). This geometric approach started with the work of Pullman [Pul71], who succeeded in rederiving the Frobenius theorem for irreducible nonnegative matrices. Naturally, this approach was also taken up in geometric spectral theory. It was found that there are close connections between the core, the peripheral spectrum, the Perron–Schaefer condition, and the distinguished faces of a K-nonnegative linear operator. This led to a revival of interest in the Perron–Schaefer condition and associated conditions for the existence of a cone K such that a preassigned matrix is K-nonnegative. (See [Bir67], [Sfr66], [Van68], [Sch81].) The study has also led to the identification of necessary and equivalent conditions for a collection of Jordan blocks to correspond to the peripheral eigenvalues of a nonnegative matrix. (See [TS94] and [McD03].) The local Perron–Schaefer condition was identified in [TS01] and has played a role in the subsequent work. In the course of this investigation, methods were found for producing invariant cones for a matrix with the Perron–Schaefer condition, see [TS94], [Tam06]. These constructions may also be useful in the study of allied fields, such as linear dynamical systems. There invariant cones for matrices are often encountered. (See, for instance, [BNS89].)

Definitions:

If P is K-nonnegative, then a nonzero P-invariant face F of K is a **distinguished face** (associated with λ) if for every P-invariant face G, with $G \subset F$, we have $\rho[G] < \rho[F]$ (and $\rho[F] = \lambda$).

If λ is an eigenvalue of $A \in \mathbf{C}^{n\times n}$, then $\ker(A - \lambda I)^k$ is denoted by $N_\lambda^k(A)$ for $k = 1, 2, \ldots$, the **index** of λ is denoted by $\nu_A(\lambda)$ (or ν_λ when A is clear), and the **generalized eigenspace** at λ is denoted by $N_\lambda^\nu(A)$. See Chapter 6.1 for more information.

Let $A \in \mathbf{C}^{n\times n}$.

The **order** of a generalized eigenvector $\mathbf{x}$ for λ is the smallest positive integer k such that $(A-\lambda I)^k\mathbf{x} = \mathbf{0}$. The maximal order of all K-semipositive generalized eigenvectors in $N_\lambda^\nu(A)$ is denoted by ord_λ.

The matrix A satisfies the **Perron–Schaefer condition** ([Sfr66], [Sch81]) if

- $\rho = \rho(A)$ is an eigenvalue of A.
- If λ is an eigenvalue of A and $|\lambda| = \rho$, then $\nu_A(\lambda) \leq \nu_A(\rho)$.

If K is a cone and P is K-nonnegative, then the set $\bigcap_{i=0}^{\infty} P^i K$, denoted by $\mathrm{core}_K(P)$, is called the **core** of P relative to K.

An eigenvalue λ of A is called **a peripheral eigenvalue** if $|\lambda| = \rho(A)$. The peripheral eigenvalues of A constitute the **peripheral spectrum** of A.

Let $\mathbf{x} \in \mathbf{C}^n$. Then A satisfies the **local Perron–Schaefer condition at x** if there is a generalized eigenvector $\mathbf{y}$ of A corresponding to $\rho_{\mathbf{x}}(A)$ that appears as a term in the representation of $\mathbf{x}$ as a sum of generalized eigenvectors of A. Furthermore, the order of $\mathbf{y}$ is equal to the maximum of the orders of the generalized eigenvectors that appear in the representation and correspond to eigenvalues with modulus $\rho_{\mathbf{x}}(A)$.

Facts:

1. [Sfr66, Chap. V] Let K be a cone in $\mathbb{R}^n$ and let P be a K-nonnegative matrix. Then P satisfies the Perron–Schaefer condition.
2. [Sch81] Let K be a cone in $\mathbb{R}^n$ and let P be a K-nonnegative matrix with spectral radius ρ. Then P has at least m linearly independent K-semipositive eigenvectors corresponding to ρ, where m is the number of Jordan blocks in the Jordan form of P of maximal size that correspond to ρ.
3. [Van68] Let $A \in \mathbb{R}^{n\times n}$. Then there exists a cone K in $\mathbb{R}^n$ such that A is K-nonnegative if and only if A satisfies the Perron–Schaefer condition.
4. [TS94] Let $A \in \mathbb{R}^{n\times n}$ that satisfies the Perron–Schaefer condition. Let m be the number of Jordan blocks in the Jordan form of A of maximal size that correspond to $\rho(A)$. Then for each positive integer k, $m \leq k \leq \dim N_\rho^1(A)$, there exists a cone K in $\mathbb{R}^n$ such that A is K-nonnegative and dim $\mathrm{span}(N_\rho^1(A) \cap K) = k$.
5. Let $A \in \mathbb{R}^{n\times n}$. Let k be a nonnegative integer and let $\omega_k(A)$ consist of all linear combinations with nonnegative coefficients of $A^k, A^{k+1}, \ldots$. The closure of $\omega_k(A)$ is a cone in its linear span if and only if A satisfies the Perron–Schaefer condition. (For this fact in the setting of complex matrices see [Sch81].)
6. Necessary and sufficient conditions involving $\omega_k(A)$ so that $A \in \mathbb{C}^{n\times n}$ has a positive (nonnegative) eigenvalue appear in [Sch81]. For the corresponding real versions, see [Tam06].
7. [Pul71], [TS94] If K is a cone and P is K-nonnegative, then $\mathrm{core}_K(P)$ is a cone in its linear span and $P(\mathrm{core}_K(P)) = \mathrm{core}_K(P)$. Furthermore, $\mathrm{core}_K(P)$ is polyhedral (or simplicial) whenever K is. So when $\mathrm{core}_K(P)$ is polyhedral, P permutes the extreme rays of $\mathrm{core}_K(P)$.
8. For a K-nonnegative matrix P, a characterization of K-irreducibility (as well as K-primitivity) of P in terms of $\mathrm{core}_K(P)$, which extends the corresponding result of Pullman for a nonnegative matrix, can be found in [TS94].
9. [Pul71] If P is an irreducible nonnegative matrix, then the permutation induced by P on the extreme rays of $\mathrm{core}_{(\mathbb{R}_0^+)^n}(P)$ is a single cycle of length equal to the number of distinct peripheral eigenvalues of P. (This fact can be regarded as a geometric characterization of the said quantity (cf. the known combinatorial characterization, see Fact 5(c) of Chapter 9.2), whereas part (b) of the next fact is its extension.)
10. [TS94, Theorem 3.14] For a K-nonnegative matrix P, if $\mathrm{core}_K(P)$ is a nonzero simplicial cone, then:

    (a) There is a one-to-one correspondence between the set of distinguished faces associated with nonzero eigenvalues and the set of cycles of the permutation τ_P induced by P on the extreme rays of $\mathrm{core}_K(P)$.

   (b) If σ is a cycle of the induced permutation τ_P, then the peripheral eigenvalues of the restriction of P to the linear span of the distinguished P-invariant face F corresponding to σ are simple and are exactly $\rho[F]$ times all the d_σth roots of unity, where d_σ is the length of the cycle σ.

11. [TS94] If P is K-nonnegative and $\text{core}_K(P)$ is nonzero polyhedral, then:

    (a) $\text{core}_K(P)$ consists of all linear combinations with nonnegative coefficients of the distinguished eigenvectors of positive powers of P corresponding to nonzero distinguished eigenvalues.

    (b) $\text{core}_K(P)$ does not contain a generalized eigenvector of any positive powers of P other than eigenvectors.

    This fact indicates that we cannot expect that the index of the spectral radius of a nonnegative linear operator can be determined from a knowledge of its core.
12. A complete description of the core of a nonnegative matrix (relative to the nonnegative orthant) can be found in [TS94, Theorem 4.2].
13. For $A \in \mathbb{R}^{n\times n}$, in order that there exists a cone K in $\mathbb{R}^n$ such that $AK = K$ and A has a K-positive eigenvector, it is necessary and sufficient that A is nonzero, diagonalizable, all eigenvalues of A are of the same modulus, and $\rho(A)$ is an eigenvalue of A. For further equivalent conditions, see [TS94, Theorem 5.9].
14. For $A \in \mathbb{R}^{n\times n}$, an equivalent condition given in terms of the peripheral eigenvalues of A so that there exists a cone K in $\mathbb{R}^n$ such that A is K-nonnegative and (a) K is polyhedral, or (b) $\text{core}_K(A)$ is polyhedral (simplicial or a single ray) can be found in [TS94, Theorems 7.9, 7.8, 7.12, 7.10].
15. [TS94, Theorem 7.12] Let $A \in \mathbb{R}^{n\times n}$ with $\rho(A) > 0$ that satisfies the Perron–Schaefer condition. Let S denote the multiset of peripheral eigenvalues of A with maximal index (i.e., $\nu_A(\rho)$), the multiplicity of each element being equal to the number of corresponding blocks in the Jordan form of A of order $\nu_A(\rho)$. Let T be the multiset of peripheral eigenvalues of A for which there are corresponding blocks in the Jordan form of A of order less than $\nu_A(\rho)$, the multiplicity of each element being equal to the number of such corresponding blocks. The following conditions are equivalent:

    (a) There exists a cone K in $\mathbb{R}^n$ such that A is K-nonnegative and $\text{core}_K(A)$ is simplicial.

    (b) There exists a multisubset $\widetilde{T}$ of T such that $S \cup \widetilde{T}$ is the multiset union of certain complete sets of roots of unity multiplied by $\rho(A)$.

16. McDonald [McD03] refers to the condition (b) that appears in the preceding result as the **Tam–Schneider condition**. She also provides another condition, called the **extended Tam–Schneider condition**, which is necessary and sufficient for a collection of Jordan blocks to correspond to the peripheral spectrum of a nonnegative matrix.
17. [TS01] If P is K-nonnegative and $\mathbf{x}$ is K-semipositive, then P satisfies the local Perron–Schaefer condition at $\mathbf{x}$.
18. [Tam06] Let A be an $n \times n$ real matrix, and let $\mathbf{x}$ be a given nonzero vector of $\mathbb{R}^n$. The following conditions are equivalent:
    (a) A satisfies the local Perron–Schaefer condition at $\mathbf{x}$.
    (b) The restriction of A to $\text{span}\{A^i\mathbf{x} : i = 0, 1, \ldots\}$ satisfies the Perron–Schaefer condition.
    (c) For every (or, for some) nonnegative integer k, the closure of $\omega_k(A, \mathbf{x})$, where $\omega_k(A, \mathbf{x})$ consists of all linear combinations with nonnegative coefficients of $A^k\mathbf{x}$, $A^{k+1}\mathbf{x}, \ldots$, is a cone in its linear span.
    (d) There is a cone C in a subspace of $\mathbb{R}^n$ containing $\mathbf{x}$ such that $AC \subseteq C$.
19. The local Perron–Schaefer condition has played a role in the work of [TS01], [TS03], and [Tam04]. Further work involving this condition and the cones $\omega_k(A, \mathbf{x})$ (defined in the preceding fact) will appear in [Tam06].
20. One may apply results on the core of a nonnegative matrix to rederive simply many known results on the limiting behavior of Markov chains. An illustration can be found in [Tam01, Sec. 4.6].

26.4 Spectral Theory of K-Reducible Matrices

In this section, we touch upon the geometric version of the extensive combinatorial spectral theory of reducible nonnegative matrices first found in [Fro12, Sect. 11] and continued in [Sch56]. Many subsequent developments are reviewed in [Sch86] and [Her99]. Results on the geometric spectral theory of reducible K-nonnegative matrices may be largely found in a series of papers by B.S. Tam, some jointly with Wu and H. Schneider ([TW89], [Tam90], [TS94], [TS01], [TS03], [Tam04]). For a review containing considerably more information than this section, see [Tam01].

In some studies, the underlying cone is lattice-ordered (for a definition and much information, see [Sfr74]) and, in some studies, the Frobenius form of a reducible nonnegative matrix is generalized; see the work by Jang and Victory [JV93] on positive eventually compact linear operators on Banach lattices. However in the geometric spectral theory the Frobenius normal form of a nonnegative reducible matrix is not generalized as the underlying cone need not be lattice-ordered. Invariant faces are considered instead of the classes that play an important role in combinatorial spectral theory of nonnegative matrices; in particular, distinguished faces and semidistinguished faces are used in place of distinguished classes and semidistinguished classes, respectively. (For definitions of the preceding terms, see [TS01].)

It turns out that the various results on a reducible nonnegative matrix are extended to a K-nonnegative matrix in different degrees of generality. In particular, the Frobenius–Victory theorem ([Fro12], [Vic85]) is extended to a K-nonnegative matrix on a general cone. The following are extended to a polyhedral cone: The Rothblum index theorem ([Rot75]), a characterization (in terms of the accessibility relation between basic classes) for the spectral radius to have geometric multiplicity 1, for the spectral radius to have index 1 ([Sch56]), and a majorization relation between the (spectral) height characteristic and the (combinatorial) level characteristic of a nonnegative matrix ([HS91b]). Various conditions are used to generalize the theorem on equivalent conditions for equality of the two characteristics ([RiS78], [HS89], [HS91a]). Even for polyhedral cones there is no complete generalization for the nonnegative-basis theorem, not to mention the preferred-basis theorem ([Rot75], [RiS78], [Sch86], [HS88]). There is a natural conjecture for the latter case ([Tam04]). The attempts to carry out the extensions have also led to the identification of important new concepts or tools. For instance, the useful concepts of semidistinguished faces and of spectral pairs of faces associated with a K-nonnegative matrix are introduced in [TS01] in proving the cone version of some of the combinatorial theorems referred to above. To achieve these ends certain elementary analytic tools are also brought in.

Definitions:

Let P be a K-nonnegative matrix.

A nonzero P-invariant face F is a **semidistinguished face** if F contains in its relative interior a generalized eigenvector of P and if F is not the join of two P-invariant faces that are properly included in F.

A K**-semipositive Jordan chain** for P of length m (corresponding to $\rho(P)$) is a sequence of m K-semipositive vectors $\mathbf{x}, (P-\rho(P)I)\mathbf{x}, \ldots, (P-\rho(P)I)^{m-1}\mathbf{x}$ such that $(P-\rho(P)I)^m\mathbf{x} = \mathbf{0}$.

A basis for $N_\rho^\nu(P)$ is called a K**-semipositive basis** if it consists of K-semipositive vectors.

A basis for $N_\rho^\nu(P)$ is called a K**-semipositive Jordan basis** for P if it is composed of K-semipositive Jordan chains for P.

The set $C(P,K) = \{\mathbf{x} \in K : (P-\rho(P)I)^i\mathbf{x} \in K$ for all positive integers $i\}$ is called the **spectral cone** of P (for K corresponding to $\rho(P)$).

Denote ν_ρ by ν.

The **height characteristic of** P is the ν-tuple $\eta(P) = (\eta_1, ..., \eta_\nu)$ given by:

$$\eta_k = \dim(N_\rho^k(P)) - \dim(N_\rho^{k-1}(P)).$$

The **level characteristic of** P is the ν-tuple $\lambda(P) = (\lambda_1, \ldots, \lambda_\nu)$ given by:

$$\lambda_k = \dim \operatorname{span}(N_\rho^k(P) \cap K) - \dim \operatorname{span}(N_\rho^{k-1}(P) \cap K).$$

The **peak characteristic of** P is the ν-tuple $\xi(P) = (\xi_1, \ldots, \xi_\nu)$ given by:

$$\xi_k = \dim(P - \rho(P)I)^{k-1}(N_\rho^k \cap K).$$

If $A \in \mathbb{C}^{n\times n}$ and $\mathbf{x}$ is a nonzero vector of $\mathbb{C}^n$, then the **order of x relative to** A, denoted by $\mathrm{ord}_A(\mathbf{x})$, is defined to be the maximum of the orders of the generalized eigenvectors, each corresponding to an eigenvalue of modulus $\rho_{\mathbf{x}}(A)$ that appear in the representation of $\mathbf{x}$ as a sum of generalized eigenvectors of A.

The ordered pair $(\rho_{\mathbf{x}}(A), \mathrm{ord}_A(\mathbf{x}))$ is called the **spectral pair of x relative to** A and is denoted by $\mathrm{sp}_A(\mathbf{x})$. We also set $\mathrm{sp}_A(\mathbf{0}) = (0, 0)$ to take care of the zero vector $\mathbf{0}$.

We use $\preceq$ to denote the lexicographic ordering between ordered pairs of real numbers, i.e., $(a, b) \preceq (c, d)$ if either $a < c$, or $a = c$ and $b \leq d$. In case $(a, b) \preceq (c, d)$ but $(a, b) \neq (c, d)$, we write $(a, b) \prec (c, d)$.

Facts:

1. If $A \in \mathbb{C}^{n\times n}$ and $\mathbf{x}$ is a vector of $\mathbb{C}^n$, then $\mathrm{ord}_A(\mathbf{x})$ is equal to the size of the largest Jordan block in the Jordan form of the restriction of A to the A-cyclic subspace generated by $\mathbf{x}$ for a peripheral eigenvalue.

   Let P be a K-nonnegative matrix.
2. In the nonnegative matrix case, the present definition of the level characteristic of P is equivalent to the usual graph-theoretic definition; see [NS94, (3.2)] or [Tam04, Remark 2.2].
3. [TS01] For any $\mathbf{x} \in K$, the smallest P-invariant face containing $\mathbf{x}$ is equal to $\Phi(\hat{\mathbf{x}})$, where $\hat{\mathbf{x}} = (I + P)^{n-1}\mathbf{x}$. Furthermore, $\mathrm{sp}_P(\mathbf{x}) = \mathrm{sp}_P(\hat{\mathbf{x}})$. In the nonnegative matrix case, the said face is also equal to F_J, where F_J is as defined in Example 1 of Section 26.1 and J is the union of all classes of P having access to $\mathrm{supp}(\mathbf{x}) = \{i : \mathbf{x}_i > 0\}$. (For definitions of classes and the accessibility relation, see Chapter 9.)
4. [TS01] For any face F of K, P-invariant or not, the value of the spectral pair $\mathrm{sp}_P(\mathbf{x})$ is independent of the choice of $\mathbf{x}$ from the relative interior of F. This common value, denoted by $\mathrm{sp}_A(F)$, is referred to as the **spectral pair of** F **relative to** A.
5. [TS01] For any faces F, G of K, we have

   (a) $\mathrm{sp}_P(F) = \mathrm{sp}_P(\hat{F})$, where $\hat{F}$ is the smallest P-invariant face of K, including F.

   (b) If $F \subseteq G$, then $\mathrm{sp}_P(F) \preceq \mathrm{sp}_P(G)$. If F, G are P-invariant faces and $F \subset G$, then $\mathrm{sp}_P(F) \preceq \mathrm{sp}_P(G)$; *viz.* either $\rho[F] < \rho[G]$ or $\rho[F] = \rho[G]$ and $\nu_{\rho[F]}[F] \leq \nu_{\rho[G]}[G]$.
6. [TS01] If K is a cone with the property that the dual cone of each of its faces is a facially exposed cone, for instance, when K is a polyhedral cone, a perfect cone, or equals $P(n)$ (see [TS01] for definitions), then for any nonzero P-invariant face G, G is semidistinguished if and only if $\mathrm{sp}_P(F) \prec \mathrm{sp}_P(G)$ for all P-invariant faces F properly included in G.
7. [Tam04] (Cone version of the Frobenius–Victory theorem, [Fro12], [Vic85], [Sch86])

   (a) For any real number λ, λ is a distinguished eigenvalue of P if and only if $\lambda = \rho[F]$ for some distinguished face F of K.

   (b) If F is a distinguished face, then there is (up to multiples) a unique eigenvector $\mathbf{x}$ of P corresponding to $\rho[F]$ that lies in F. Furthermore, $\mathbf{x}$ belongs to the relative interior of F.

   (c) For each distinguished eigenvalue λ of P, the extreme vectors of the cone $N_\lambda^1(P) \cap K$ are precisely all the distinguished eigenvectors of P that lie in the relative interior of certain distinguished faces of K associated with λ.
8. Let P be a nonnegative matrix. The Jordan form of P contains only one Jordan block corresponding to $\rho(P)$ if and only if any two basic classes of P are comparable (with respect to the accessibility relation); all Jordan blocks corresponding to $\rho(P)$ are of size 1 if and only if no two basic classes are comparable ([Schn56]). An extension of these results to a K-nonnegative matrix on a class of cones that contains all polyhedral cones can be found in [TS01, Theorems 7.2 and 7.1].

9. [Tam90, Theorem 7.5] If K is polyhedral, then:

   (a) There is a K-semipositive Jordan chain for P of length ν_ρ; thus, there is a K-semipositive vector in $N_\rho^\nu(P)$ of order ν_ρ, *viz.* $\mathrm{ord}_\rho = \nu_\rho$.

   (b) The Perron space $N_\rho^\nu(P)$ has a basis consisting of K-semipositive vectors.

   However, when K is nonpolyhedral, there need not exist a K-semipositive vector in $N_\rho^\nu(P)$ of order ν_ρ, *viz.* $\mathrm{ord}_\rho < \nu_\rho$. For a general distinguished eigenvalue λ, we always have $\mathrm{ord}_\lambda \leq \nu_\lambda$, no matter whether K is polyhedral or not.
10. Part (b) of the preceding fact is not yet a complete cone version of the nonnegative-basis theorem, as the latter theorem guarantees the existence of a basis for the Perron space that consists of semipositive vectors that satisfy certain combinatorial properties. For a conjecture on a cone version of the nonnegative-basis theorem, see [Tam04, Conj. 9.1].
11. [TS01, Theorem 5.1] (Cone version of the (combinatorial) generalization of the Rothblum index theorem, [Rot75], [HS88]).

    Let K be a polyhedral cone. Let λ be a distinguished eigenvalue of P for K. Then there is a chain $F_1 \subset F_2 \subset \ldots \subset F_k$ of $k = \mathrm{ord}_\lambda$ distinct semidistinguished faces of K associated with λ, but there is no such chain with more than ord_λ members. When K is a general cone, the maximum cardinality of a chain of semidistinguished faces associated with a distinguished eigenvalue λ may be less than, equal to, or greater than ord_λ; see [TS01, Ex. 5.3, 5.4, 5.5].
12. For $K = (\mathbb{R}_0^+)^n$, *viz.* P is a nonnegative matrix, characterizations of different types of P-invariant faces (in particular, the distinguished and semidistinguished faces) are given in [TS01] (in terms of the concept of an initial subset for P; see [HS88] or [TS01] for definition of an initial subset).
13. [Tam04] The spectral cone $C(P, K)$ is always invariant under $P - \rho(P)I$ and satisfies:

    $$N_\rho^1(P) \cap K \subseteq C(A, K) \subseteq N_\rho^\nu(P) \cap K.$$

    If K is polyhedral, then $C(A, K)$ is a polyhedral cone in $N_\rho^\nu(P)$.
14. (Generalization of corresponding results on nonnegative matrices, [NS94]) We always have $\xi_k(P) \leq \eta_k(P)$ and $\xi_k(P) \leq \lambda_k(P)$ for $k = 1, \ldots, \nu_\rho$.
15. [Tam04, Theorem 5.9] Consider the following conditions:

    (a) $\eta(P) = \lambda(P)$.

    (b) $\eta(P) = \xi(P)$.

    (c) For each k, $k = 1, \ldots, \nu_\rho$, $N_\rho^k(P)$ contains a K-semipositive basis.

    (d) There exists a K-semipositive Jordan basis for P.

    (e) For each k, $k = 1, \ldots, \nu_\rho$, $N_\rho^k(P)$ has a basis consisting of vectors taken from $N_\rho^k(P) \cap C(P, K)$.

    (f) For each k, $k = 1, \ldots, \nu_\rho$, we have

    $$\eta_k(P) = \dim(P - \rho(P)I)^{k-1}[N_\rho^k(P) \cap C(P, K)].$$

    Conditions (a) to (c) are equivalent and so are conditions (d) to (f). Moreover, we always have (a)$\Longrightarrow$(d), and when K is polyhedral, conditions (a) to (f) are all equivalent.
16. As shown in [Tam04], the level of a nonzero vector $\mathbf{x} \in N_\rho^\nu(P)$ can be defined to be the smallest positive integer k such that $\mathbf{x} \in \mathrm{span}(N_\rho^k(P) \cap K)$; when there is no such k the level is taken to be ∞. Then the concepts of K-semipositive level basis, height-level basis, peak vector, etc., can be introduced and further conditions can be added to the list given in the preceding result.
17. [Tam04, Theorem 7.2] If K is polyhedral, then $\lambda(P) \preceq \eta(P)$.
18. Cone-theoretic proofs for the preferred-basis theorem for a nonnegative matrix and for a result about the nonnegativity structure of the principal components of a nonnegative matrix can be found in [Tam04].

26.5 Linear Equations over Cones

Given a K-nonnegative matrix P and a vector $\mathbf{b} \in K$, in this section we consider the solvability of following two linear equations over cones and some consequences:

$$(\lambda I - P)\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \in K \tag{26.1}$$

and

$$(P - \lambda I)\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \in K. \tag{26.2}$$

Equation (26.1) has been treated by several authors in finite-dimensional as well as infinite-dimensional settings, and several equivalent conditions for its solvability have been found. (See [TS03] for a detailed historical account.) The study of Equation (26.2) is relatively new. A treatment of the equation by graph-theoretic arguments for the special case when $\lambda = \rho(P)$ and $K = (\mathbb{R}_0^+)^n$ can be found in [TW89]. The general case is considered in [TS03]. It turns out that the solvability of Equation (26.2) is a more delicate problem. It depends on whether λ is greater than, equal to, or less than $\rho_{\mathbf{b}}(P)$.

Facts:

Let P be a K-nonnegative matrix, let $\mathbf{0} \neq \mathbf{b} \in \mathbf{K}$, and let λ be a given positive real number.

1. [TS03, Theorem 3.1] The following conditions are equivalent:
   (a) Equation (26.1) is solvable.
   (b) $\rho_{\mathbf{b}}(P) < \lambda$.
   (c) $\lim_{m\to\infty} \sum_{j=0}^{m} \lambda^{-j} P^j \mathbf{b}$ exists.
   (d) $\lim_{m\to\infty} (\lambda^{-1} P)^m \mathbf{b} = \mathbf{0}$.
   (e) $\langle \mathbf{z}, \mathbf{b} \rangle = 0$ for each generalized eigenvector $\mathbf{z}$ of P^T corresponding to an eigenvalue with modulus greater than or equal to λ.
   (f) $\langle \mathbf{z}, \mathbf{b} \rangle = 0$ for each generalized eigenvector $\mathbf{z}$ of P^T corresponding to a distinguished eigenvalue of P for K that is greater than or equal to λ.
2. For a fixed λ, the set $(\lambda I - P)K \cap K$, which consists of precisely all vectors $\mathbf{b} \in K$ for which Equation (26.1) has a solution, is equal to $\{\mathbf{b} \in K : \rho_{\mathbf{b}}(P) < \lambda\}$ and is a face of K.
3. For a fixed λ, the set $(P - \lambda I)K \cap K$, which consists of precisely all vectors $\mathbf{b} \in K$ for which Equation (26.2) has a solution, is, in general, not a face of K.
4. [TS03, Theorem 4.1] When $\lambda > \rho_{\mathbf{b}}(P)$, Equation (26.2) is solvable if and only if λ is a distinguished eigenvalue of P for K and $\mathbf{b} \in \Phi(N_\lambda^1(P) \cap K)$.
5. [TS03, Theorem 4.5] When $\lambda = \rho_{\mathbf{b}}(P)$, if Equation (26.2) is solvable, then
   $\mathbf{b} \in (P - \rho_{\mathbf{b}}(P)I)\Phi$
   $(N^{\nu}_{\rho_{\mathbf{b}}(P)}(P) \cap K)$.
6. [TS03, Theorem 4.19] Let r denote the largest real eigenvalue of P less than $\rho(P)$. (If no such eigenvalues exist, take $r = -\infty$.) Then for any λ, $r < \lambda < \rho(P)$, we have

   $$\Phi((P - \lambda I)K \cap K) = \Phi(N_\rho^{\nu}(P) \cap K).$$

   Thus, a necessary condition for Equation (26.2) to have a solution is that $\mathbf{b} \overset{K}{\leq} \mathbf{u}$ for some $\mathbf{u} \in N_\rho^{\nu}(P) \cap K$.
7. [TS03, Theorem 5.11] Consider the following conditions:
   (a) $\rho(P) \in \Sigma_1(P^T)$.
   (b) $N_\rho^{\nu}(P) \cap K = N_\rho^1(P) \cap K$, and P has no eigenvectors in $\Phi(N_\rho^1(P) \cap K)$ corresponding to an eigenvalue other than $\rho(P)$.

(c) $K \cap (P - \rho(P)I)K = \{\mathbf{0}\}$ (equivalently, $\mathbf{x} \geq^K \mathbf{0}$, $P\mathbf{x} \geq^K \rho(P)\mathbf{x}$ imply that $P\mathbf{x} = \rho(P)\mathbf{x}$).

We always have (a)$\Longrightarrow$(b)$\Longrightarrow$(c). When K is polyhedral, conditions (a), (b), and (c) are equivalent. When K is nonpolyhedral, the missing implications do not hold.

26.6 Elementary Analytic Results

In geometric spectral theory, besides the linear-algebraic method and the cone-theoretic method, certain elementary analytic methods have also been called into play; for example, the use of Jordan form or the components of a matrix. This approach may have begun with the work of Birkhoff [Bir67] and it was followed by Vandergraft [Van68] and Schneider [Sch81]. Friedland and Schneider [FS80] and Rothblum [Rot81] have also studied the asymptotic behavior of the powers of a nonnegative matrix, or their variants, by elementary analytic methods. The papers [TS94] and [TS01] in the series also need a certain kind of analytic argument in their proofs; more specifically, they each make use of the K-nonnegativity of a certain matrix, either itself a component or a matrix defined in terms of the components of a given K-nonnegative matrix (see Facts 3 and 4 in this section). In [HNR90], Hartwig, Neumann, and Rose offer a (linear) algebraic-analytic approach to the Perron–Frobenius theory of a nonnegative matrix, one which utilizes the resolvent expansion, but does not involve the Frobenius normal form. Their approach is further developed by Neumann and Schneider ([NS92], [NS93], [NS94]). By employing the concept of spectral cone and combining the cone-theoretic methods developed in the earlier papers of the series with this algebraic-analytic method, Tam [Tam04] offers a unified treatment to reprove or extend (or partly extend) several well-known results in the combinatorial spectral theory of nonnegative matrices. The proofs given in [Tam04] rely on the fact that if K is a cone in $\mathbb{R}^n$, then the set $\pi(K)$ that consists of all K-nonnegative matrices is a cone in the matrix space $\mathbb{R}^{n\times n}$ and if, in addition, K is polyhedral, then so is $\pi(K)$ ([Fen53, p. 22], [SV70], [Tam77]). See [Tam01, Sec. 6.5] and [Tam04, Sec. 9] for further remarks on the use of the cone $\pi(K)$ in the study of the spectral properties of K-nonnegative matrices.

In this section, we collect a few elementary analytic results (whose proofs rely on the Jordan form), which have proved to be useful in the study of the geometric spectral theory. In particular, Facts 3, 4, and 5 identify members of $\pi(K)$. As such, they can be regarded as nice results, which are difficult to come by for the following reason: If K is nonsimplicial, then $\pi(K)$ must contain matrices that are not nonnegative linear combinations of its rank-one members ([Tam77]). However, not much is known about such matrices ([Tam92]).

Definitions:

Let P be a K-nonnegative matrix. Denote ν_ρ by ν.

The **principal eigenprojection** of P, denoted by $Z_P^{(0)}$, is the projection of $\mathbb{C}^n$ onto the Perron space N_ρ^ν along the direct sum of other generalized eigenspaces of P.

For $k = 0, \ldots, \nu$, the k**th principal component of** P is given by

$$Z_P^{(k)} = (P - \rho(P))^k Z_P^{(0)}.$$

The kth component of P corresponding to an eigenvalue λ is defined in a similar way.

For $k = 0, \ldots, \rho$, the k**th transform principal component of** P is given by:

$$J_P^{(k)}(\varepsilon) = Z_P^{(k)} + Z_P^{(k+1)}/\varepsilon + \cdots + Z_P^{(\nu-1)}/\varepsilon^{\nu-k-1} \quad \text{for all } \varepsilon \in \mathbb{C}\backslash\{0\}.$$

Facts:

Let P be a K-nonnegative matrix. Denote ν_ρ by ν.

1. [Kar59], [Sch81] $Z_P^{(\nu-1)}$ is K-nonnegative.
2. [TS94, Theorem 4.19(i)] The sum of the νth components of P corresponding to its peripheral eigenvalues is K-nonnegative; it is the limit of a convergent subsequence of $((\nu-1)!P^k/[\rho^{k-\nu+1}k^{\nu-1}])$.

3. [Tam04, Theorem 3.6(i)] If K is a polyhedral cone, then for $k = 0, \ldots, \nu - 1$, $J_P^{(k)}(\varepsilon)$ is K-nonnegative for all sufficiently small positive ε.

26.7 Splitting Theorems and Stability

Splitting theorems for matrices have played a large role in the study of convergence of iterations in numerical linear algebra; see [Var62]. Here we present a cone version of a splitting theorem which is proven in [Sch65] and applied to stability (inertia) theorems for matrices. A closely related result is generalized to operators on a partially ordered Banach space in [DH03] and [Dam04]. There it is used to describe stability properties of (stochastic) control systems and to derive non-local convergence results for Newton's method applied to nonlinear operator equation of Riccati type. We also discuss several kinds of positivity for operators involving a cone that are relevant to the applications mentioned.

For recent splitting theorems involving cones, see [SSA05]. For applications of theorems of the alternative for cones to the stability of matrices, see [CHS97]. Cones occur in many parts of stability theory; see, for instance, [Her98].

Definitions:

Let K be a cone in $\mathbb{R}^n$ and let $A \in \mathbb{R}^{n\times n}$.

A is **positive stable** if spec$(A) \subseteq \mathbb{C}^+$ *viz.* the spectrum of A is contained in the open right half-plane.

A is K**-inverse nonnegative** if A is nonsingular and A^{-1} is K-nonnegative.

A is K**-resolvent nonnegative** if there exists an $\alpha_0 \in \mathbb{R}$ such that, for all $\alpha > \alpha_0$, $\alpha I - A$ is K-inverse nonnegative.

A is **cross-positive** on K if for all $\mathbf{x} \in K$, $\mathbf{y} \in K^*$, $\mathbf{y}^T\mathbf{x} = 0$ implies $\mathbf{y}^T A\mathbf{x} \geq 0$.

A is a $\boldsymbol{Z}$**-matrix** if all of its off-diagonal entries are nonpositive.

Facts:

Let K be a cone in $\mathbb{R}^n$.

1. A is K-resolvent nonnegative if and only if A is cross-positive on K. Other equivalent conditions and also Perron-Frobenius type theorems for the class of cross-positive matrices can be found in [Els74], [SV70] or [BNS89].
2. When K is $(\mathbb{R}_0^+)^n$, A is cross-positive on K if and only if $-A$ is a Z-matrix.
3. [Sch65], [Sch97]. Let $T = R - P$ where $R, P \in \mathbb{R}^{n\times n}$ and suppose that P is K-nonnegative. If R satisfies $R(\text{int}K) \supseteq \text{int}K$ or $R(\text{int}K) \cap \text{int}K = \emptyset$, then the following are equivalent:
   (a) T is K-inverse nonnegative.
   (b) For all $\mathbf{y} >^K \mathbf{0}$ there exists (unique) $\mathbf{x} >^K \mathbf{0}$ such that $\mathbf{y} = T\mathbf{x}$.
   (c) There exists $\mathbf{x} >^K \mathbf{0}$ such that $T\mathbf{x} >^K \mathbf{0}$.
   (d) There exists $\mathbf{x} \geq^K \mathbf{0}$ such that $T\mathbf{x} >^K \mathbf{0}$.
   (e) R is K-inverse nonnegative and $\rho(R^{-1}P) < 1$.
4. Let $T \in \mathbb{R}^{n\times n}$. If $-T$ is K-resolvent nonnegative, then T satisfies $T(\text{int}K) \supseteq \text{int}K$ or $T(\text{int}K) \cap \text{int}K = \emptyset$. But the converse is false, see Example 1 below.
5. [DH03, Theorem 2.11], [Dam04, Theorem 3.2.10]. Let T, R, P be as given in Fact 3. If $-R$ is K-resolvent nonnegative, then conditions (a)–(e) of Fact 3 are equivalent. Moreover, the following are additional equivalent conditions:
   (f) T is positive stable.
   (g) R is positive stable and $\rho(R^{-1}P) < 1$.
6. If K is $(\mathbb{R}_0^+)^n$, $R = \alpha I$ and P is a nonnegative matrix, then $T = R - P$ is a Z-matrix. It satisfies the equivalent conditions (a)–(g) of Facts 3 and 5 if and only if it is an M-matrix [BP79, Chapter 6].

7. (Special case of Fact 5 with $P = 0$). Let $T \in \mathbb{R}^{n \times n}$. If $-T$ is K-resolvent nonnegetive, then conditions (a)–(d) of Fact 3 and conditions (f) of Fact 5 are equivalent.
8. In [GT06] a matrix T is called a Z-tranformation on K if $-T$ is cross-positive on K. Many properties on Z-matrices, such as being a P-matrix, a Q-matrix (which has connection with the linear complementarity problem), an inverse-nonnegative matrix, a positive stable matrix, a diagonally stable matrix, etc., are extended to Z-transformations. For a Z-transformation, the equivalence of these properties is examined for various kinds of cones, particularly for symmetric cones in Euclidean Jordan algebras.
9. [Schn65], [Schn97]. (Special case of Fact 3 with K equal to the cone of positive semi-definite matrices in the real space of $n \times n$ Hermitian matrices, and $R(H) = AHA^*$, $P(H) = \Sigma_{k=1}^{s} C_k H C_K^*$). Let $A, C_k, k = 1, \ldots, s$ be complex $n \times n$ matrices which can be simultaneously upper triangularized by similarity. Then there exists a natural correspondence $\alpha_i, \gamma_i^{(k)}$ of the eigenvalues of $A, C_k, k = 1, \ldots s$. For Hermitian H, let $T(H) = AHA^* - \Sigma_{k=1}^{s} C_k H C_k^*$. Then the following are equivalent:

   (a) $|\alpha_i|^2 - \Sigma_{k=1}^{s} |\gamma_i^{(k)}|^2 > 0$, $i = 1, \ldots, n$.

   (b) For all positive definite G there exists a (unique) positive definite H such that $T(H) = G$.

   (c) There exists a positive definite H such that $T(H)$ is positive definite.

10. Gantmacher-Lyapunov [Gan59, Chapter XV] (Special case of Fact 9 with A replaced by $A + I, s = 2, C_1 = A, C_2 = I$, and special case of Fact 7 with K equal to the cone of positive semi-definite matrices in the real space of $n \times n$ Hermitian matrices and $T(H) = AH + HA^*$).
    Let $A \in \mathbb{C}^{n \times n}$. The following are equivalent:

    (a) For all positive definite G there exists a (unique) positive definite H such that $AH + HA^* = G$.

    (b) There exists a positive definite H such that $AH + HA^*$ is positive definite.

    (c) A is positive stable.

11. Stein [Ste52](Special case of Fact 9 with $A = I, s = 1, C_1 = C$, and special case of Fact 7 with $T(H) = H - CHC^*$).
    Let $C \in \mathbb{C}^{n \times n}$. The following are equivalent:

    (a) There exists a positive definite H such that $H - CHC^*$ is positive definite.

    (b) The spectrum of C is contained in the open unit disk.

Examples:

Let $K = (\mathbb{R}_0^+)^2$ and take $T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Then $TK = K$ and so $T(\text{int}K) \supseteq \text{int}K$. Note that $\langle T\mathbf{e_1}, \mathbf{e_2} \rangle = 1 > 0$ whereas $\langle \mathbf{e_1}, \mathbf{e_2} \rangle = 0$; so $-T$ is not cross-positive on K and hence not K-resolvent nonnegative. Since the eigenvalues of T are -1 and 1, T is not positive stable. This example tells us that the converse of Fact 4 is false. It also shows that to the list of equivalent conditions of Fact 3 we cannot add condition (f) of Fact 5.

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